

Abstract Linear Algebra

Prof. Lanier

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Please pick up a copy of the handout.

Try those warm-up problems.

For you to do ASAP:

- Read through the syllabus and course docs
- Fill out the start-of-course survey by tonight
- Read sections 1-4 of Chapter 1 of *LADW*
- Form a study group
- Explore the Canvas site some
- Come to office hours! - tonight on Zoom, 7:00-7:30pm

Today in class:

- Discuss material from Ch 1.1-4
- Finish overview of the main ideas of the course
- Discuss flex

distribution

$$k\left(\binom{a}{b} + \binom{c}{d}\right) \stackrel{?}{=} k\binom{a}{b} + k\binom{c}{d}$$

distribution

$$k\left(\begin{pmatrix} a \\ b \end{pmatrix} + \begin{pmatrix} c \\ d \end{pmatrix}\right) \stackrel{?}{=} k\begin{pmatrix} a \\ b \end{pmatrix} + k\begin{pmatrix} c \\ d \end{pmatrix}$$

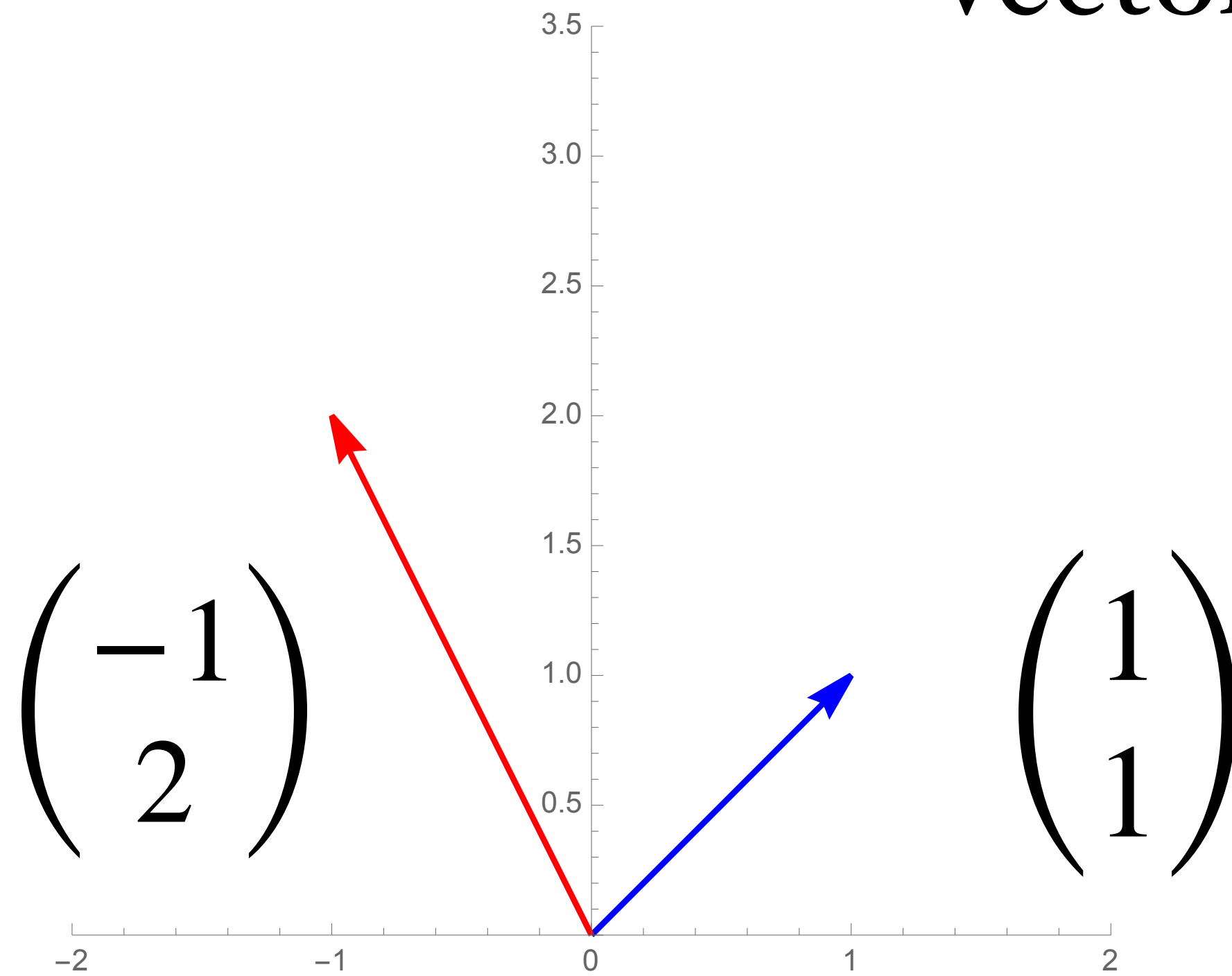
$$\begin{pmatrix} ka + kc \\ kb + kd \end{pmatrix}$$



linear combination

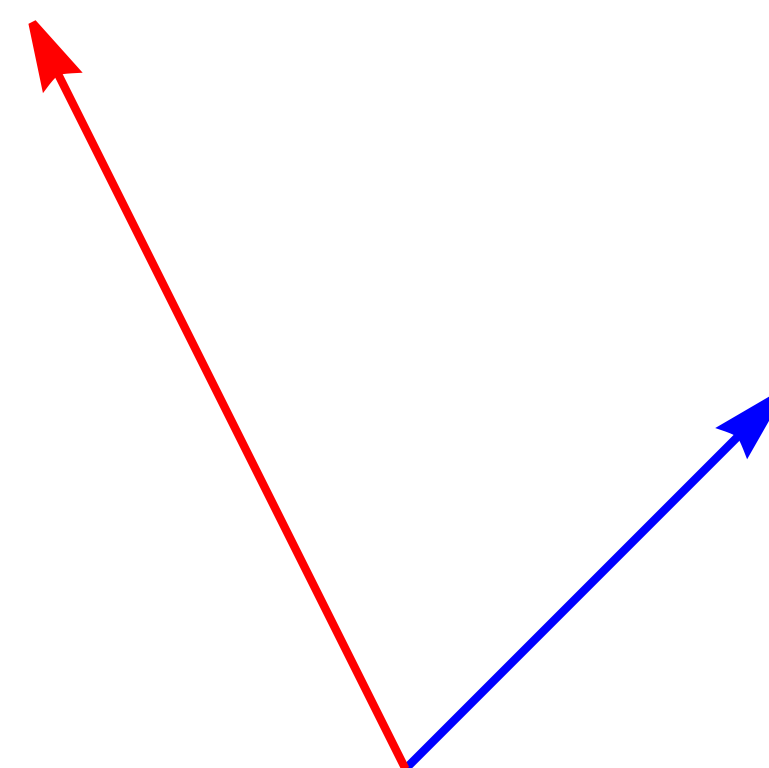
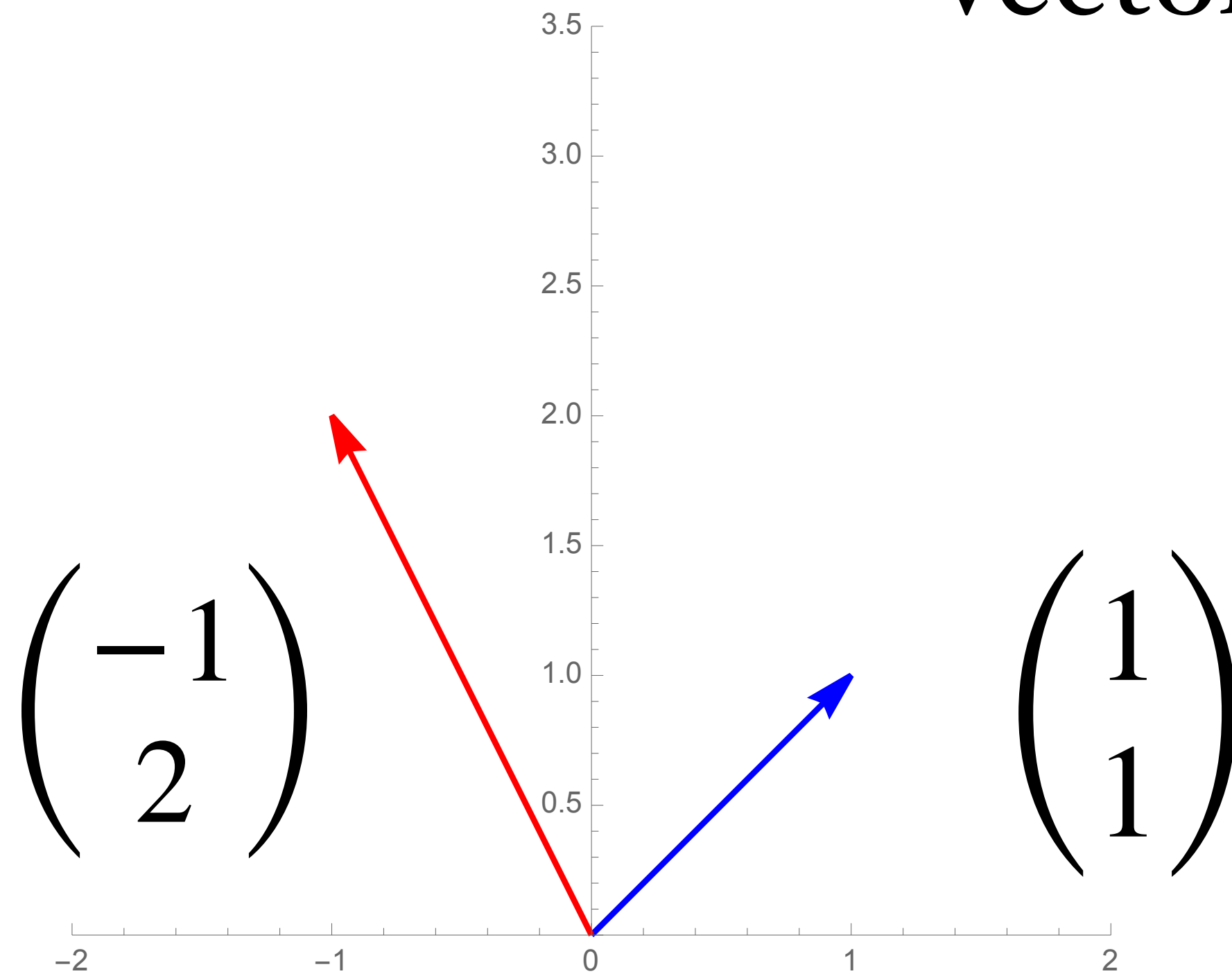
$$u, v \in V \quad \rightsquigarrow \quad au + bv$$

vectors in $\mathbb{R}^2$



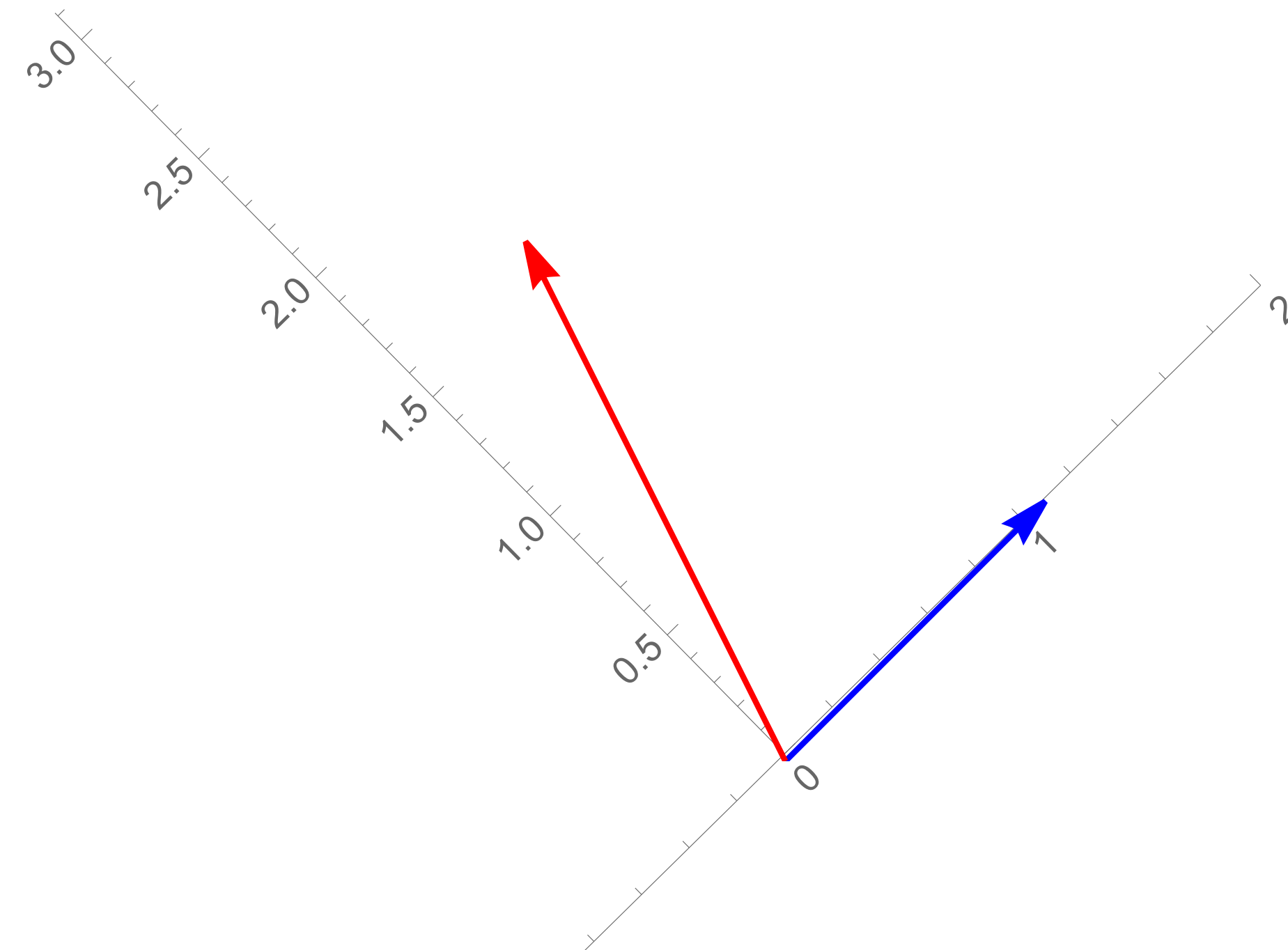
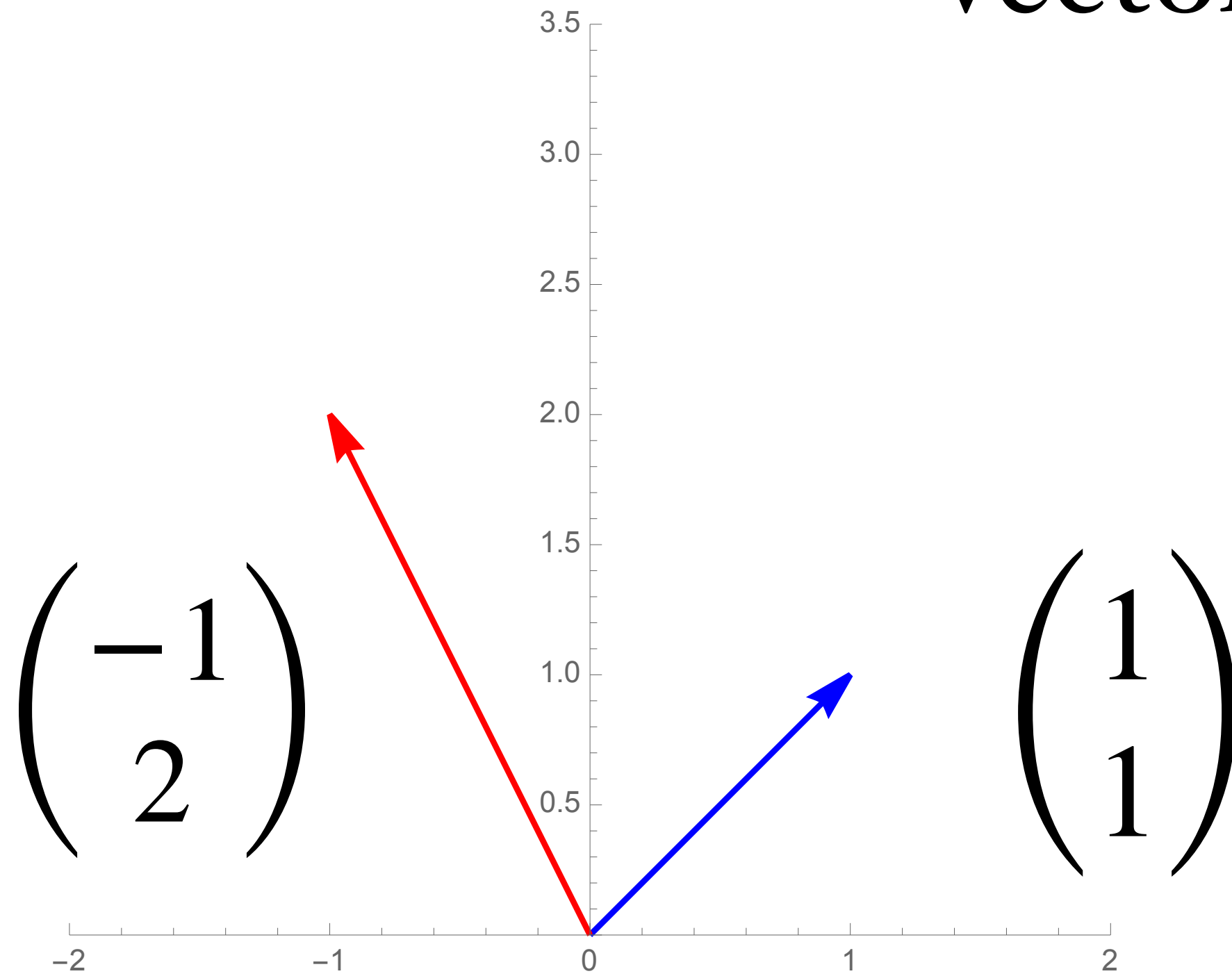
coordinates vs. coordinate-free

vectors in $\mathbb{R}^2$



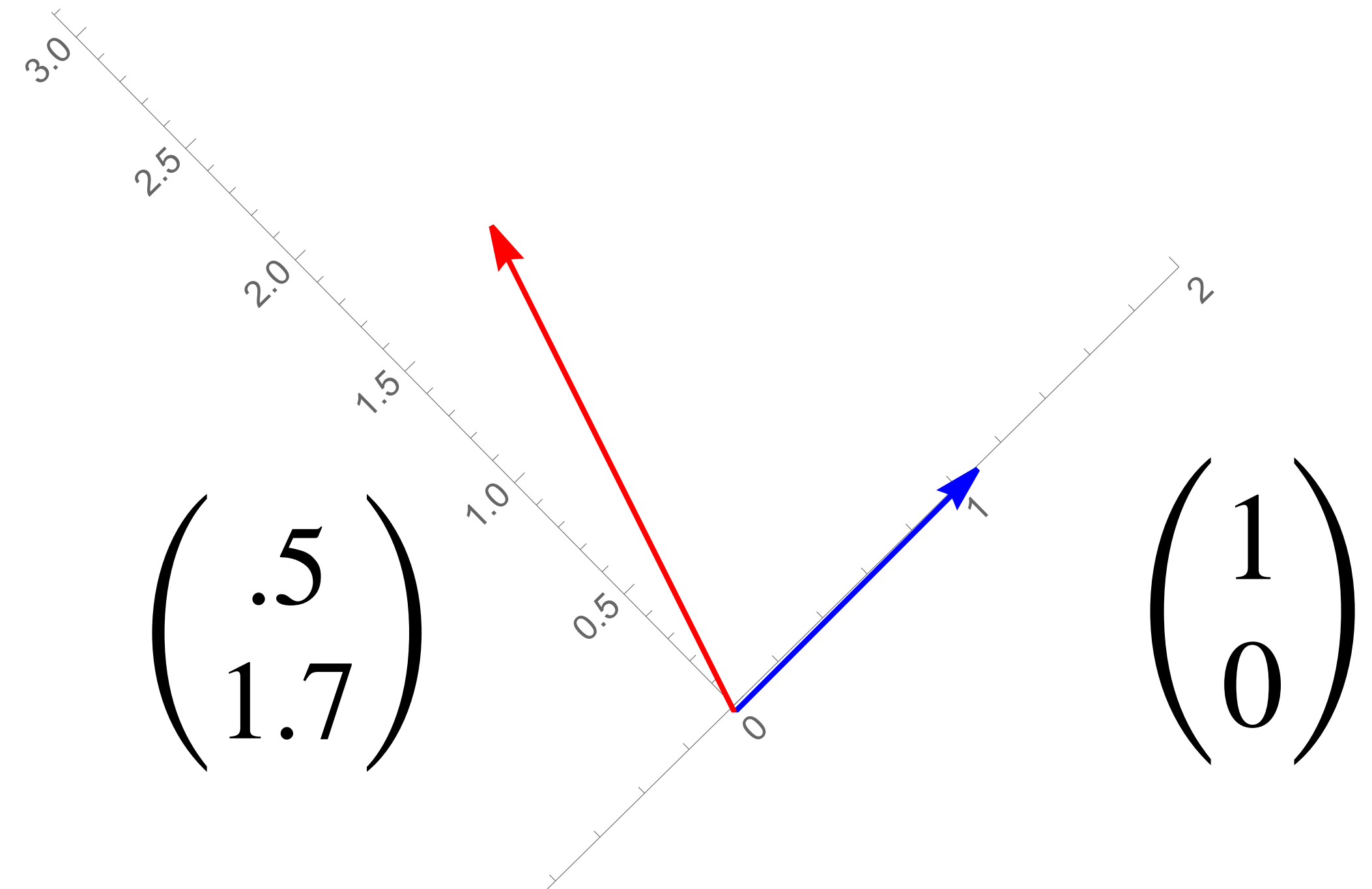
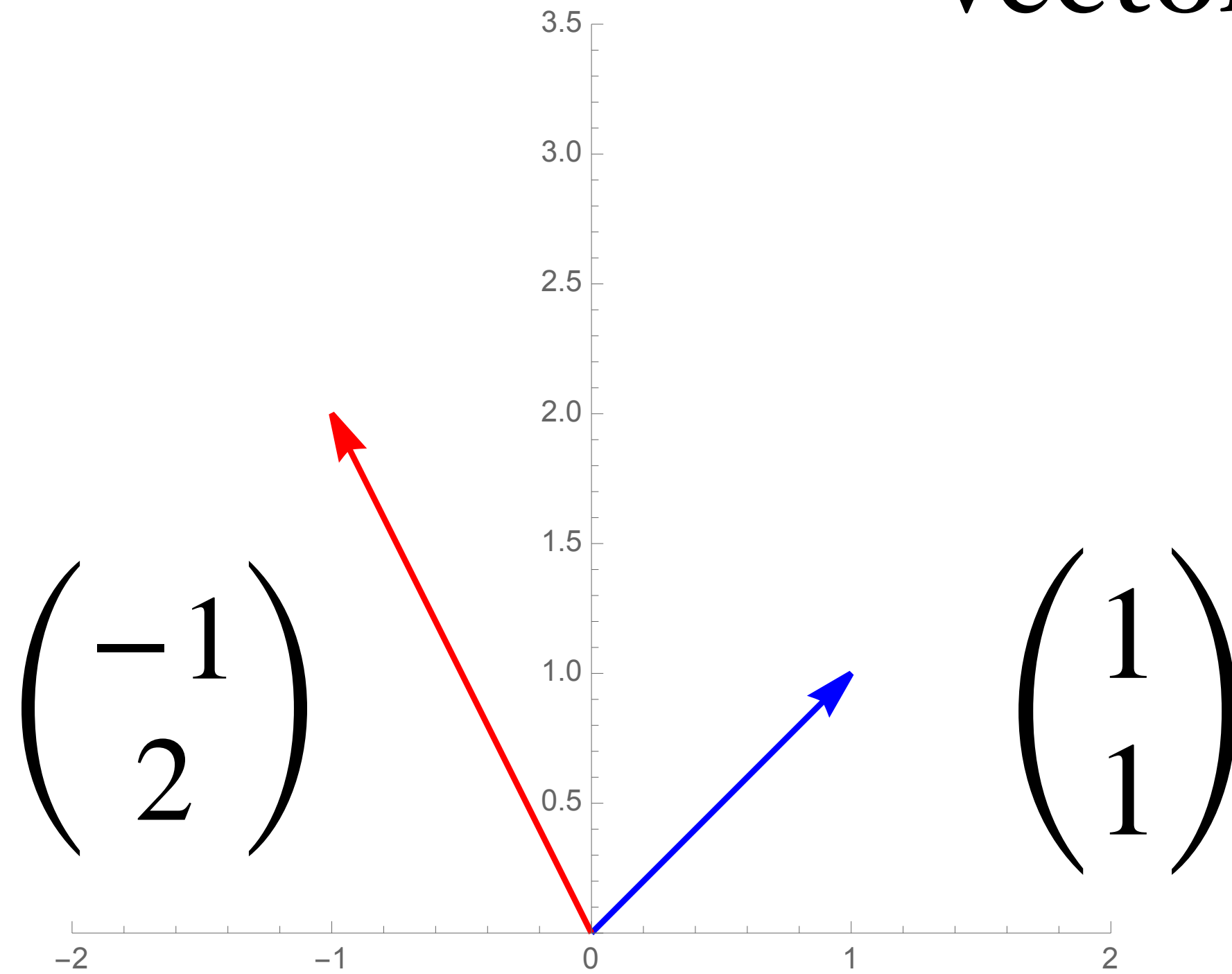
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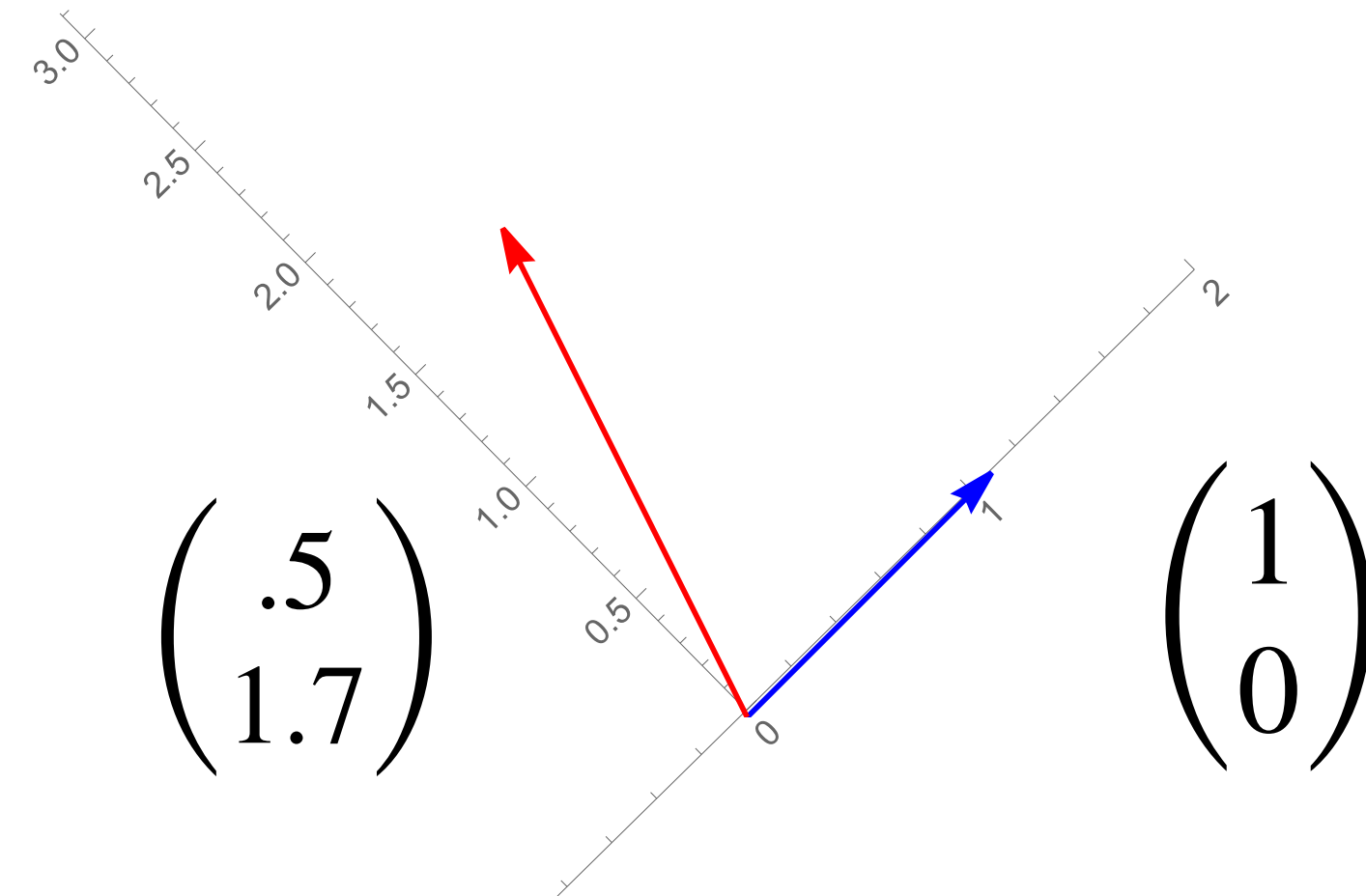
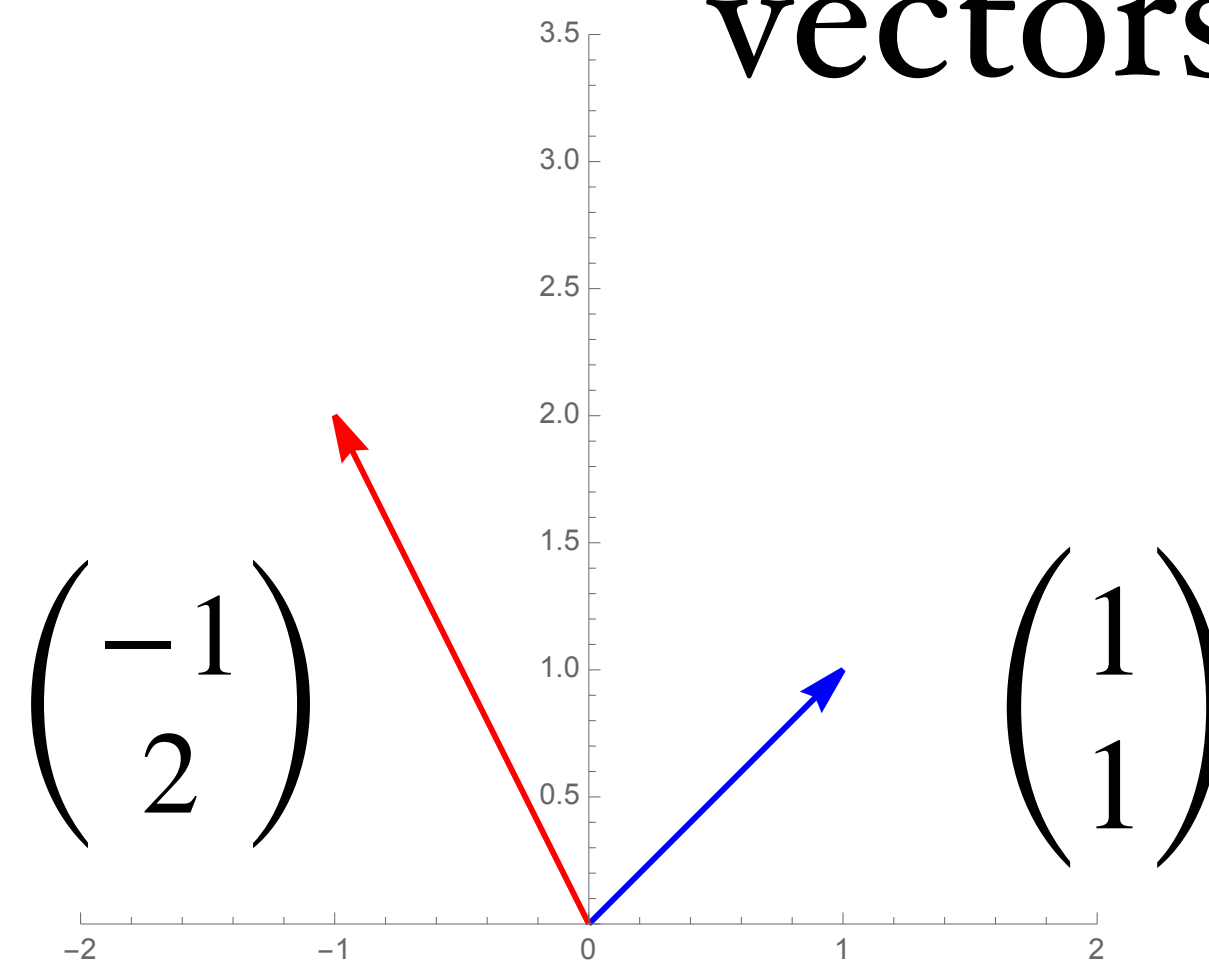
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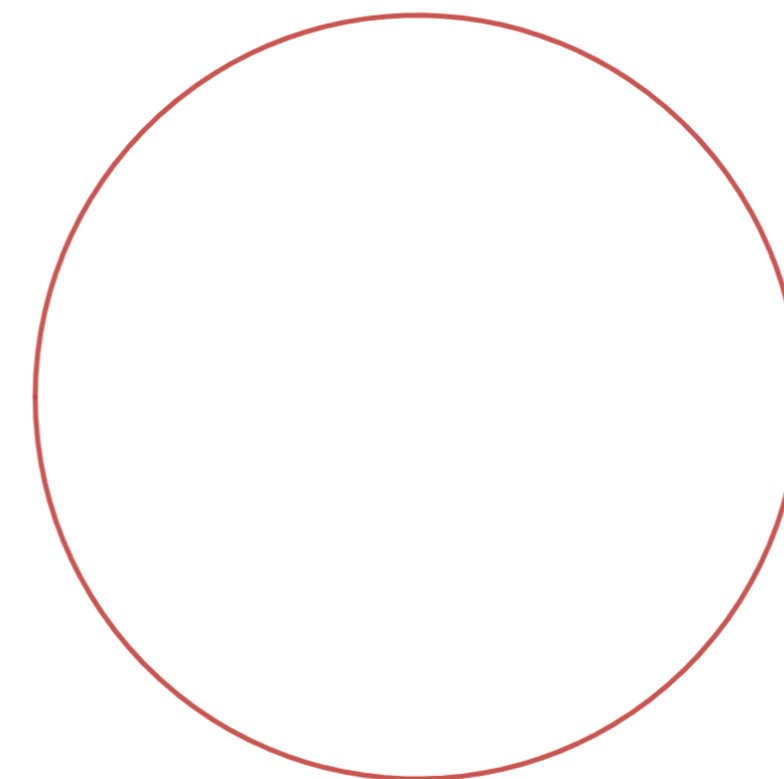
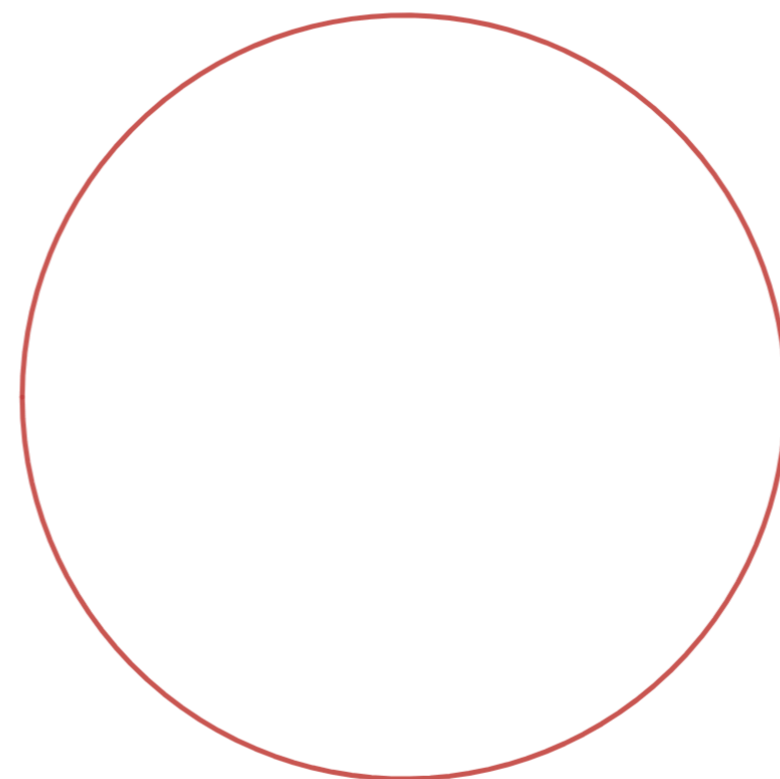
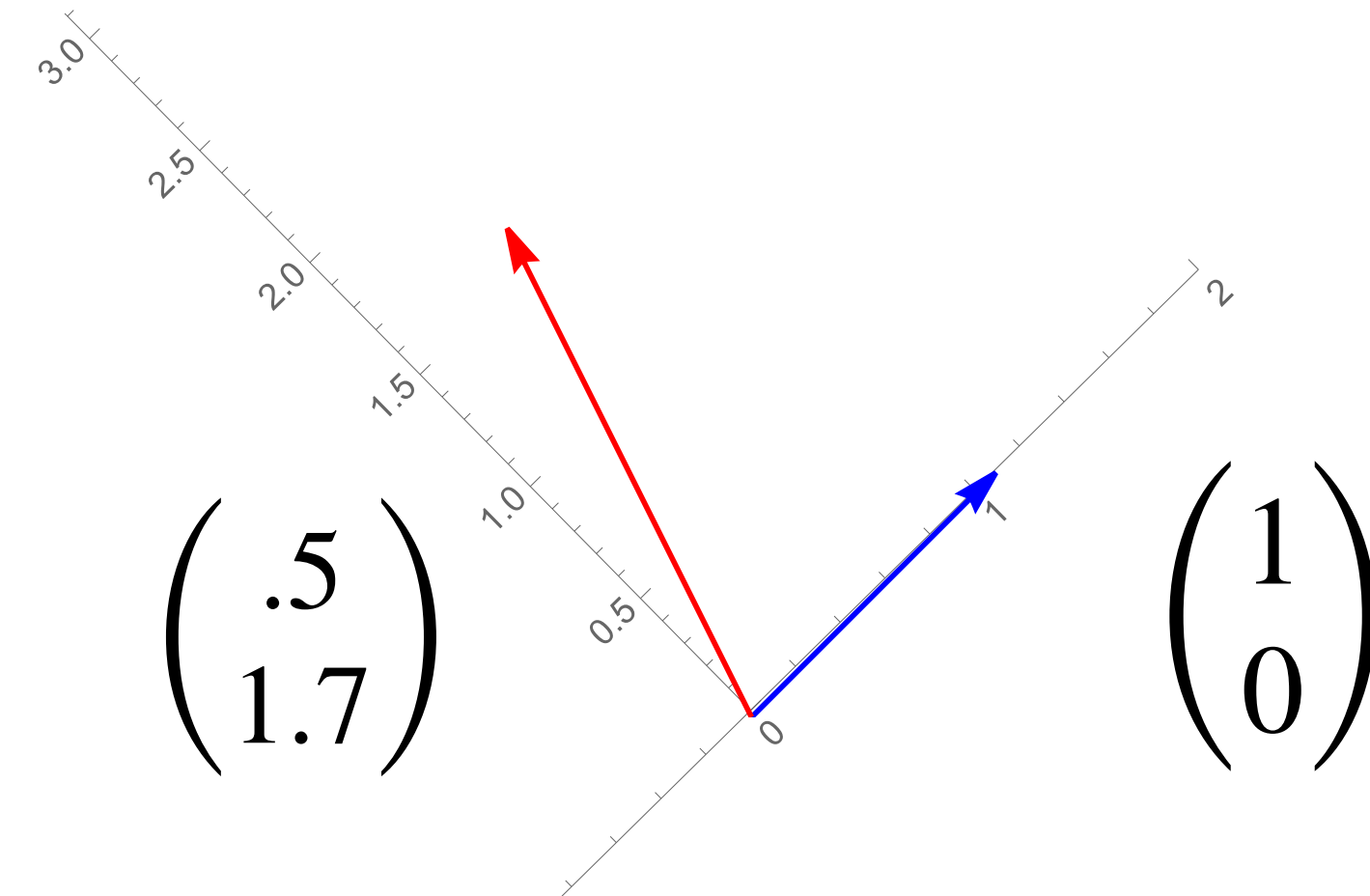
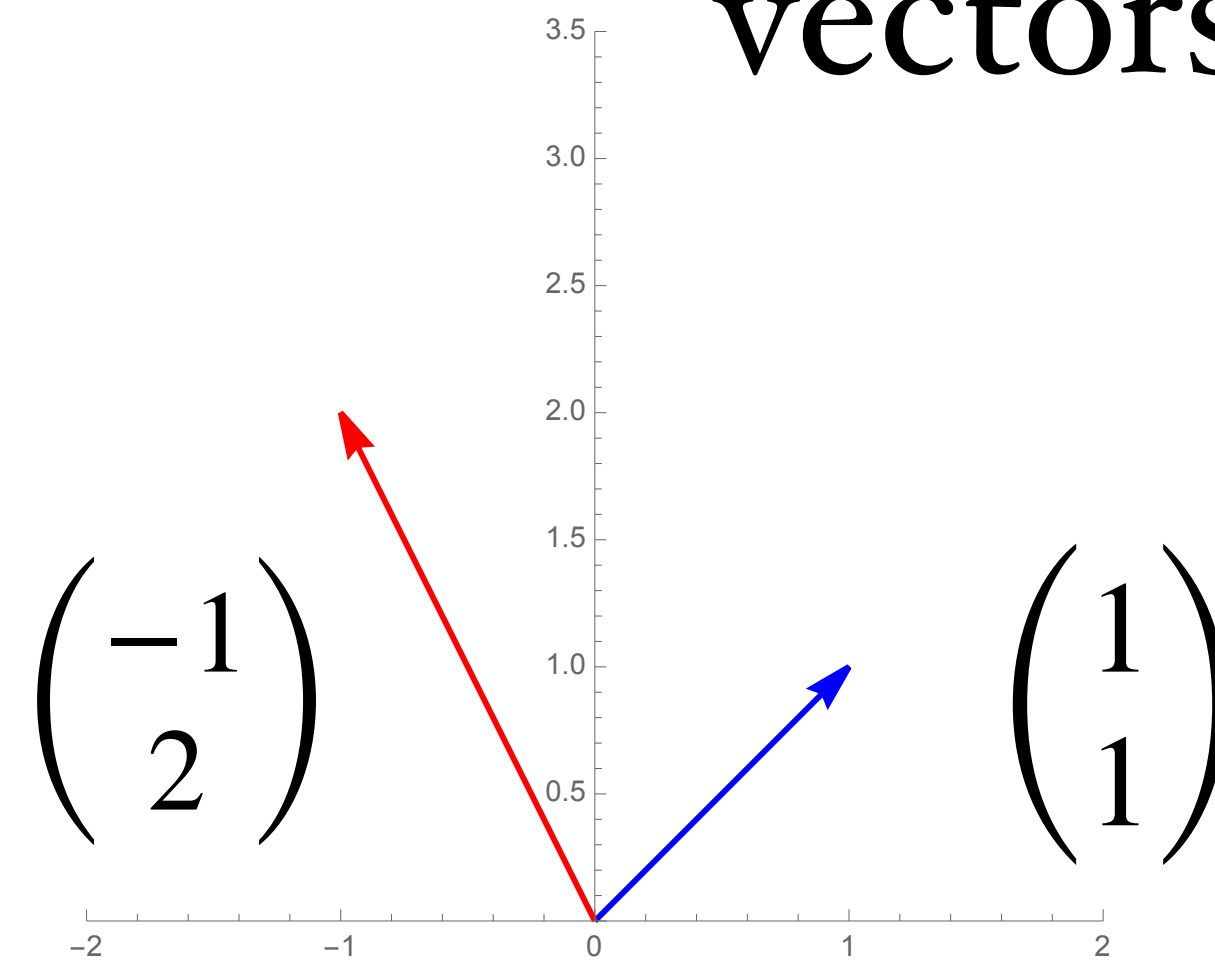
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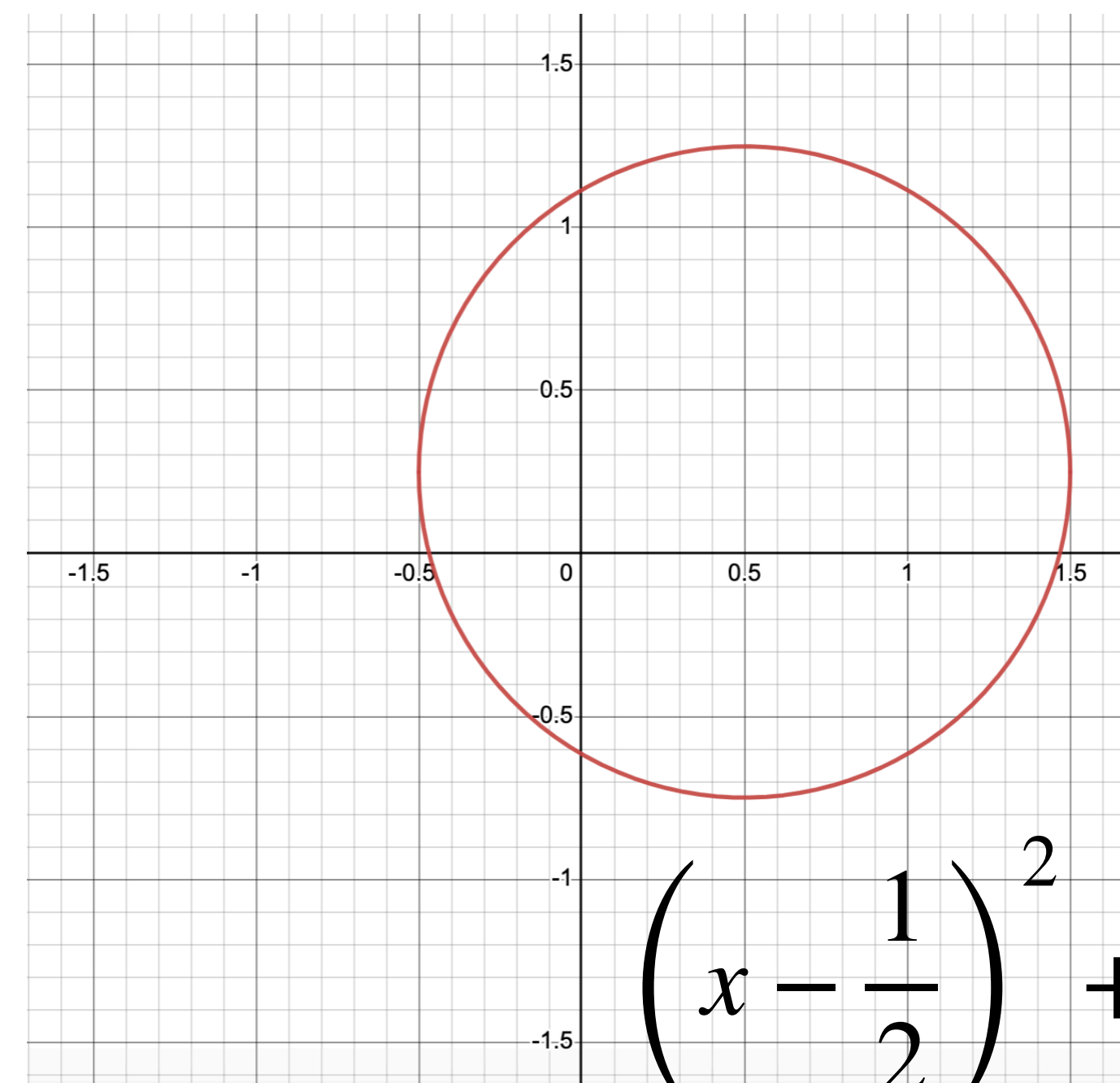
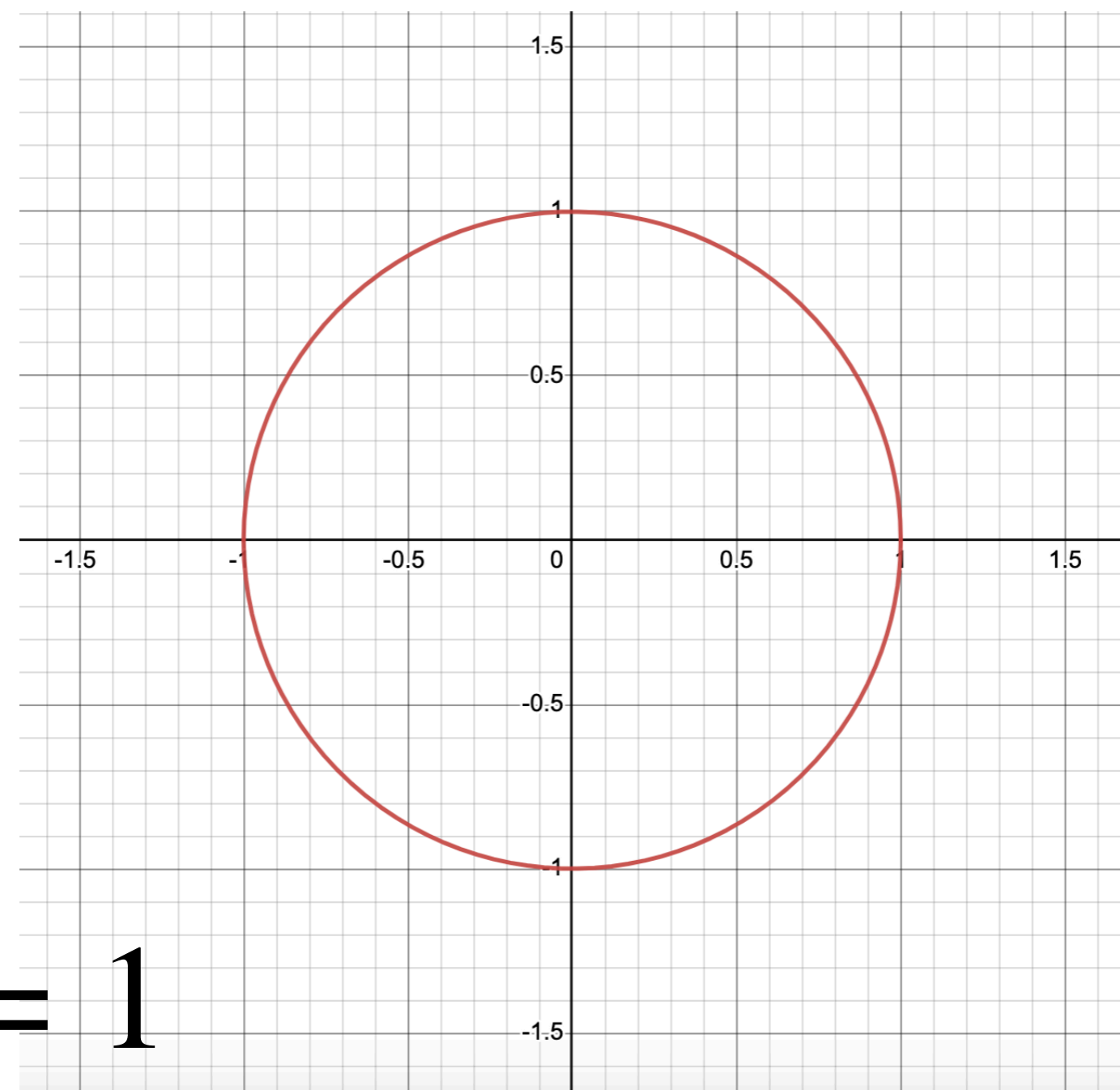
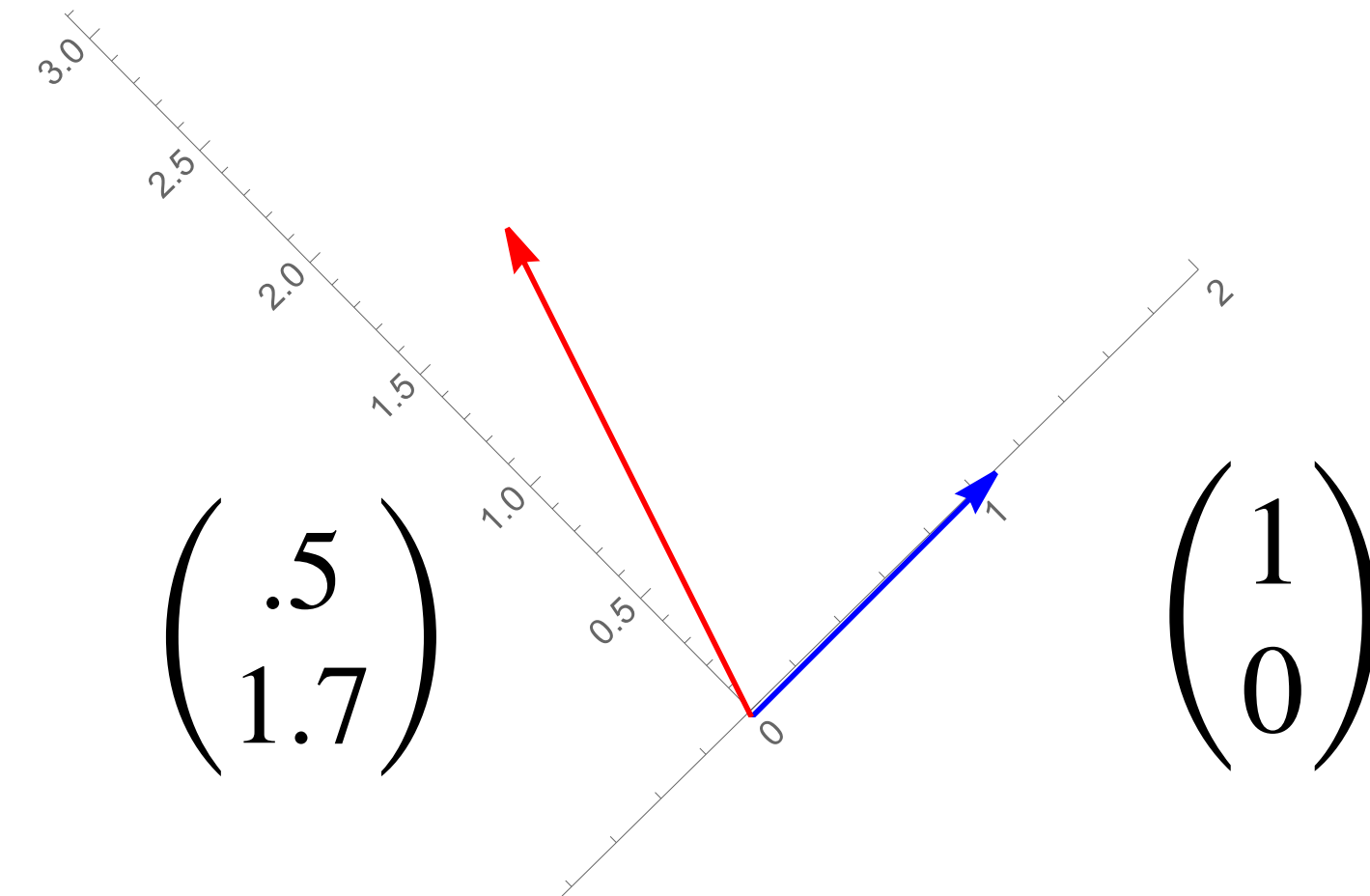
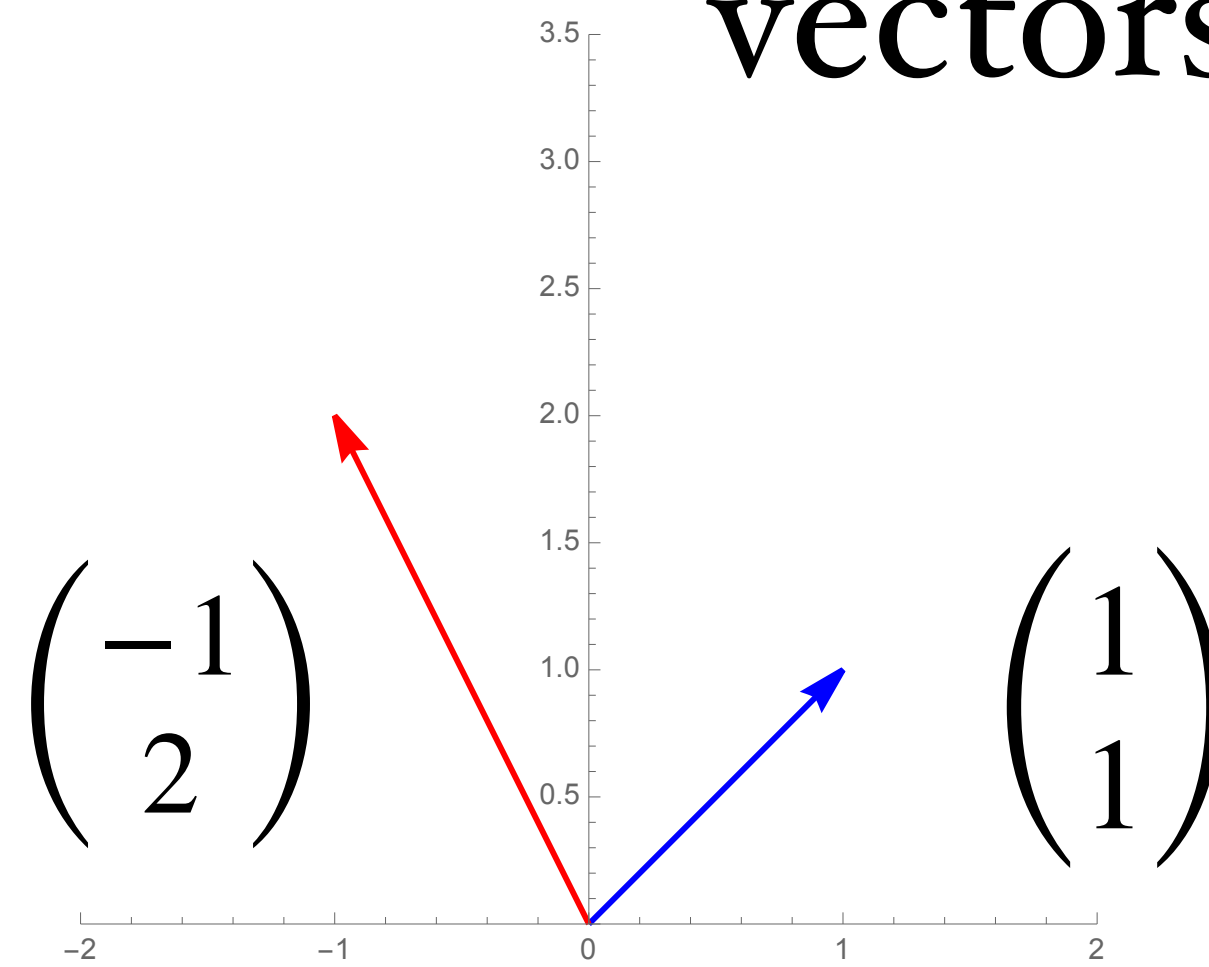
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coordinates vs. coordinate-free

vectors in $\mathbb{R}^2$



$$x^2 + y^2 = 1$$

$$\left(x - \frac{1}{2}\right)^2 + \left(y - \frac{1}{4}\right)^2 = 1$$

coordinates vs. coordinate-free

vector space:

vectors you can add and subtract with each other,

a zero vector, inverse (opposite) vectors;

and a way to scale the vectors that “respects” the addition

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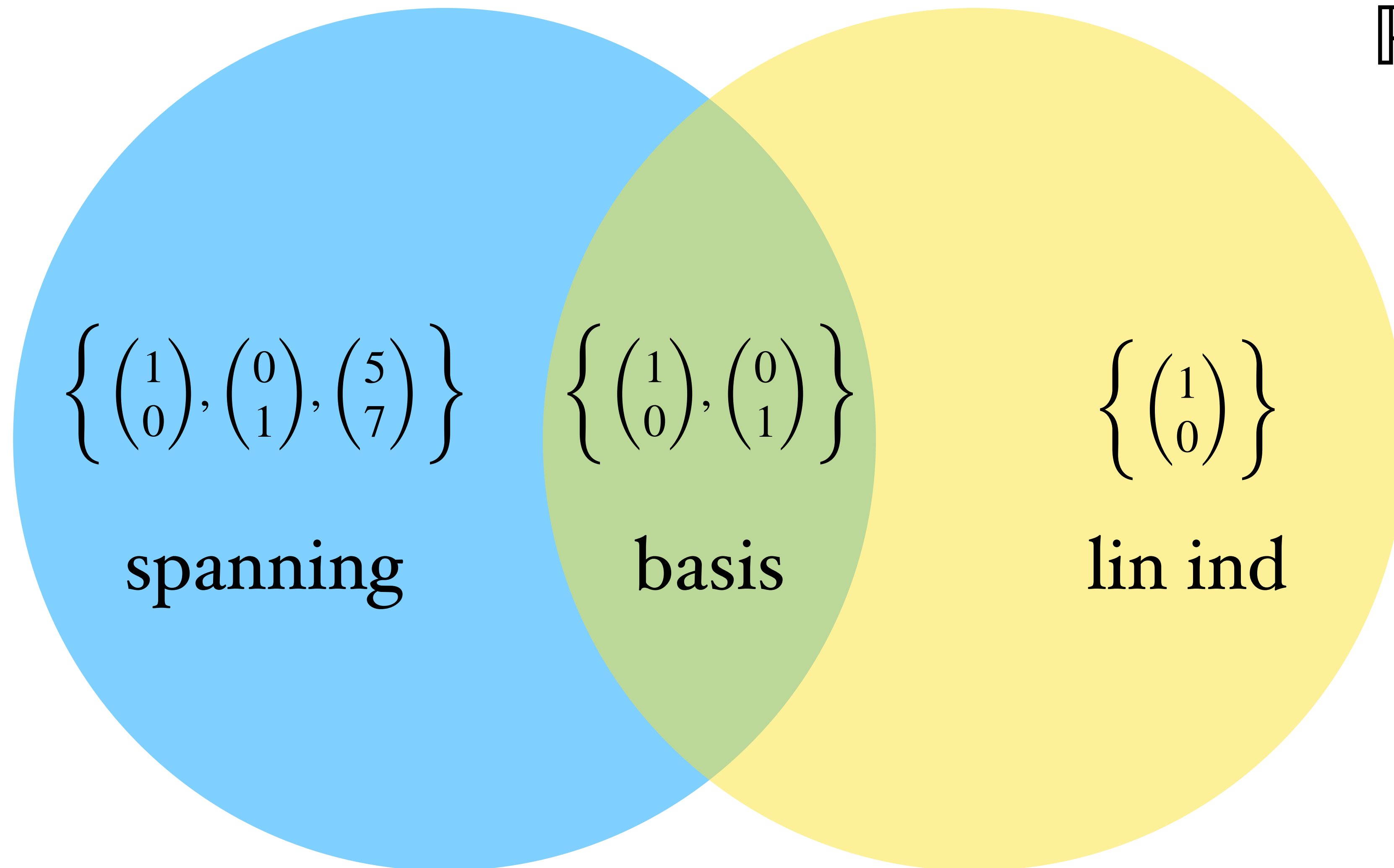
a group of vectors and a field of scalars

$$\mathbb{R}^2$$

$$\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 5 \\ 2 \end{pmatrix} \right\} \quad \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\} \quad \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}$$

$$\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \end{pmatrix} \right\}$$

$\mathbb{R}^2$



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$\mathbb{R}^2$

$$\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 5 \\ 7 \end{pmatrix} \right\}$$

spanning

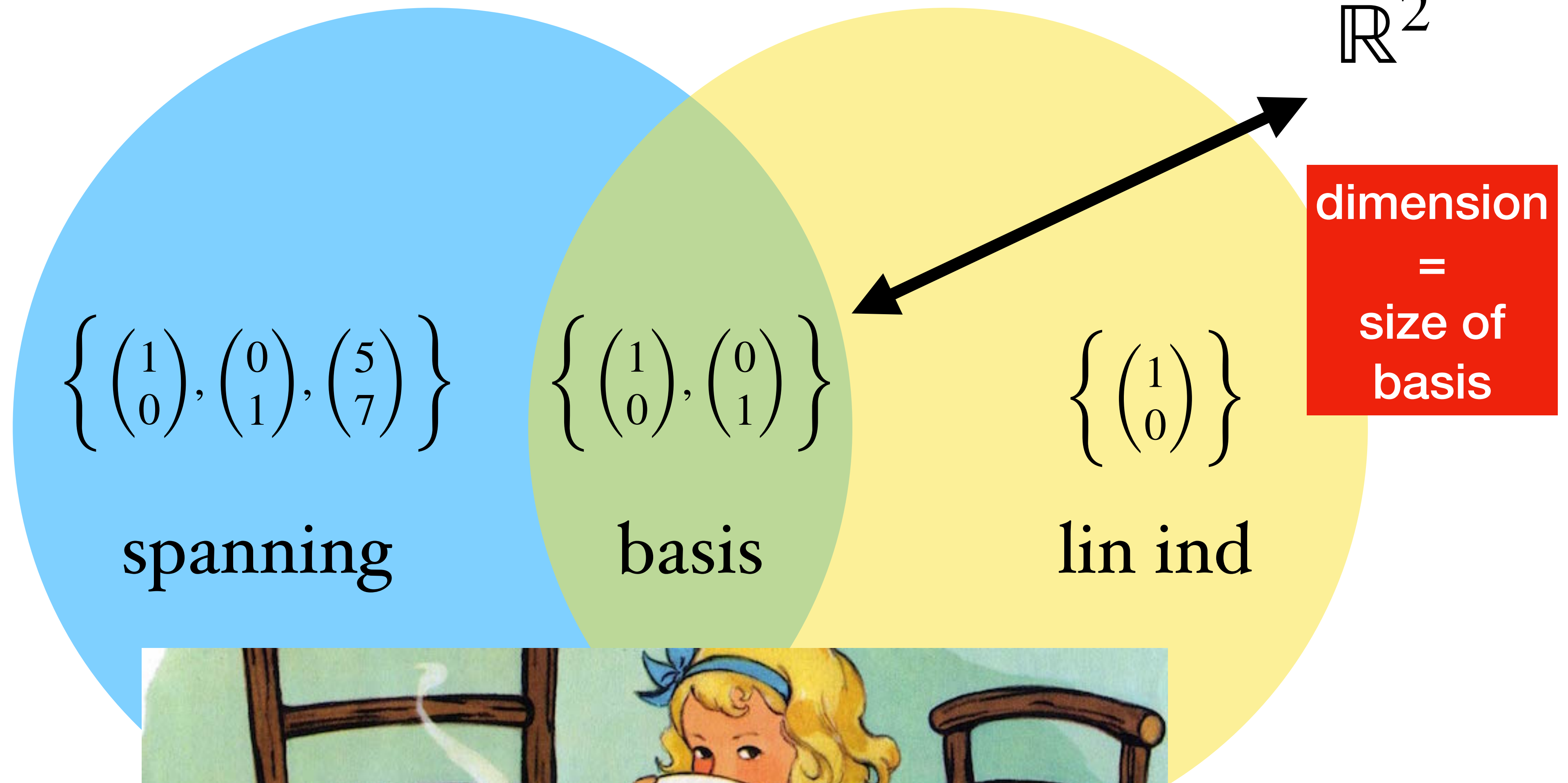
$$\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$$

basis

$$\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}$$

lin ind





From Monday's warm-up:

What are some solutions to ...

$$w - 2x + y + z = 0$$

$$y + z = 0$$

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The solutions to a (homogeneous) system
of linear equations form a vector space!

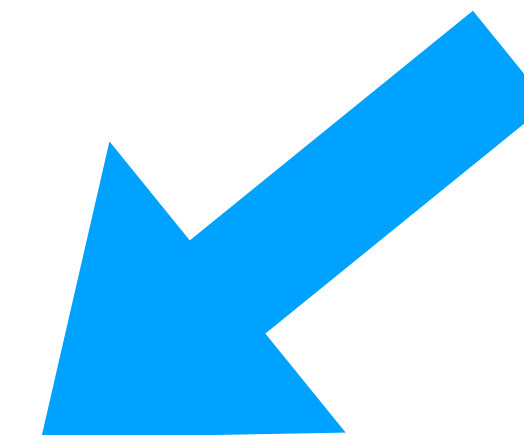
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b/c need to
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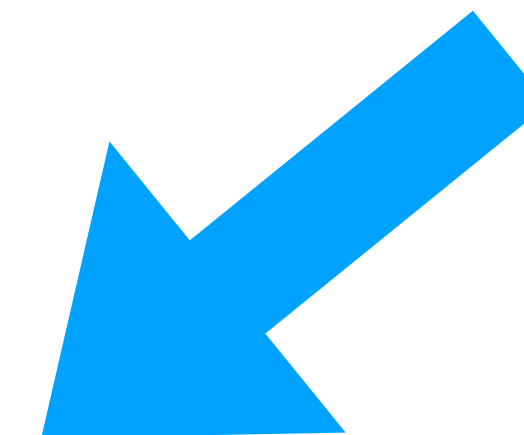
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What is the
dimension of the
solution set?

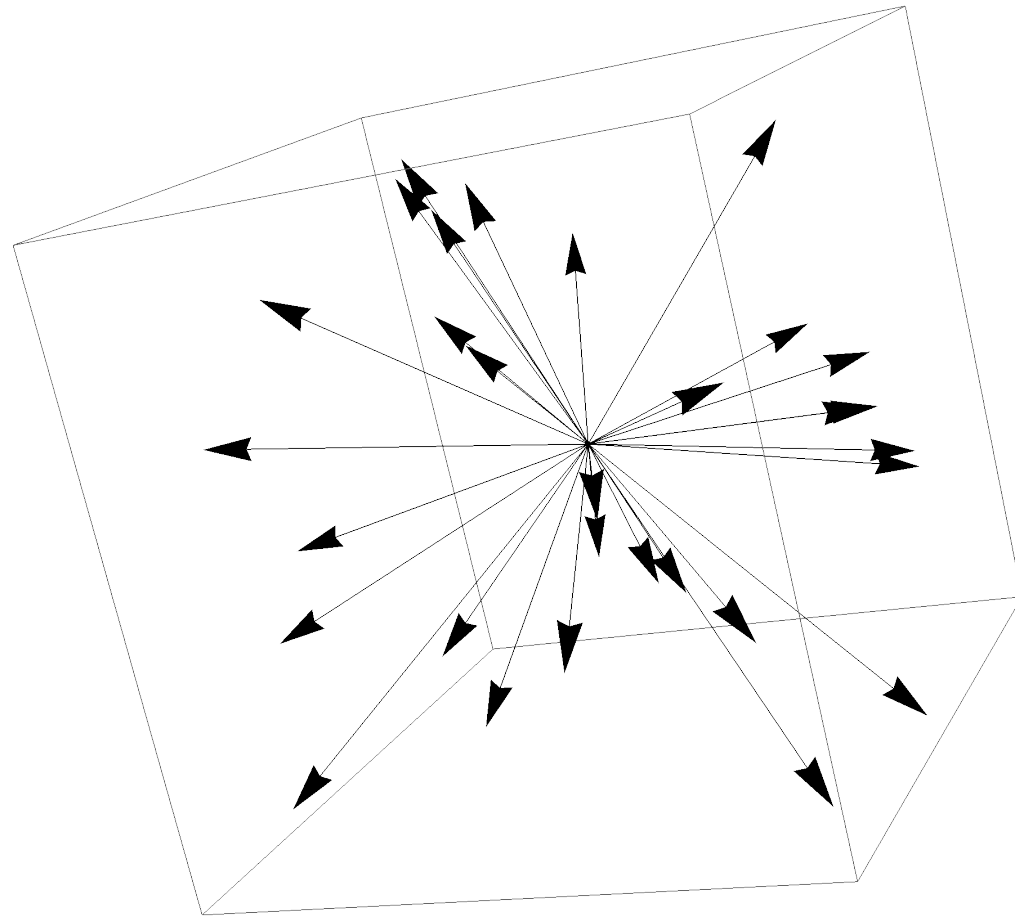
What is a basis
for it?

b/c need to
have the zero vector



The solutions to a (homogeneous) system
of linear equations form a vector space!

Abstract Linear Algebra



Proof. Suppose the system $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$ is linearly dependent. Then there exist scalars α_k , $\sum_{k=1}^p |\alpha_k| \neq 0$ such that

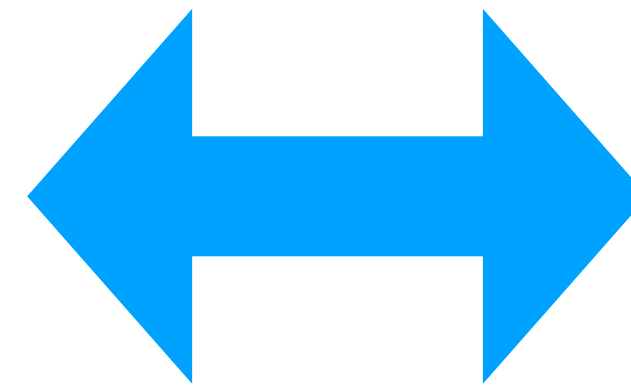
$$\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \dots + \alpha_p \mathbf{v}_p = \mathbf{0}.$$

Let k be the index such that $\alpha_k \neq 0$. Then, moving all terms except $\alpha_k \mathbf{v}_k$ to the right side we get

$$\alpha_k \mathbf{v}_k = - \sum_{\substack{j=1 \\ j \neq k}}^p \alpha_j \mathbf{v}_j.$$

Vector spaces
and linear transformations
and proofs

ABSTRACT



$$\begin{aligned} 2x + 3y &= 0 \\ 5x - 3y &= 0 \end{aligned}$$

$$\left(\begin{array}{cc|c} 2 & 3 & 0 \\ 5 & -3 & 0 \end{array} \right)$$

Systems of equations
and matrices
and computations

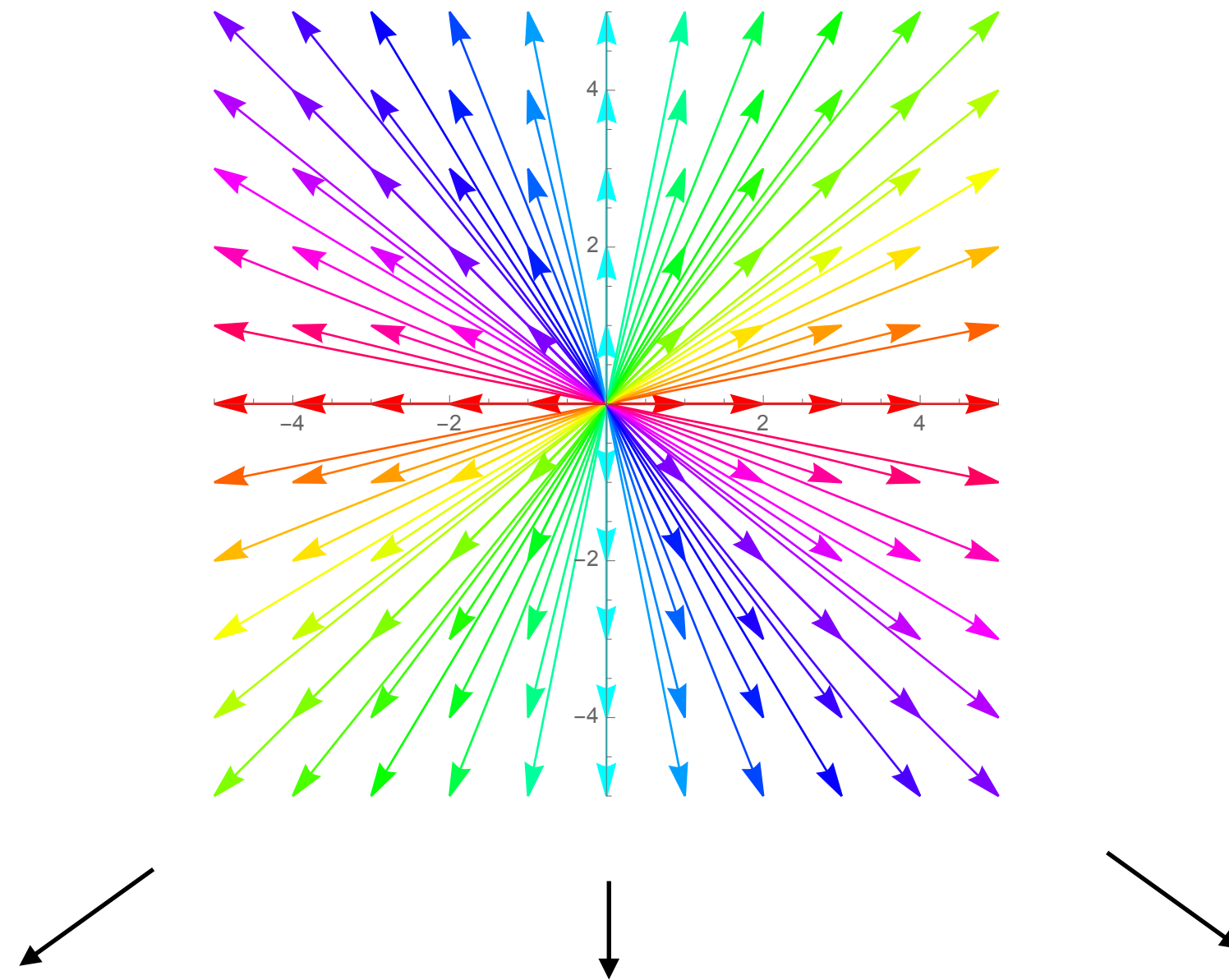
CONCRETE

- Ch 1: vector spaces, bases, linear transformations, matrices
- Ch 2: solving linear systems, row reduction, inverse matrices
- Ch 3: determinants
- Ch 4: eigenvalues and eigenvectors
- Ch 5: inner products, orthogonality
- Ch 6 & 9: matrix decompositions

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Linear transformations

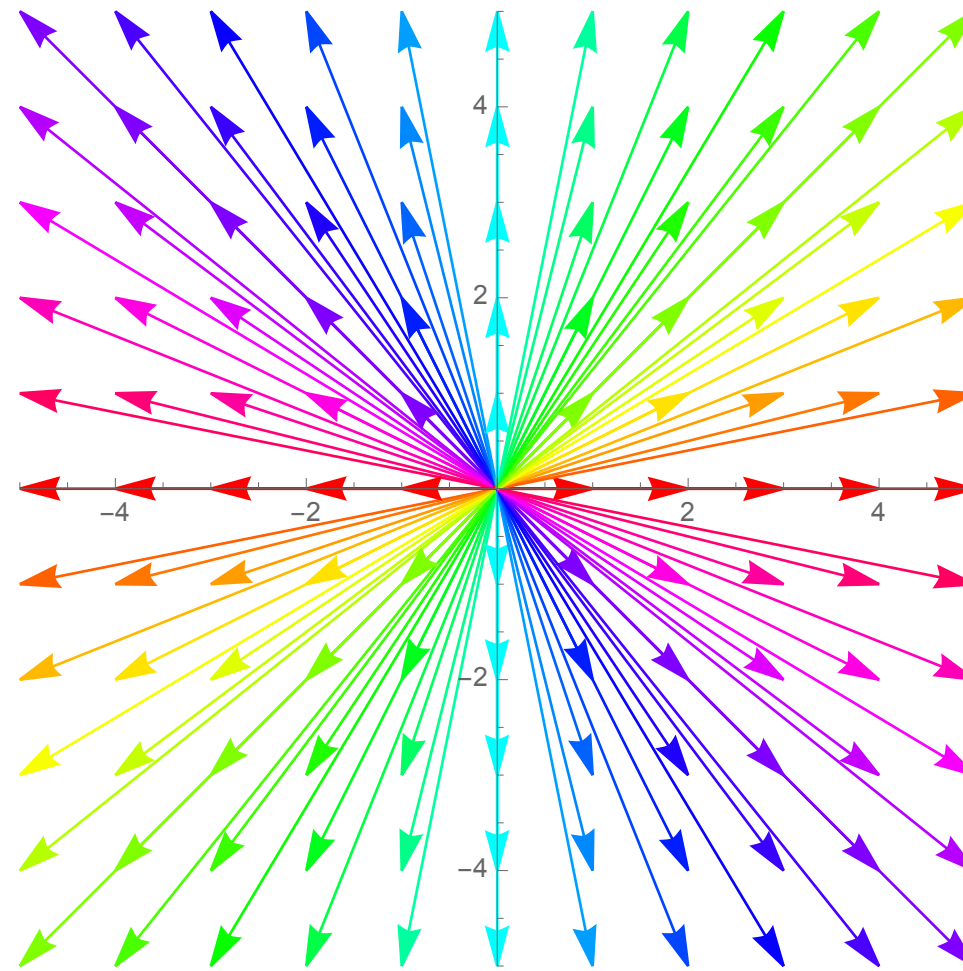


$$T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

Linear transformations

geometry

- fix the zero vector (origin)
- map lines to lines

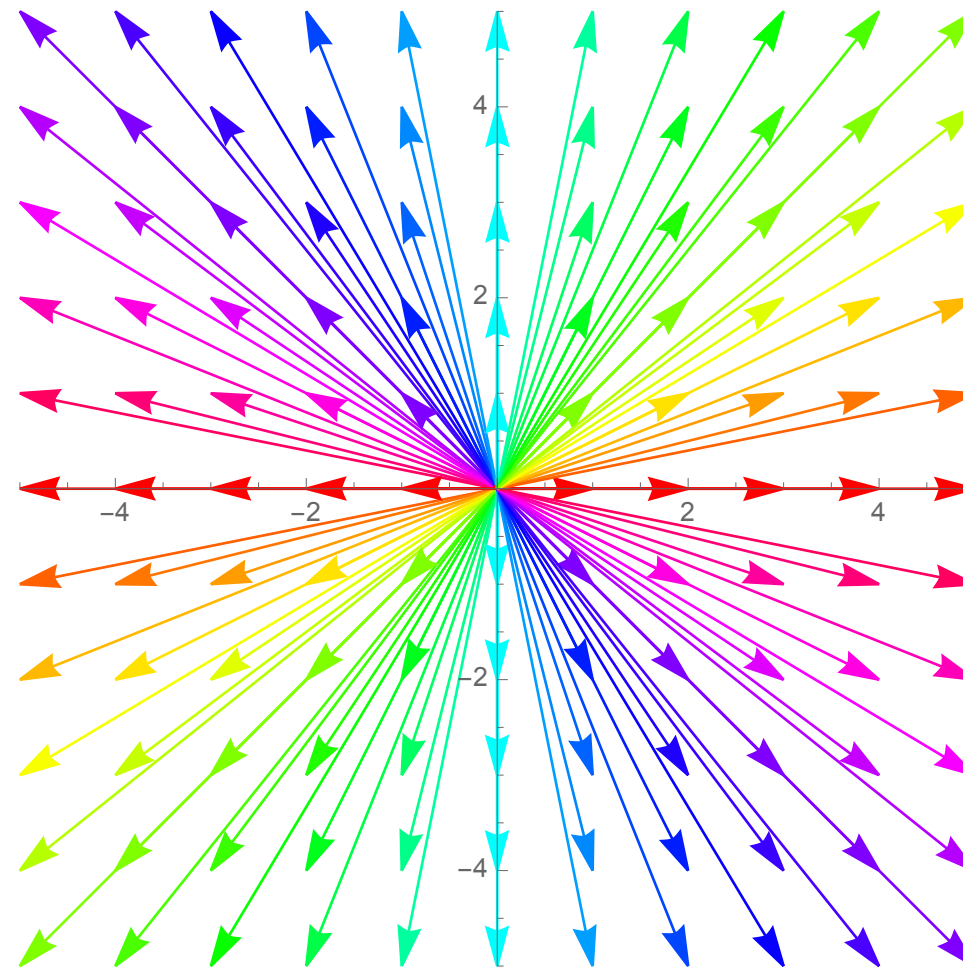


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algebra

$$\begin{aligned} \odot T(v + w) &= T(v) + T(w) \\ \odot T(av) &= aT(v) \end{aligned}$$

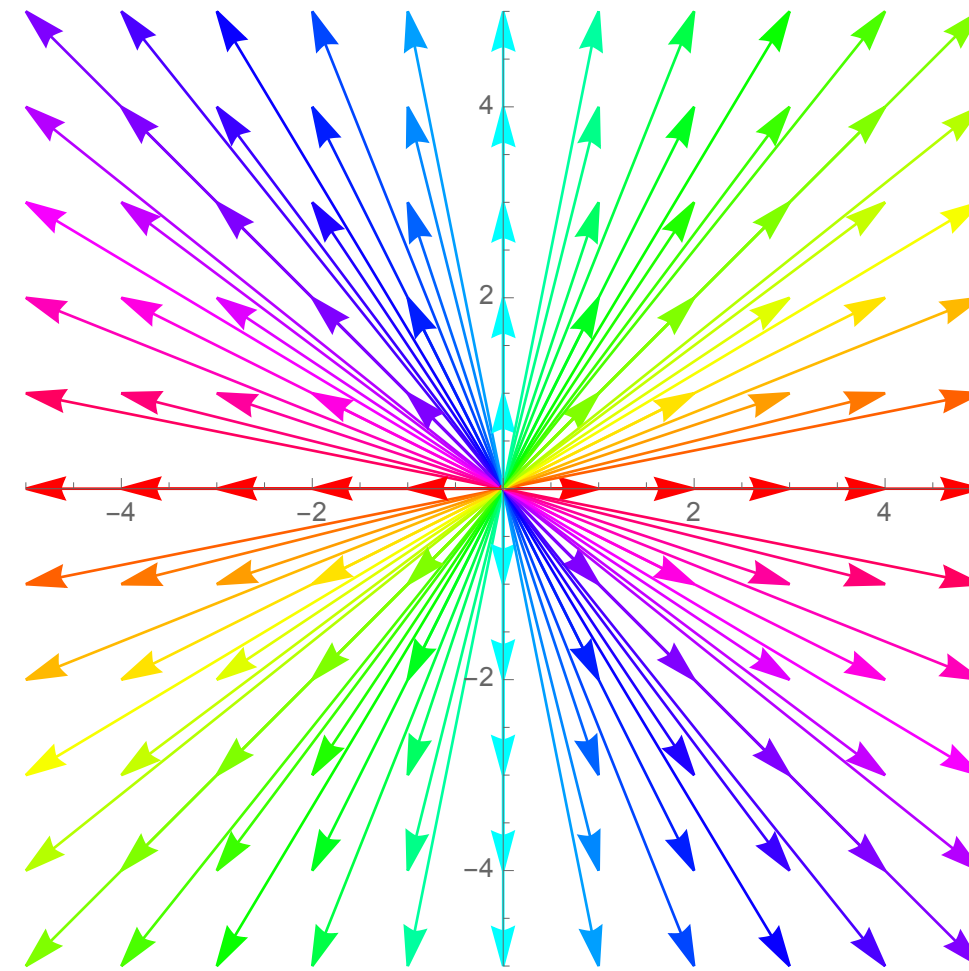
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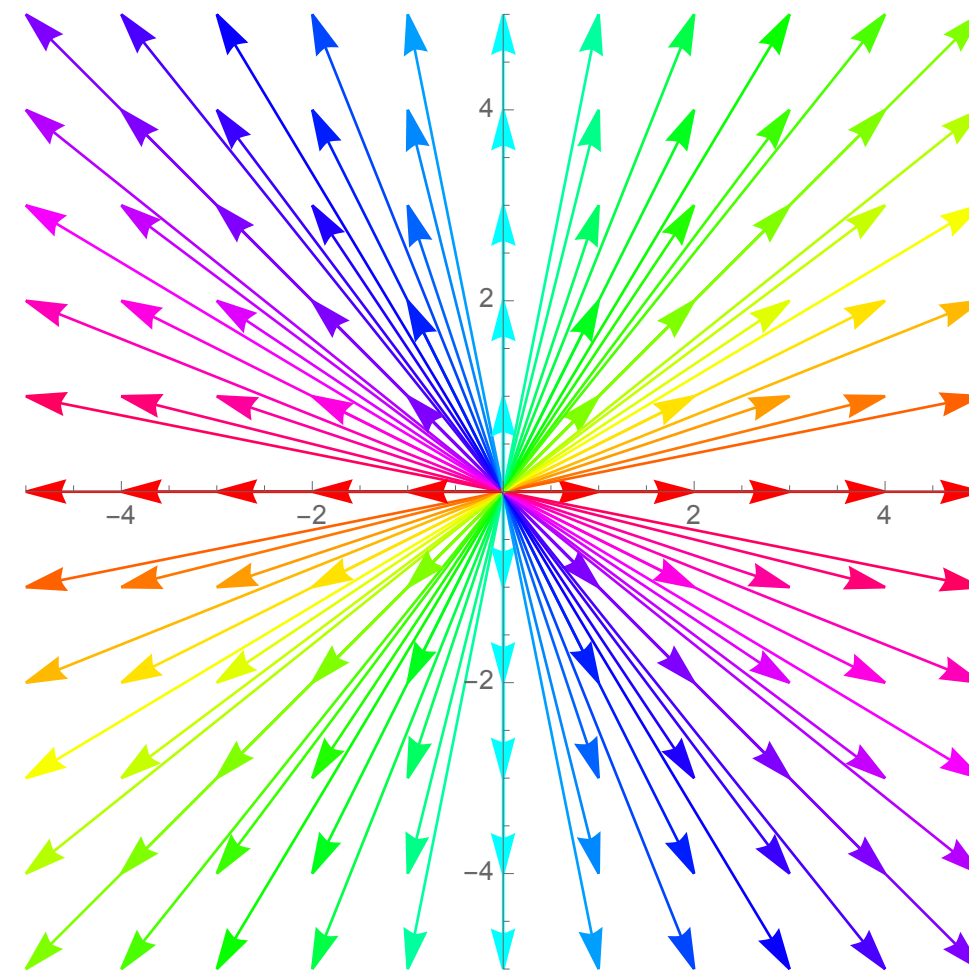
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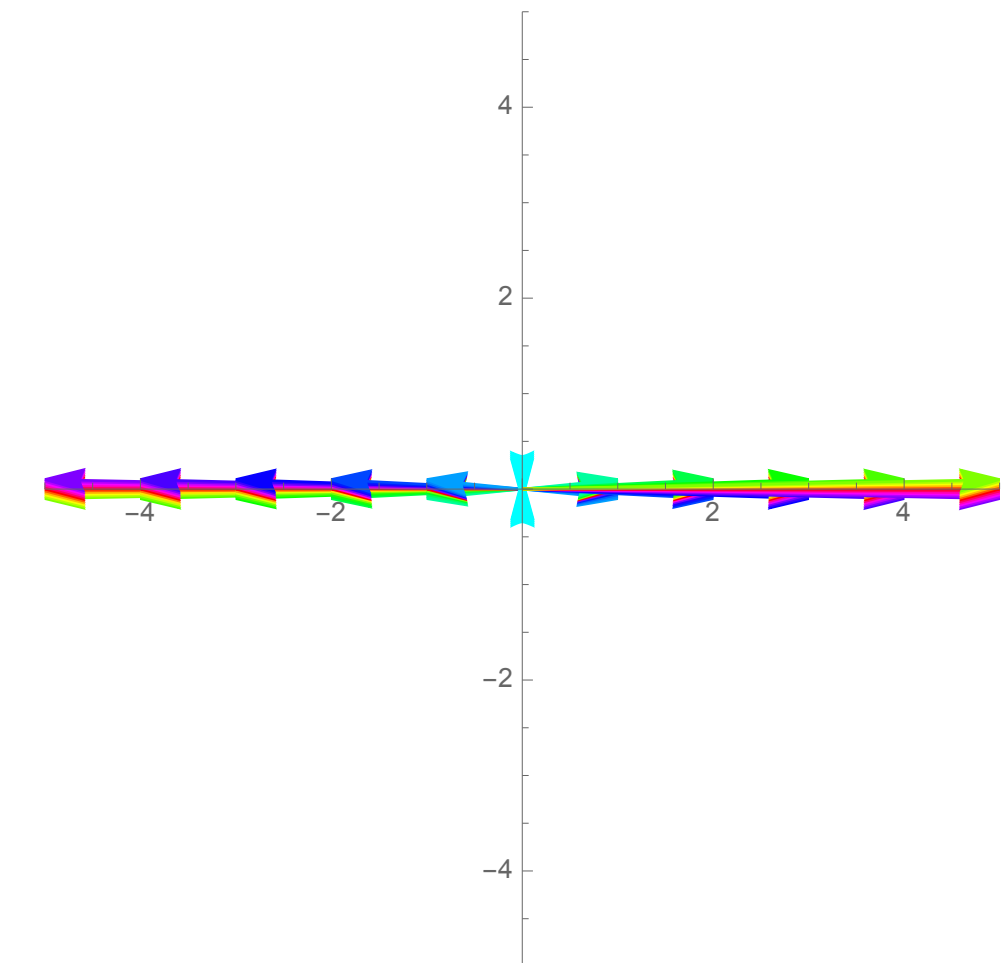
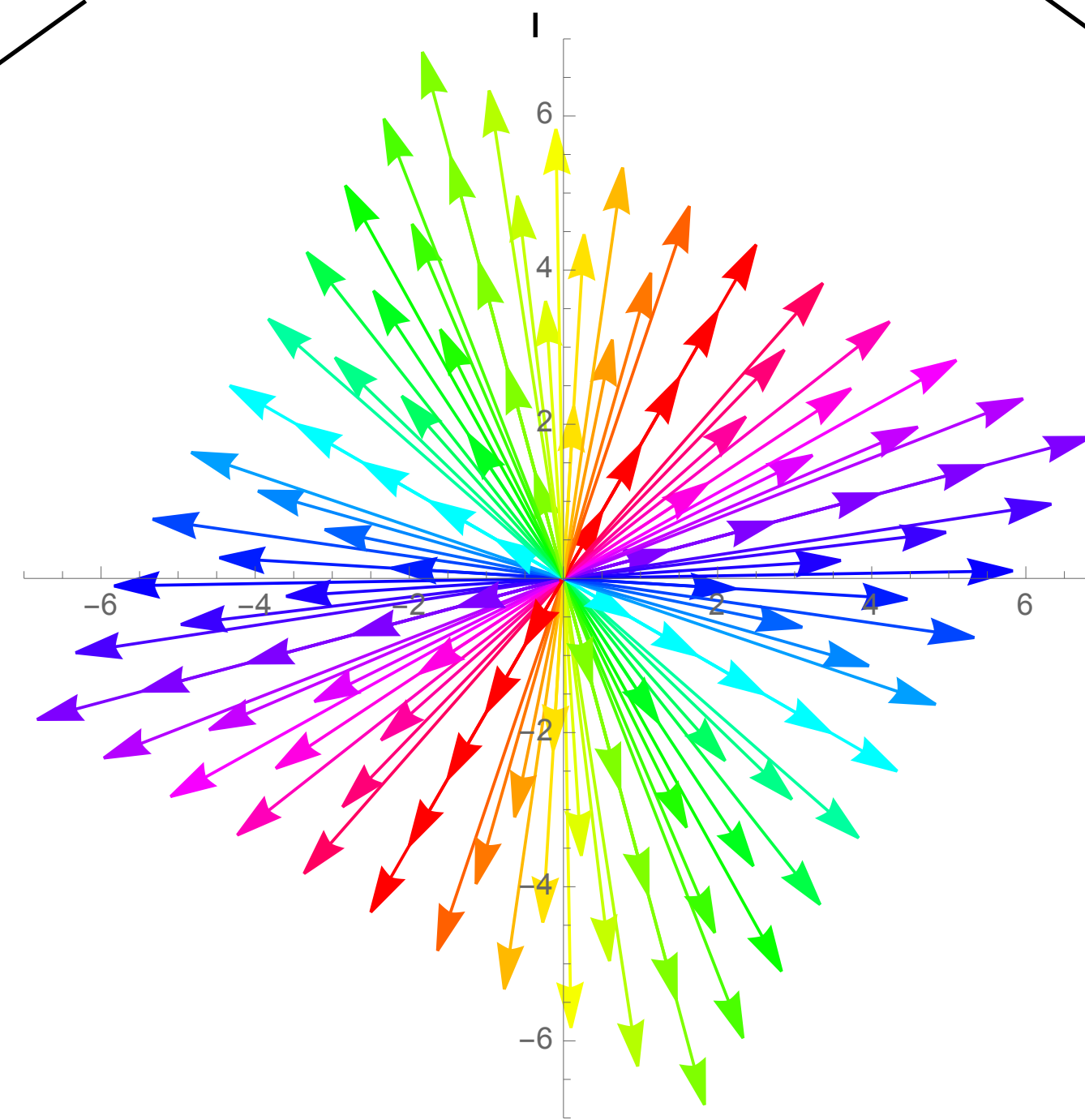
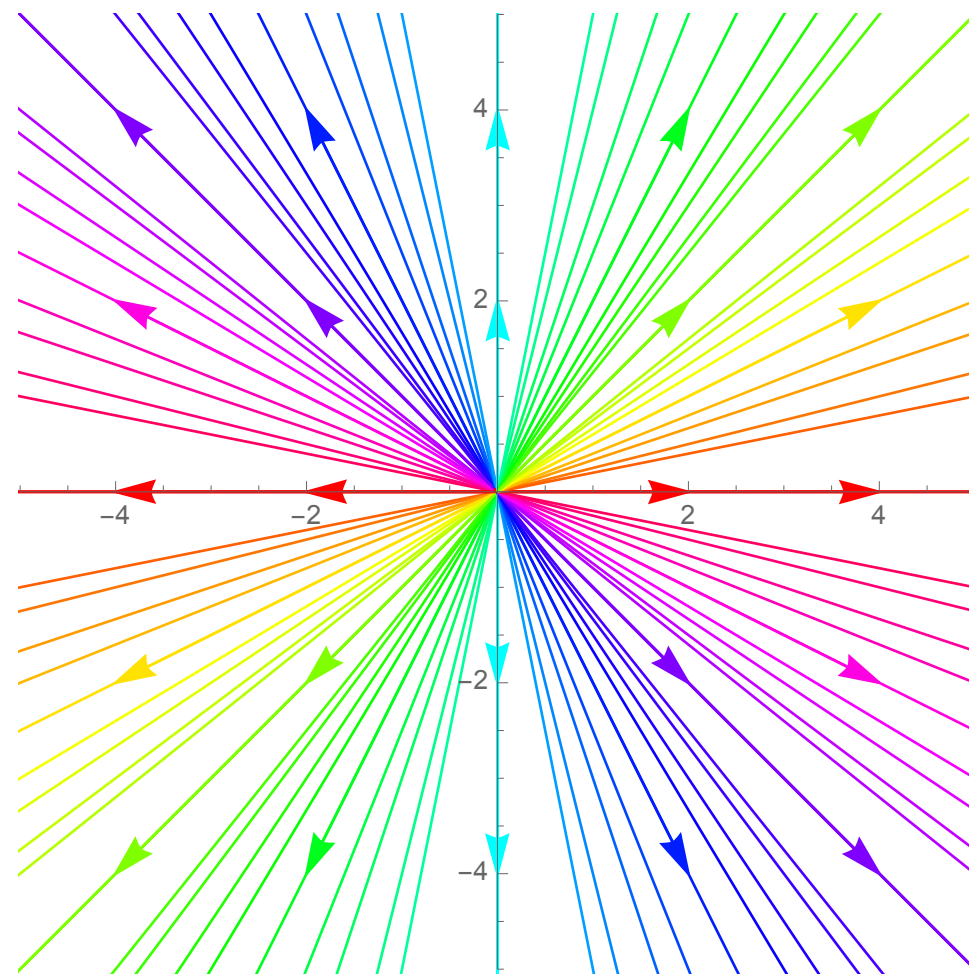
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algebra

- $T(v + w) = T(v) + T(w)$
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Linear transformations

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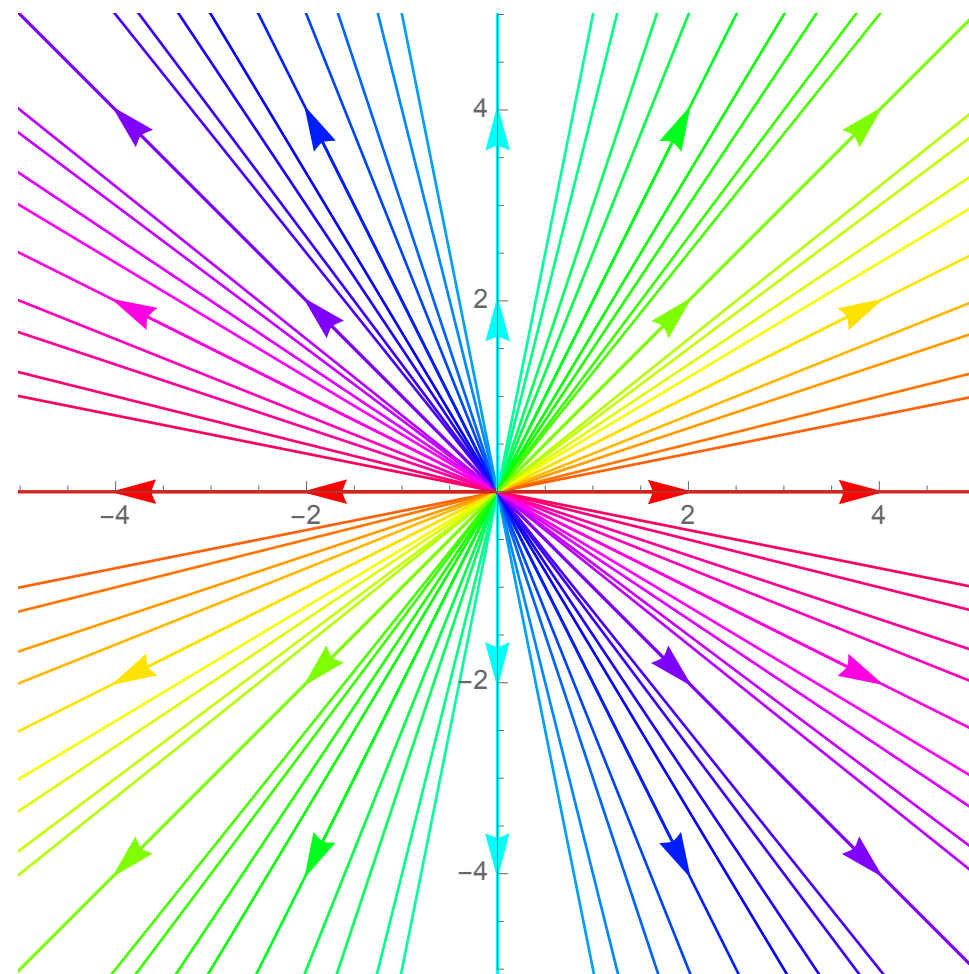
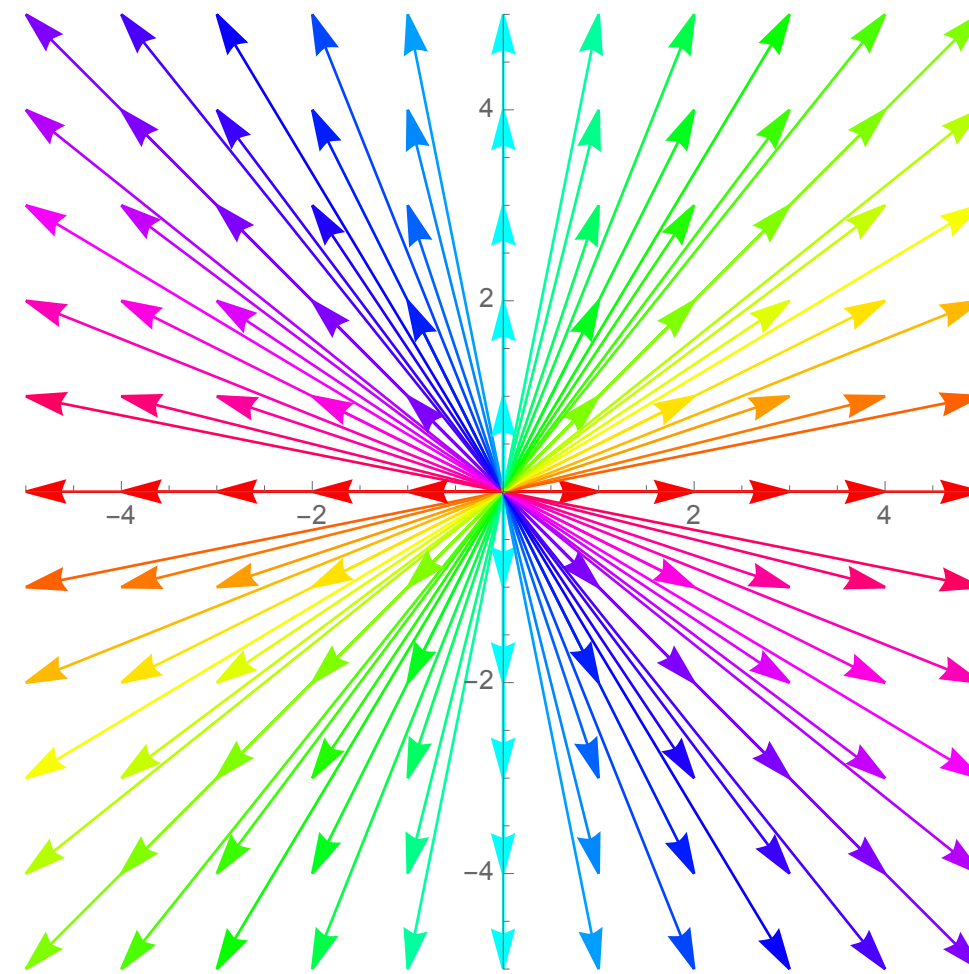
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algebra

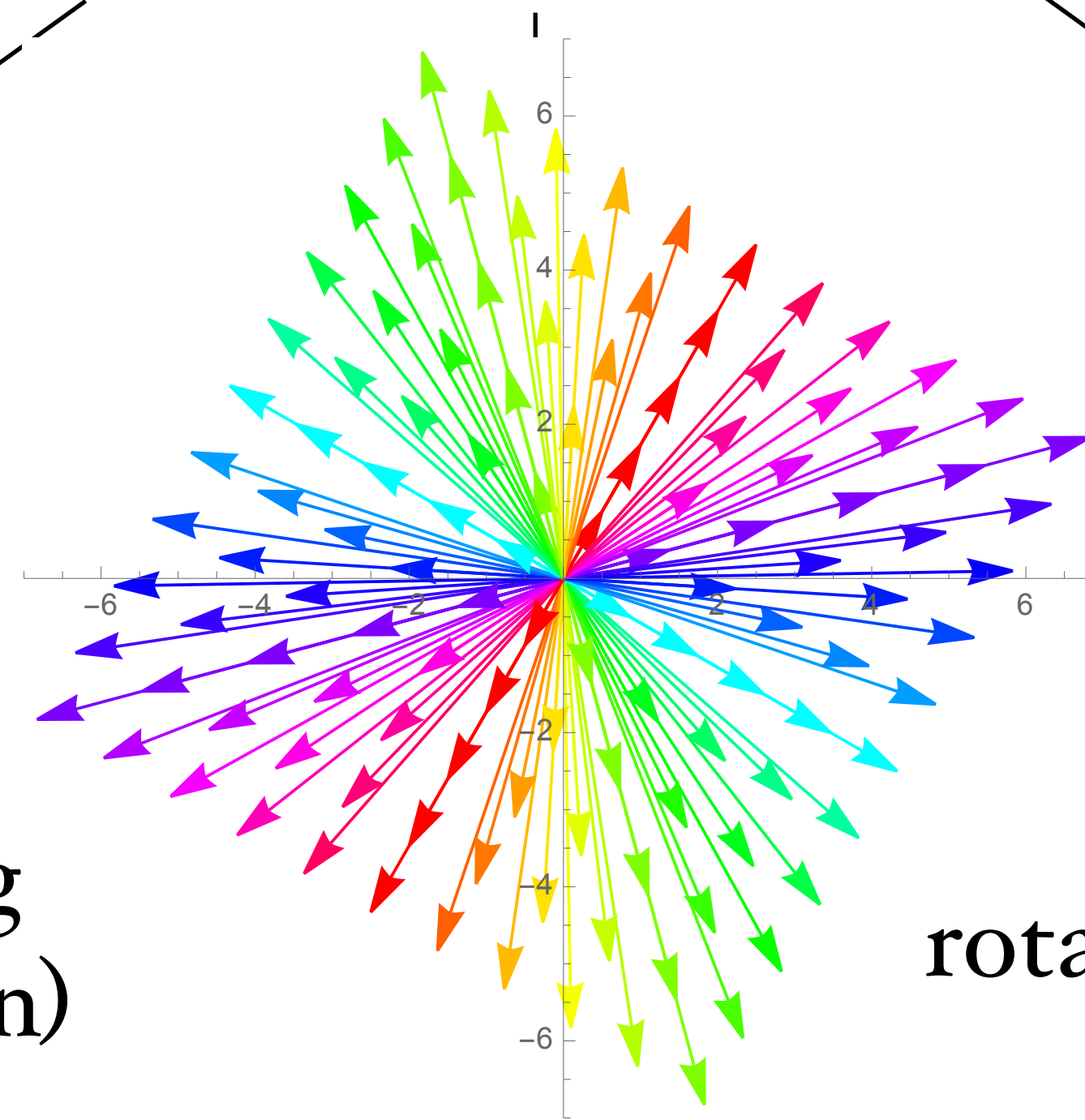
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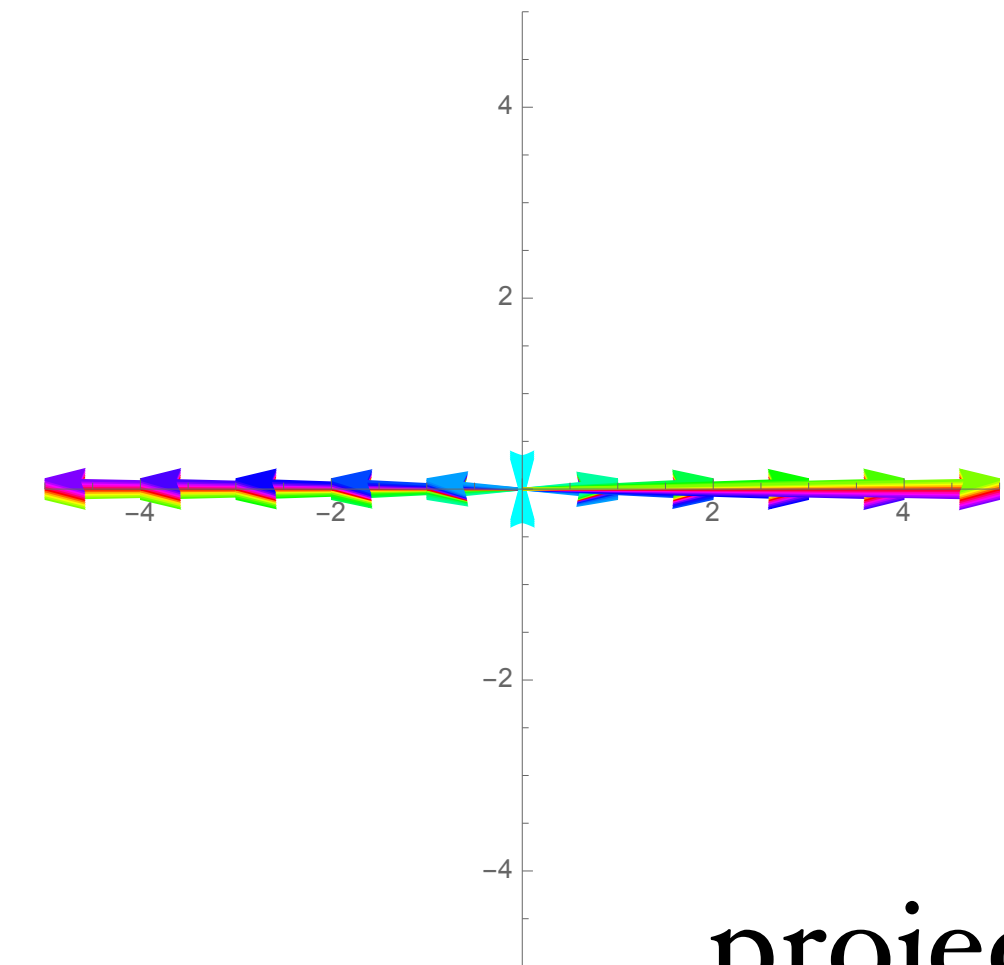
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scaling
(dilation)

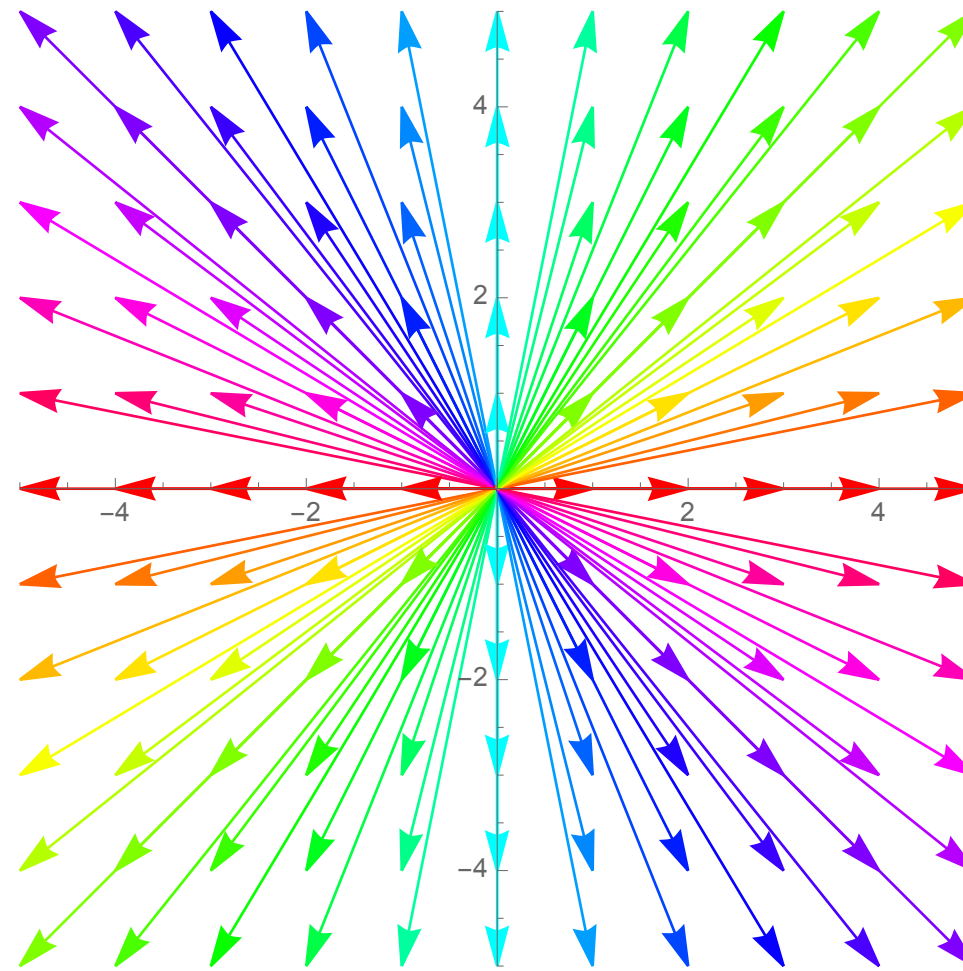


rotation



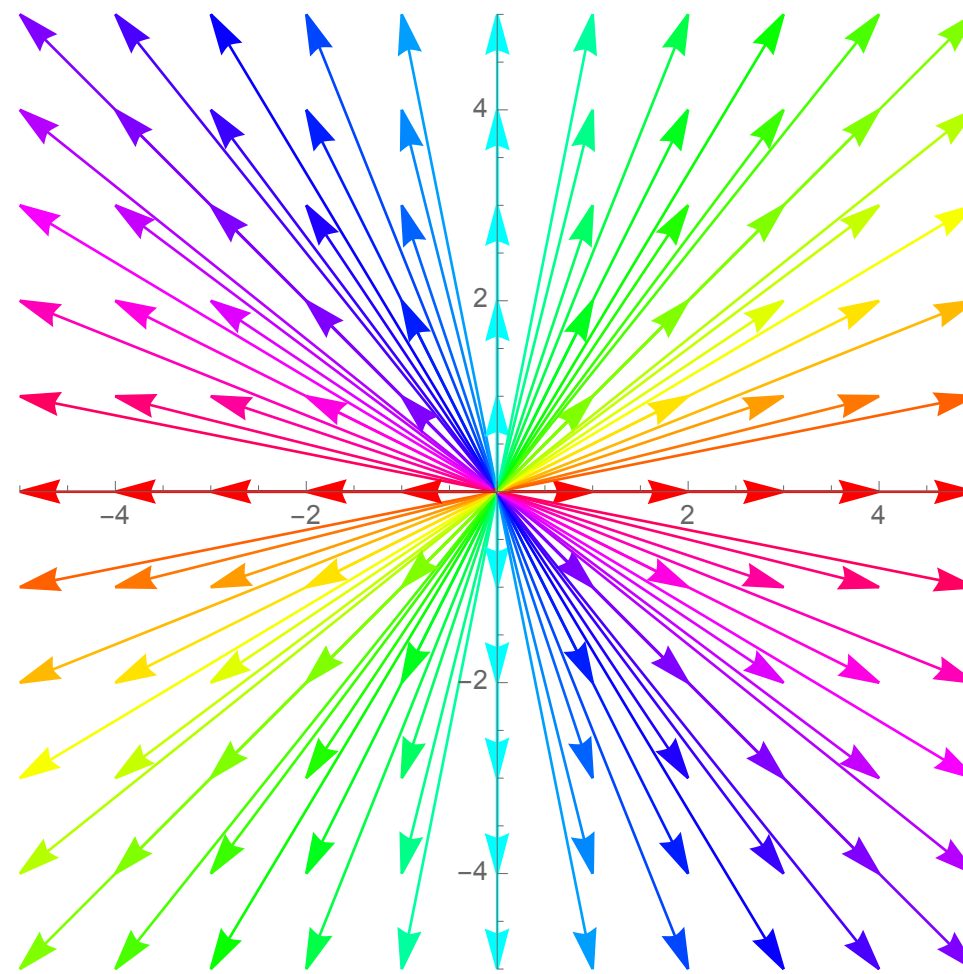
projection

Linear transformations

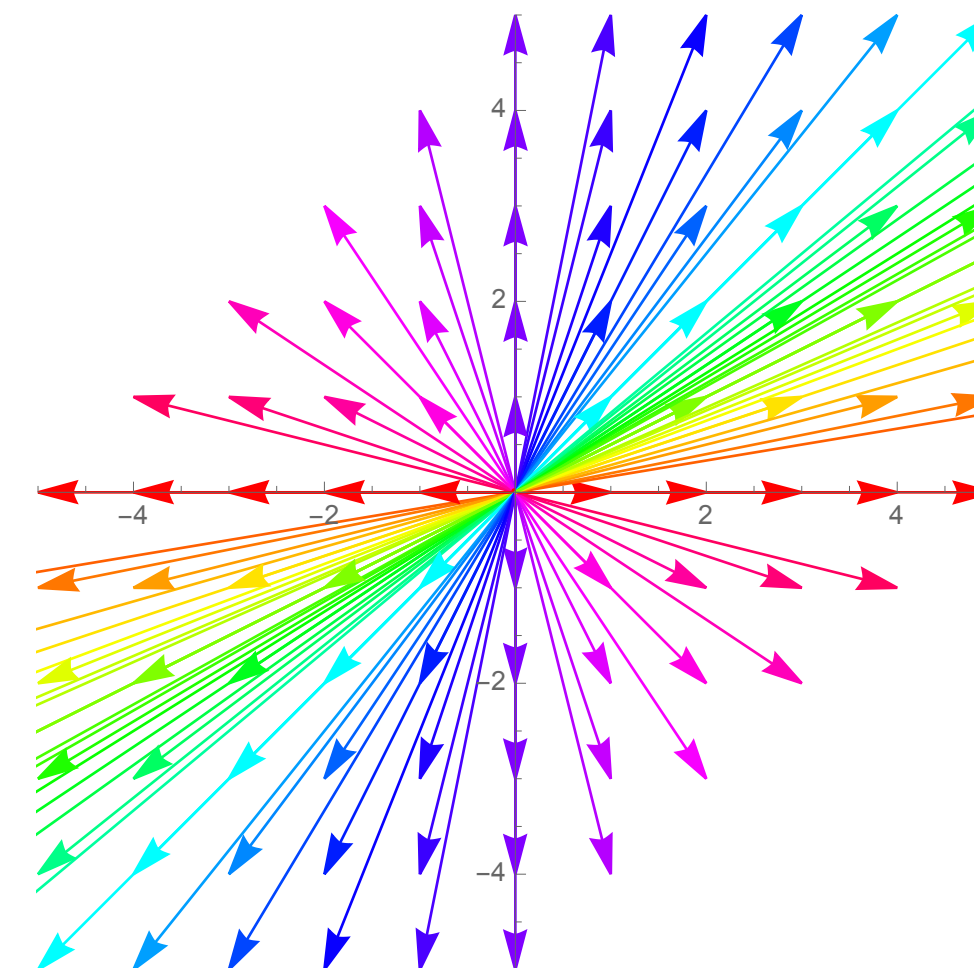
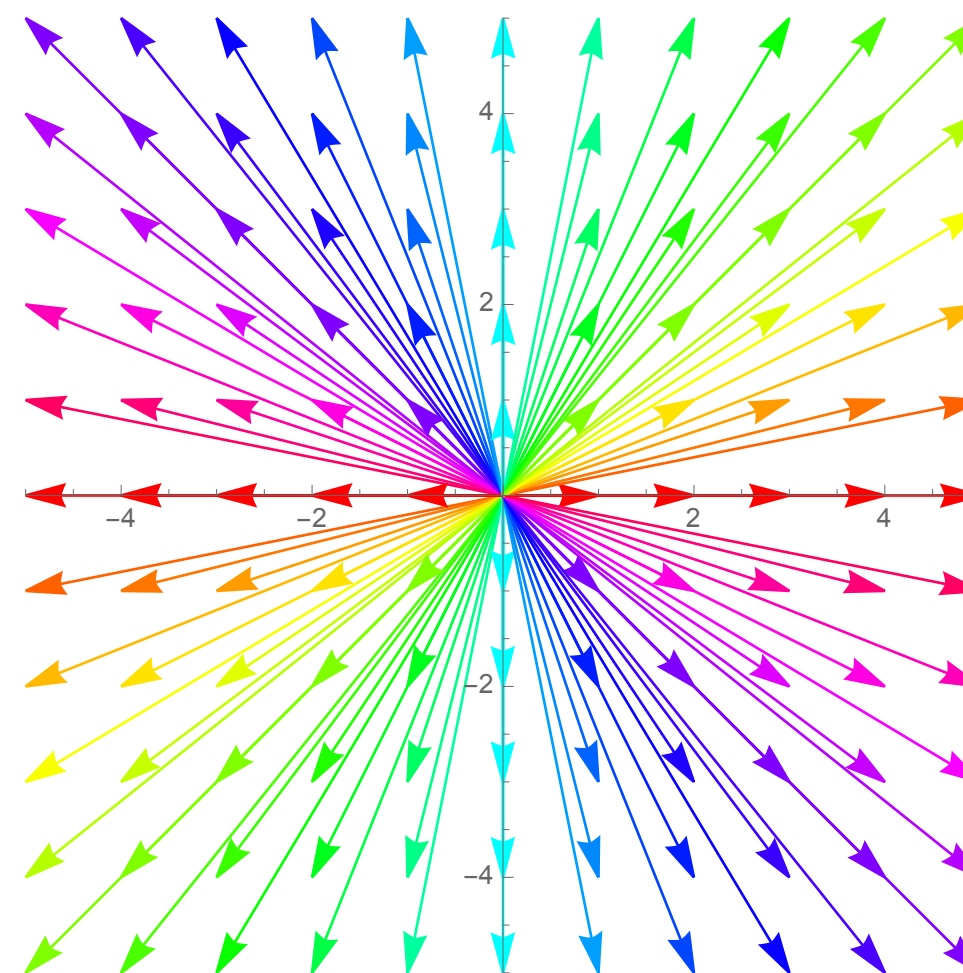
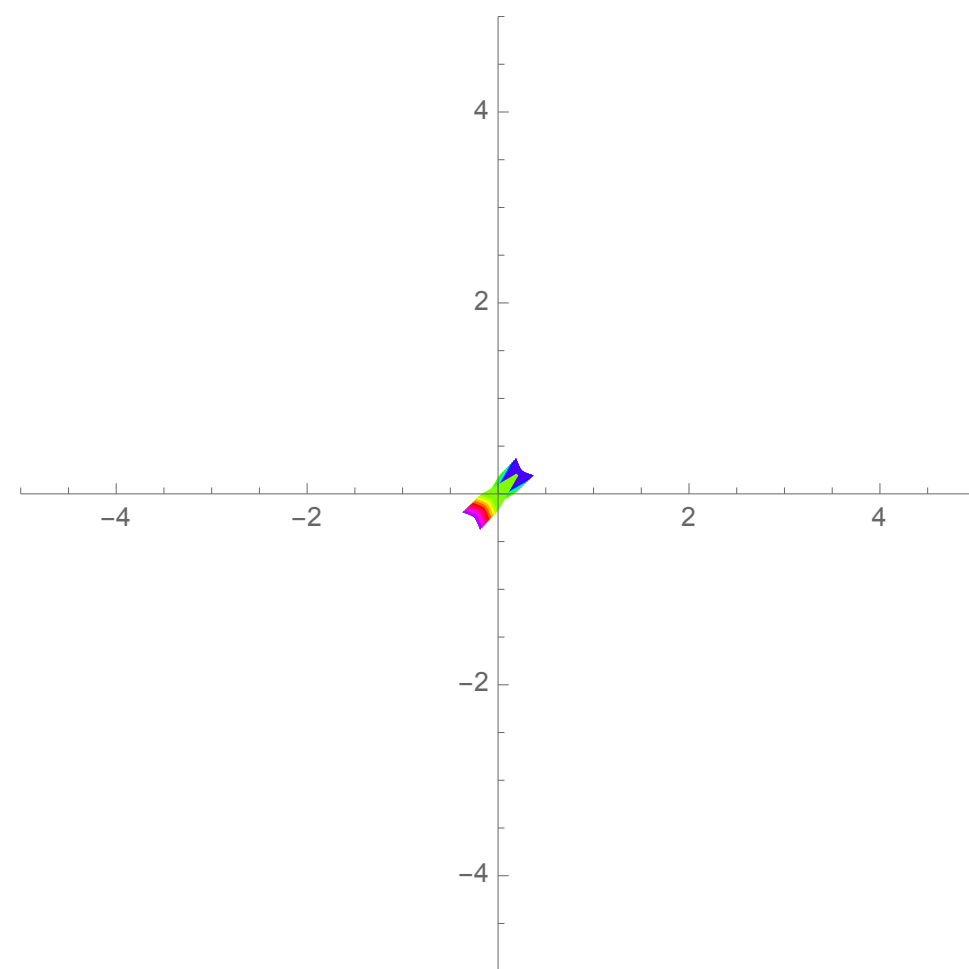
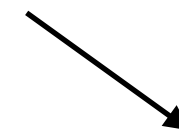
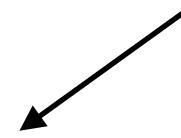


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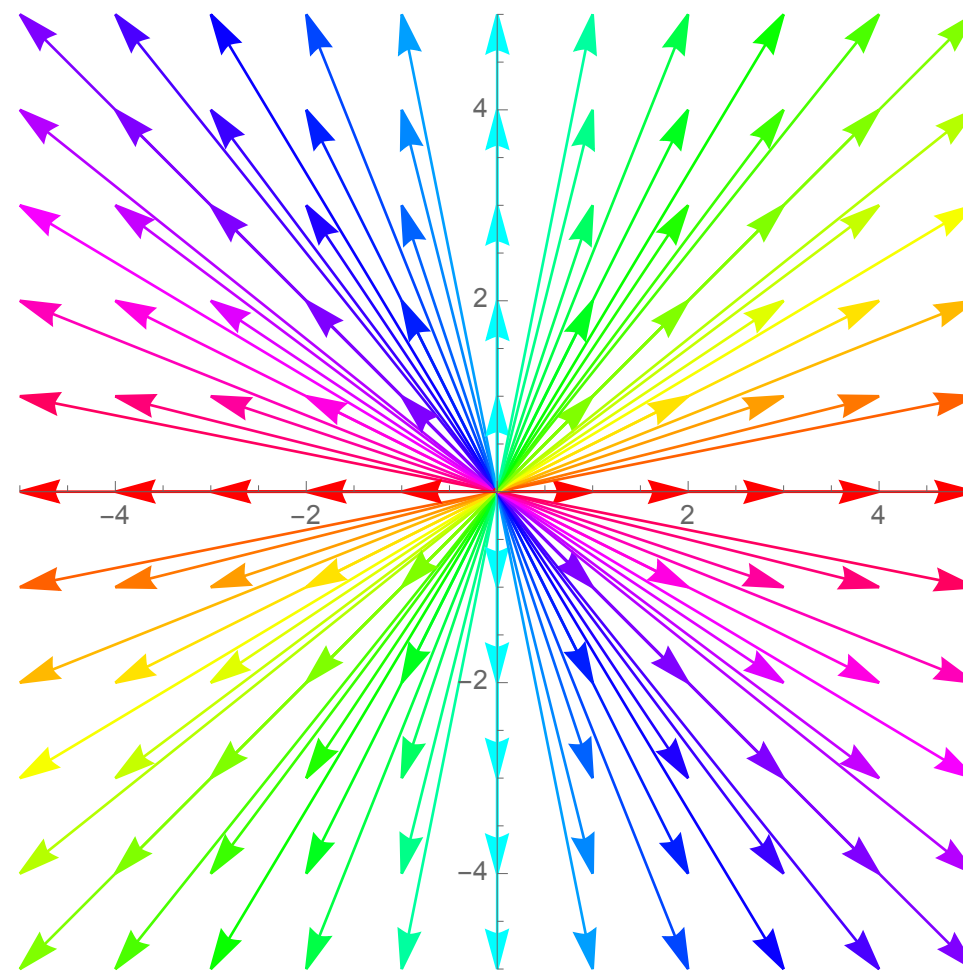
Linear transformations



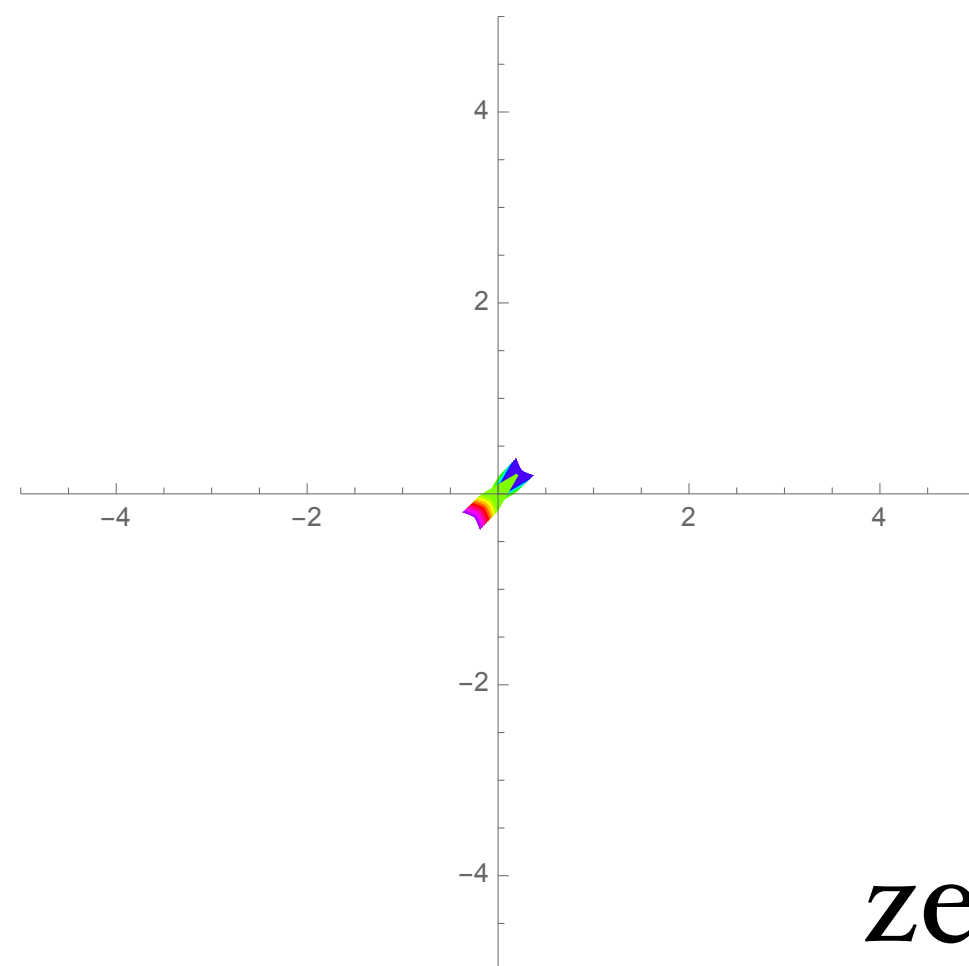
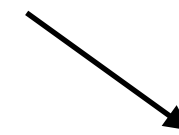
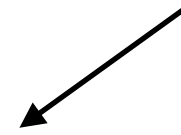
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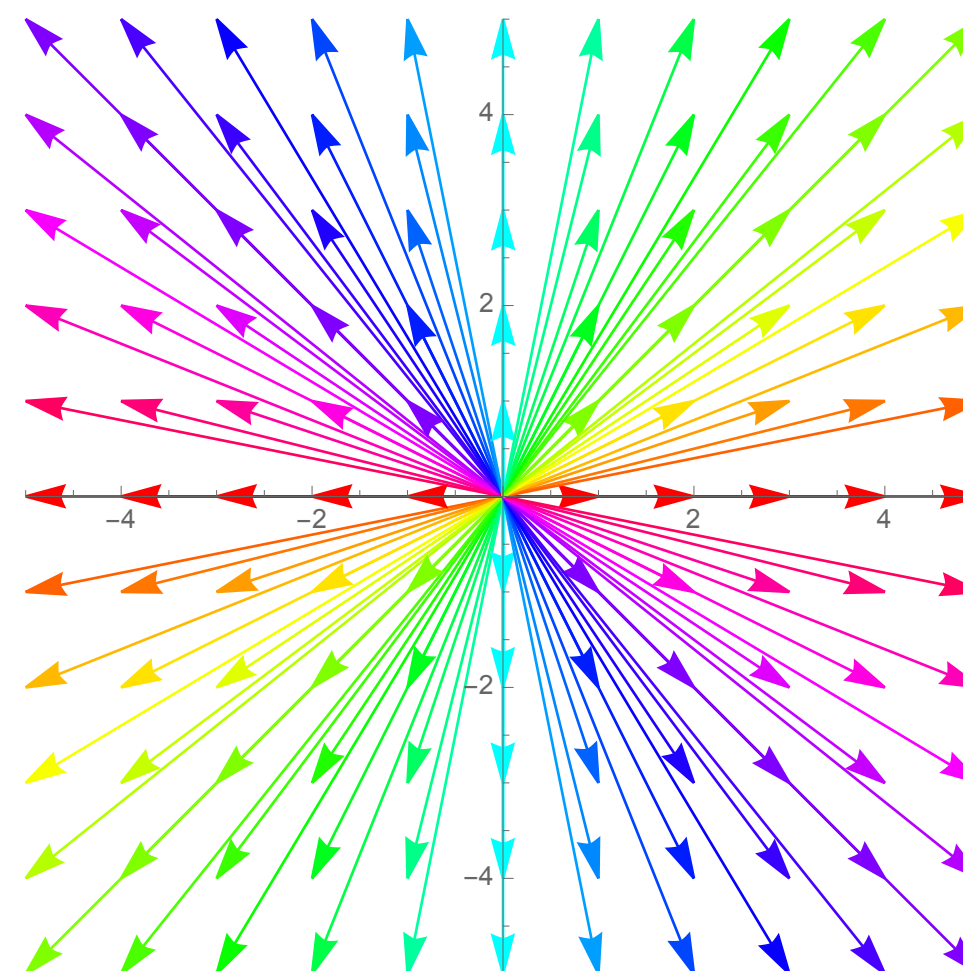
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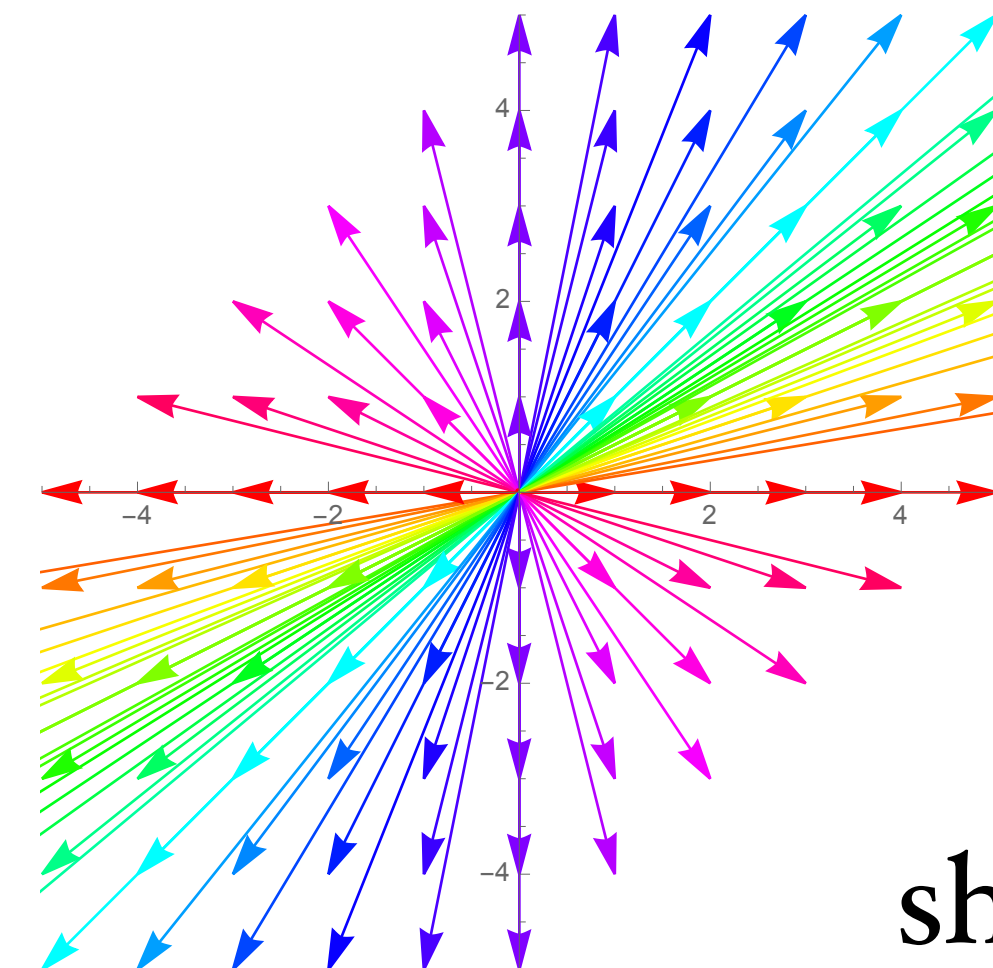
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zero



identity



shear

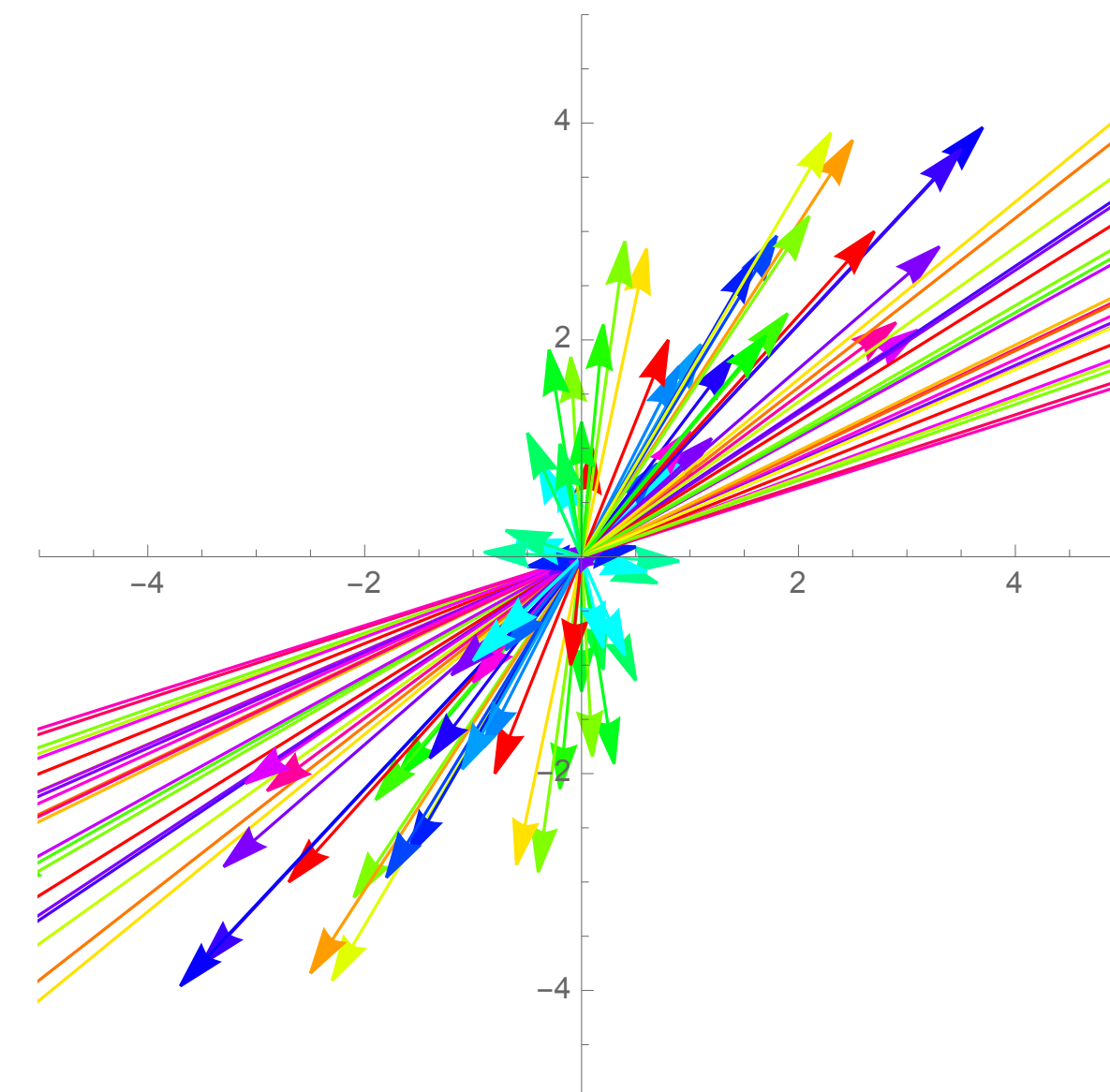
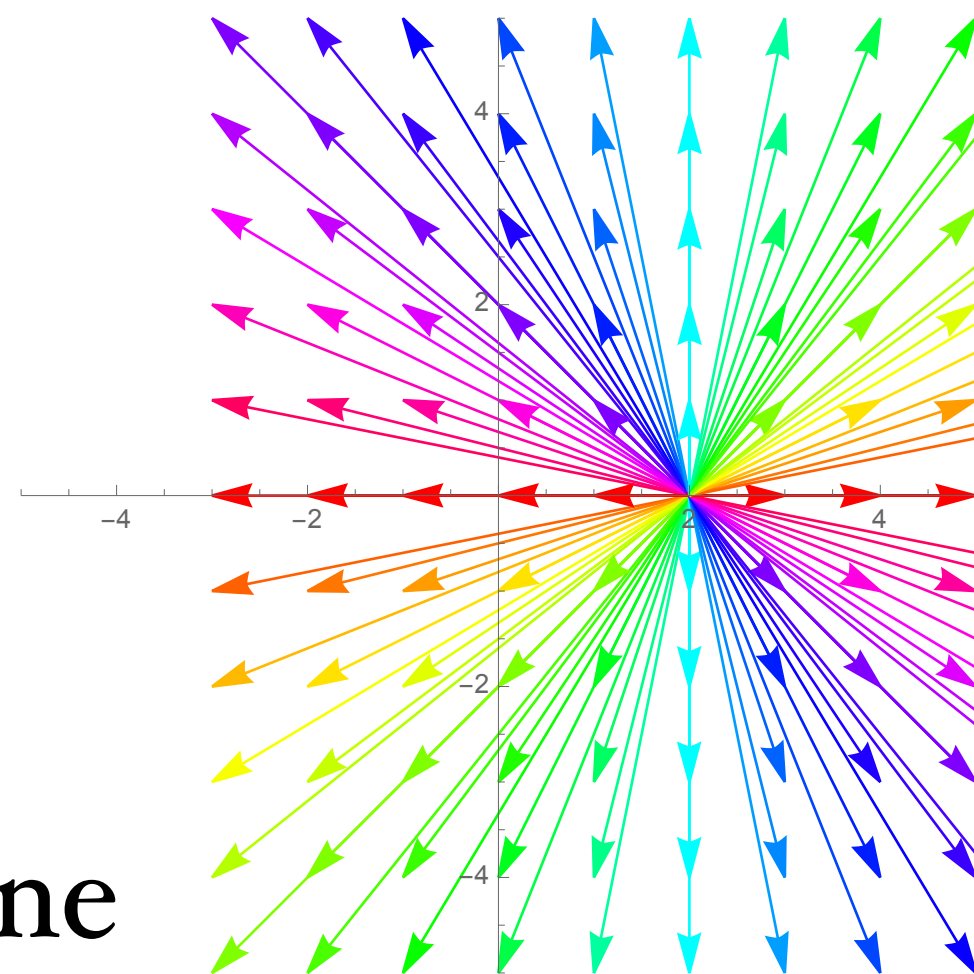
Linear transformations

non-examples

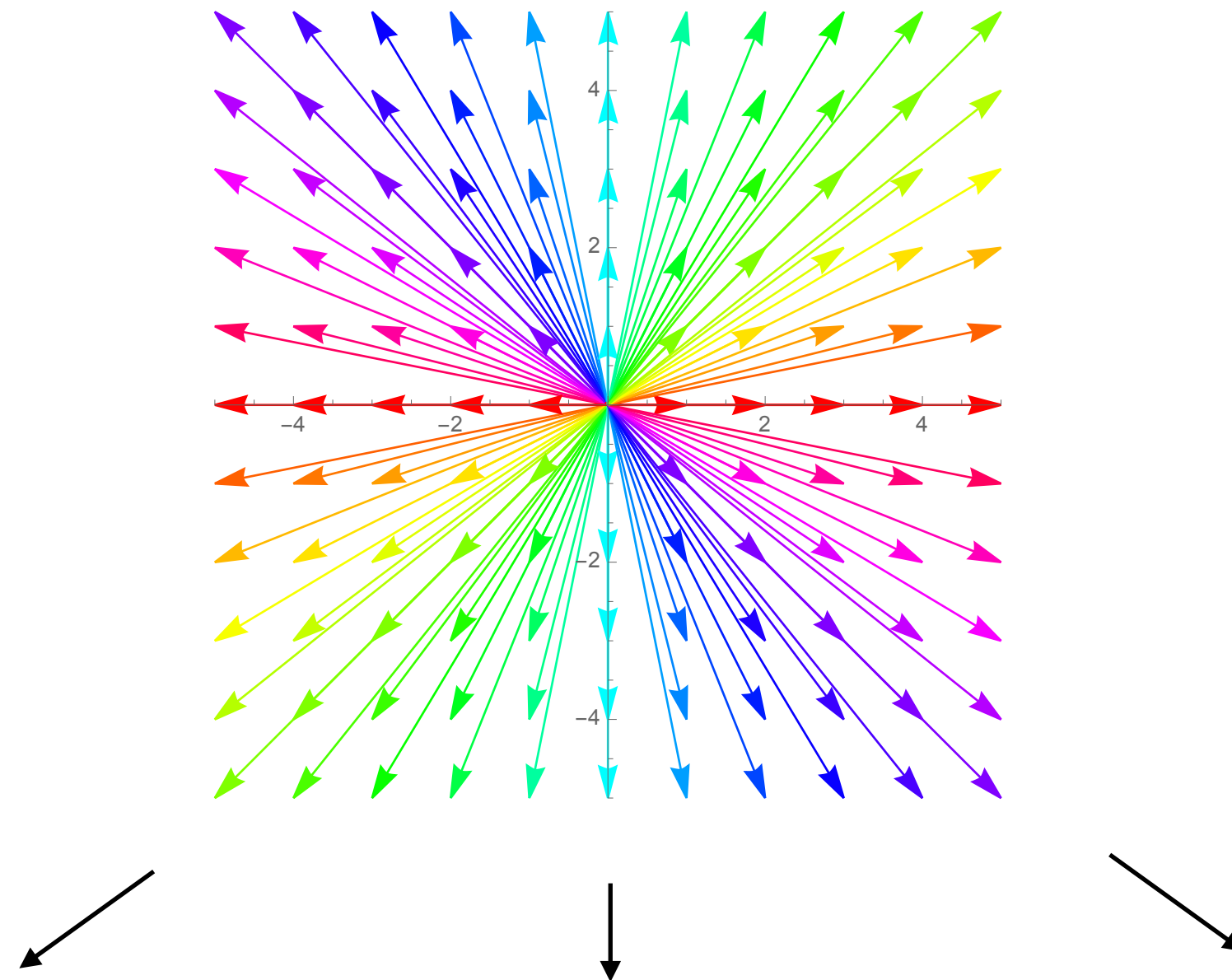
- doesn't fix the zero vector

- doesn't map lines to lines

affine

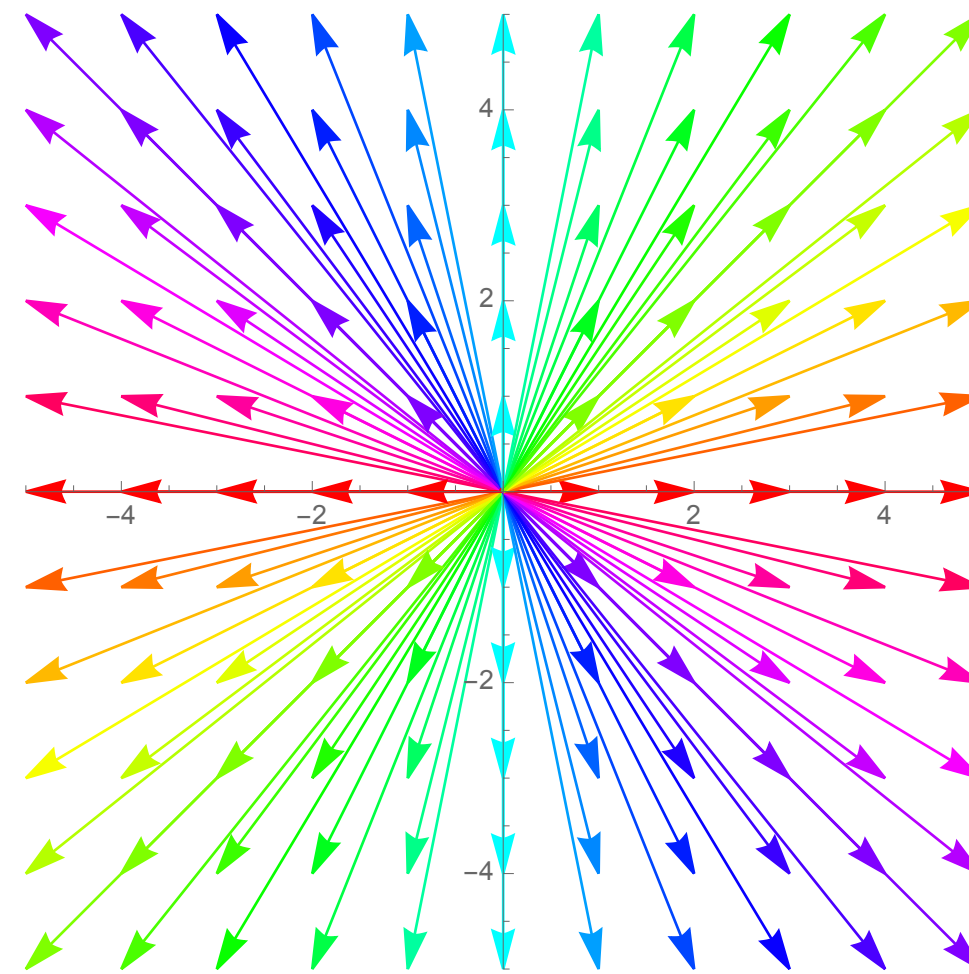


Linear transformations

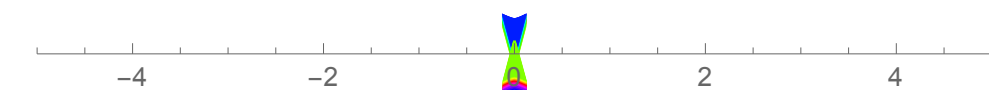
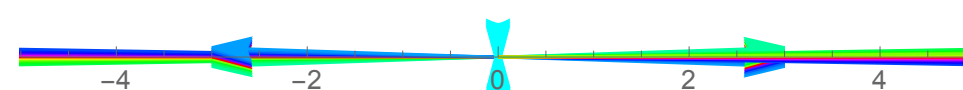
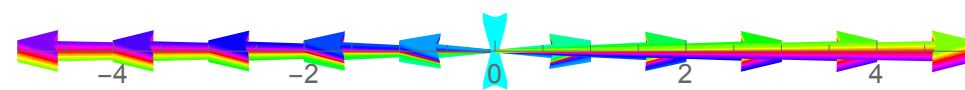
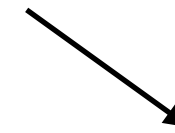
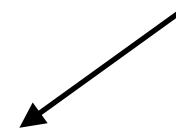


$$\mathbb{R}^2 \rightarrow \mathbb{R}^1$$

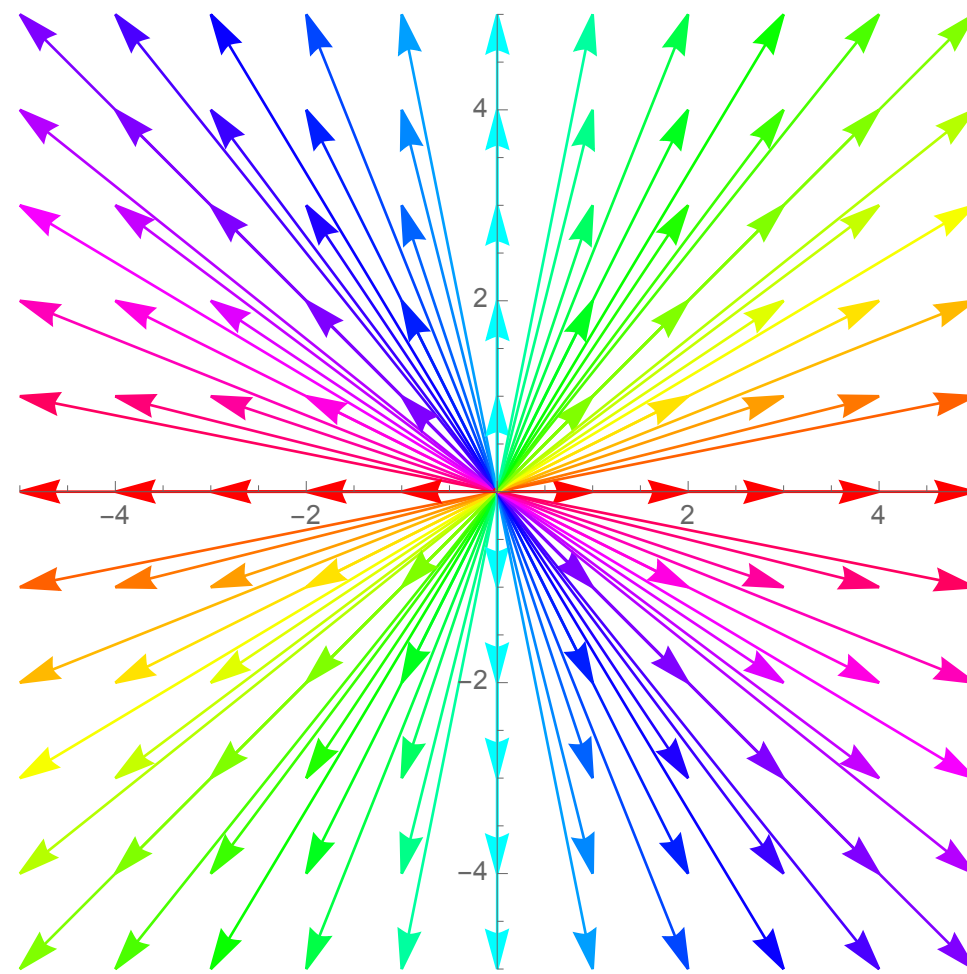
Linear transformations



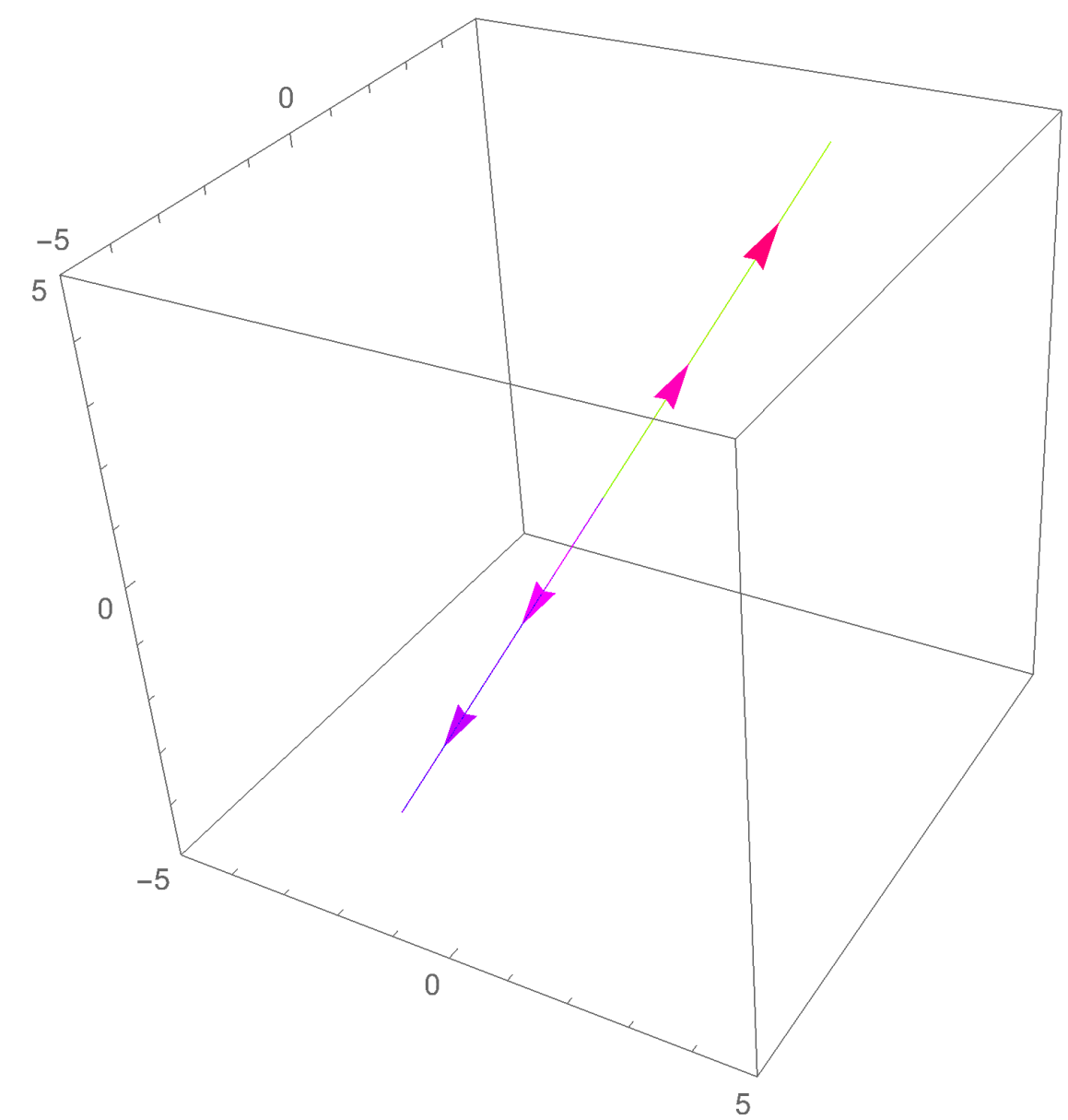
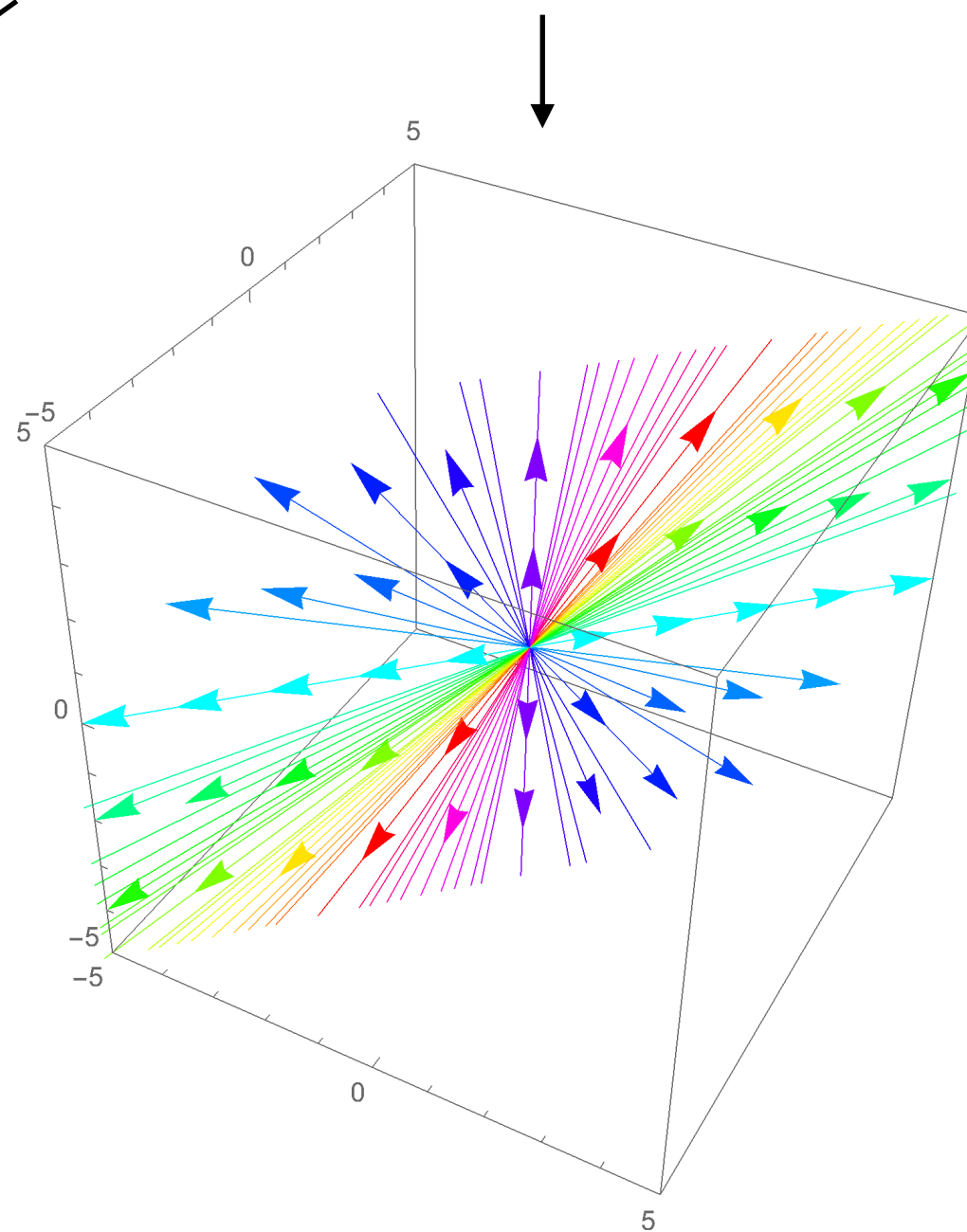
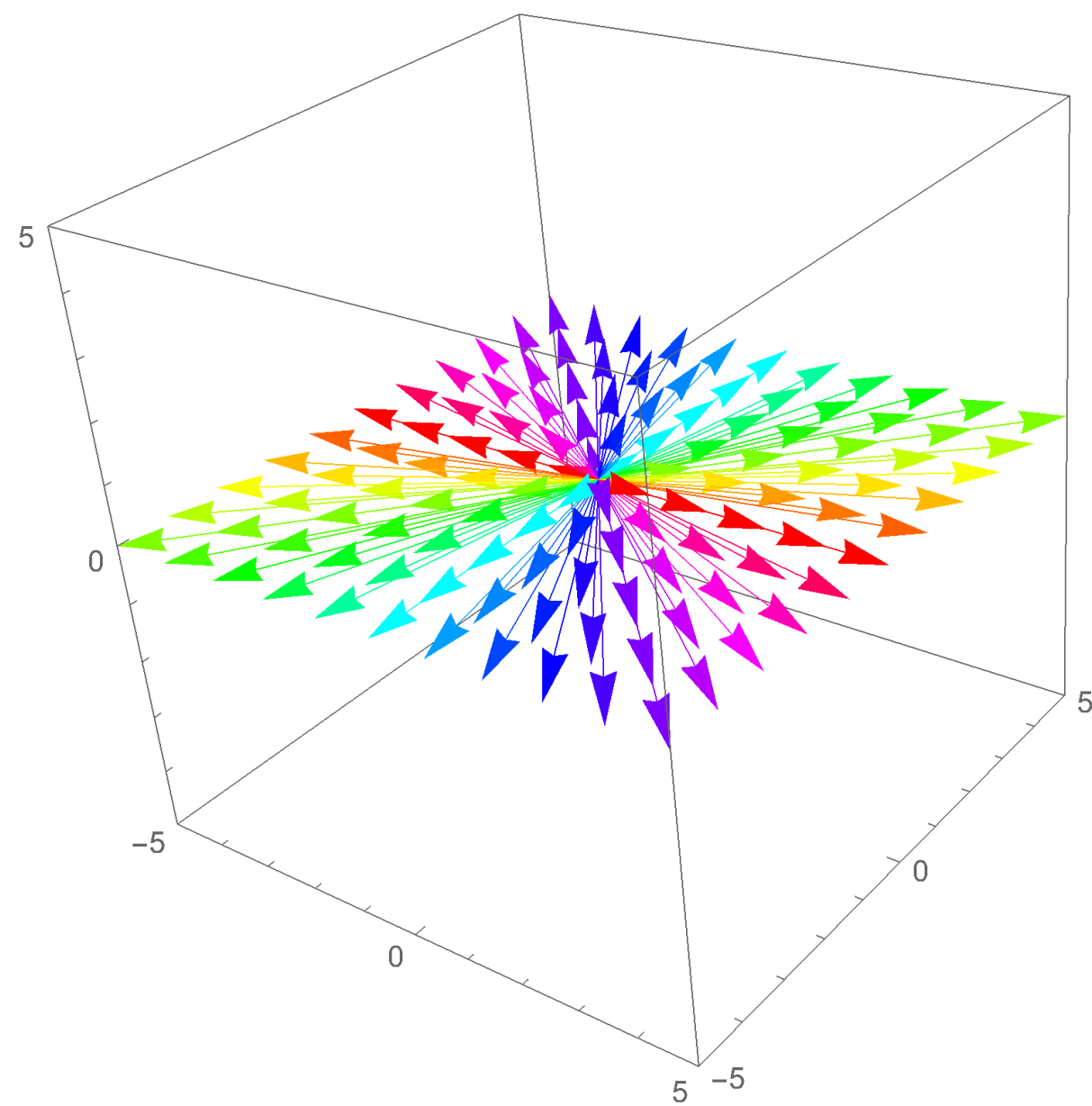
$$\mathbb{R}^2 \rightarrow \mathbb{R}^1$$



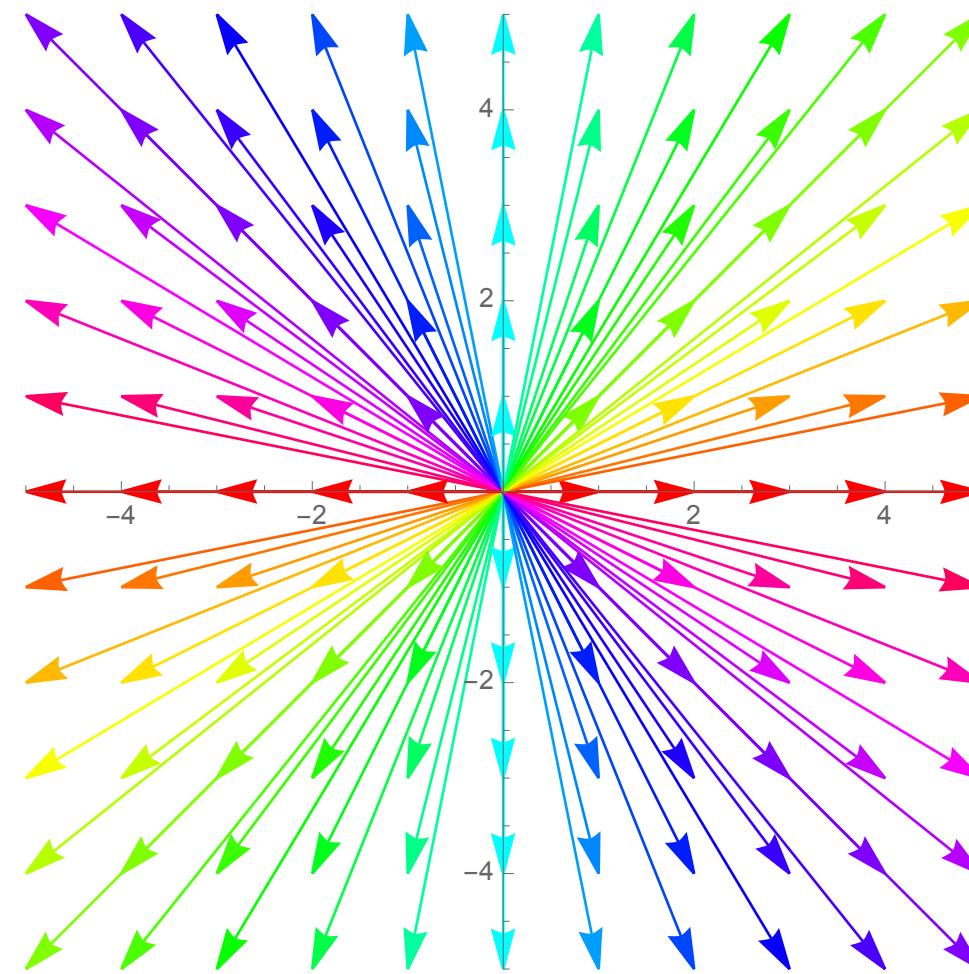
Linear transformations



$$\mathbb{R}^2 \rightarrow \mathbb{R}^3$$

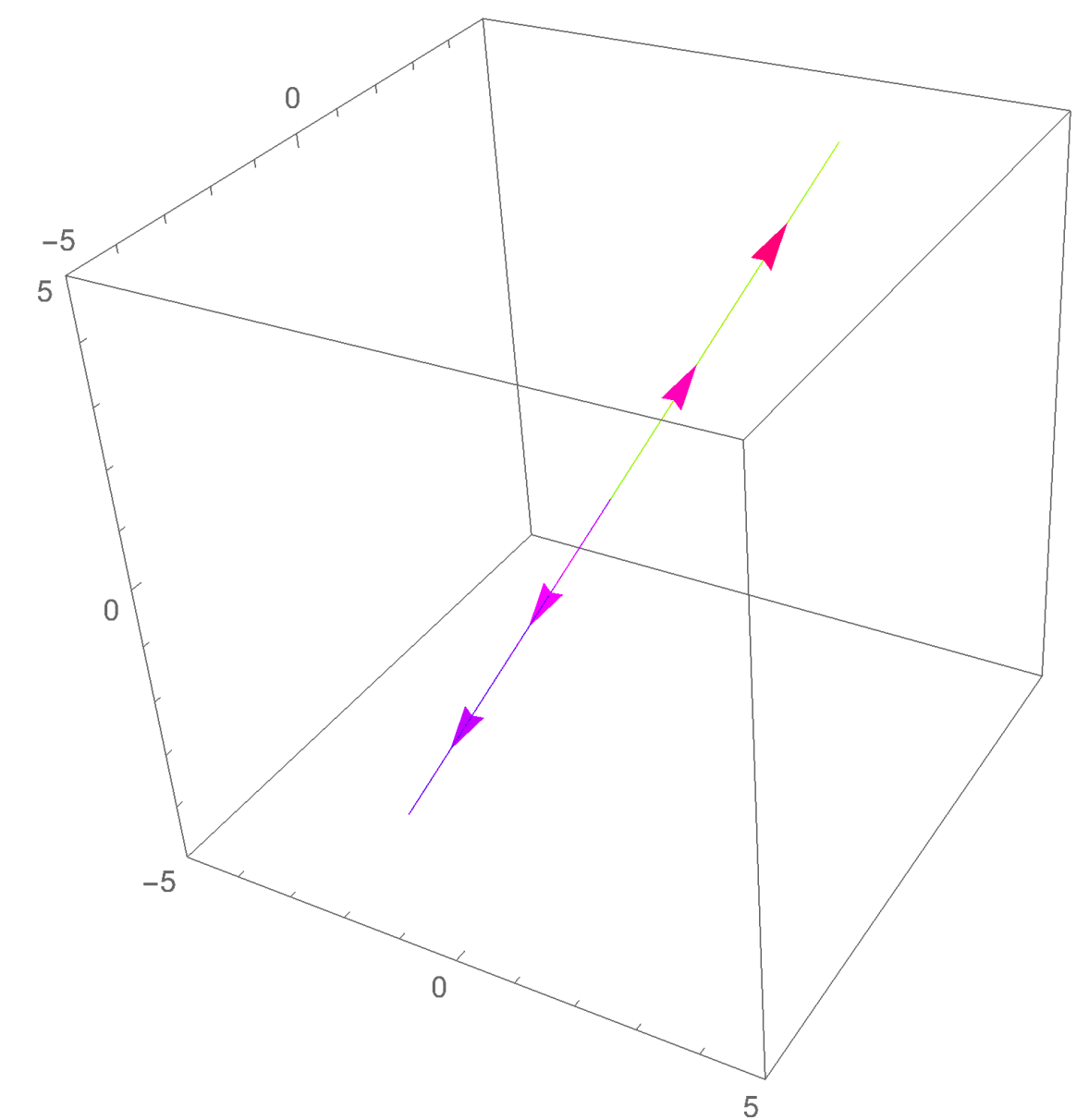
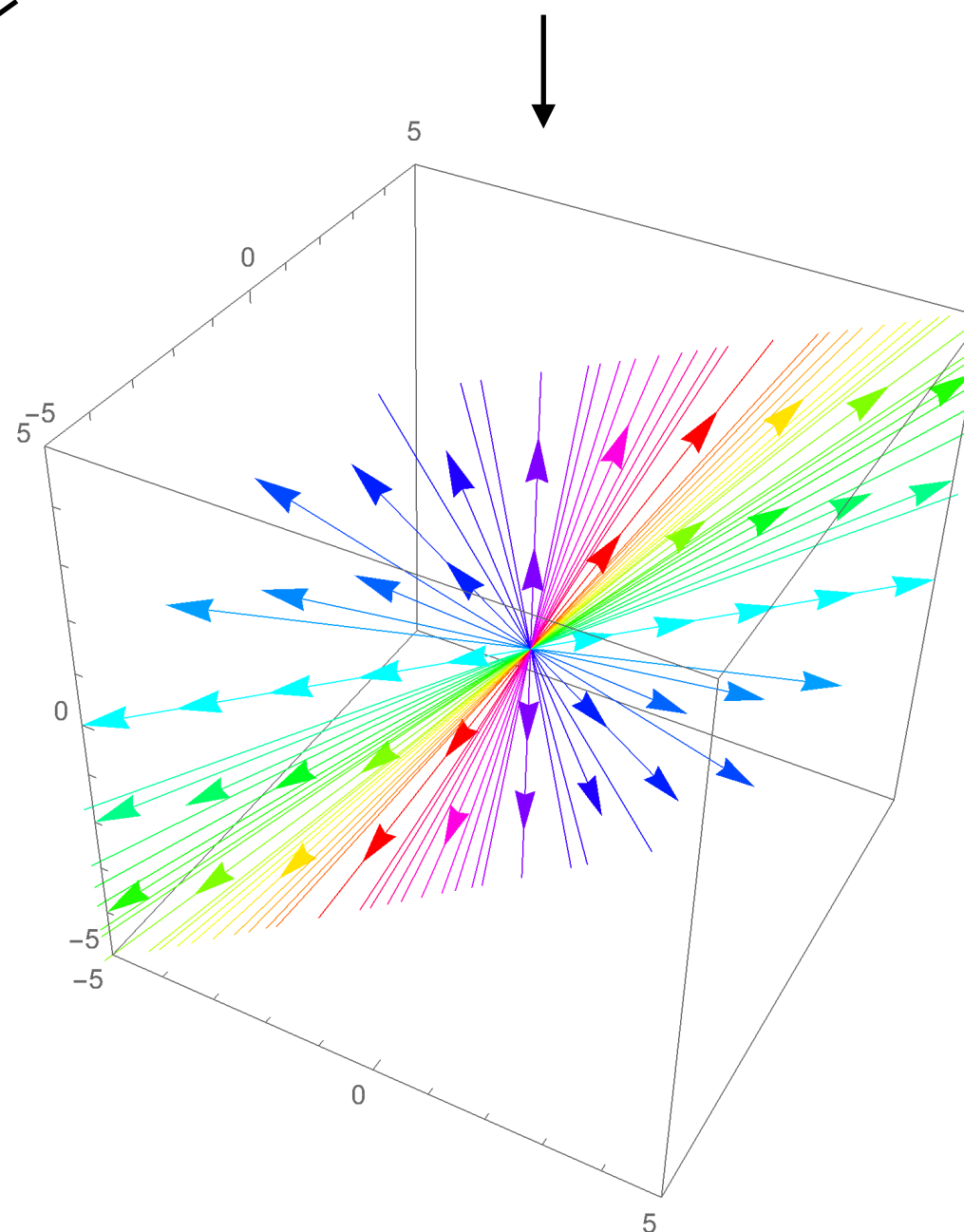
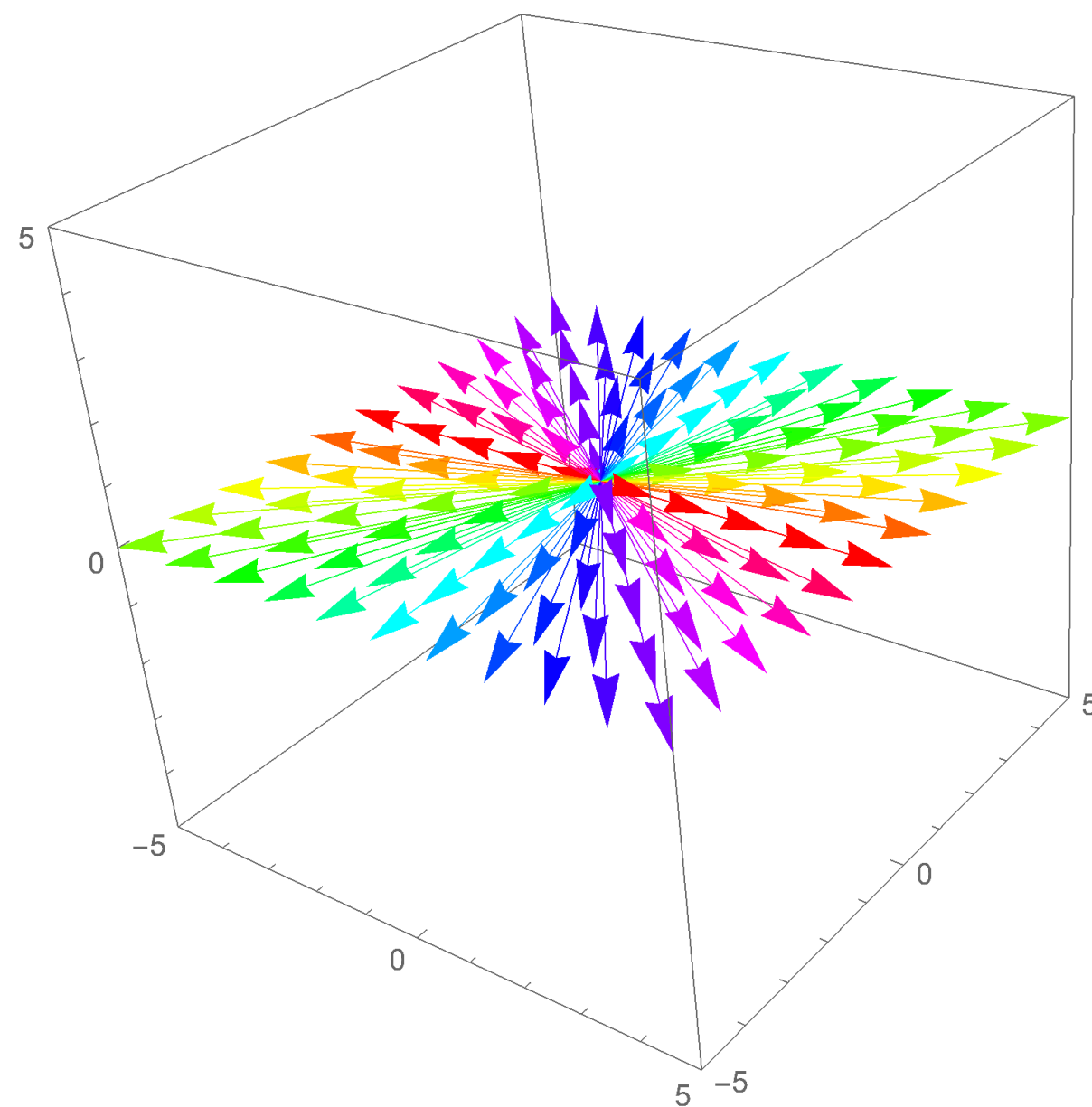


Linear transformations



How would
we ever be able
to write down
a linear transformation?

$$\mathbb{R}^2 \rightarrow \mathbb{R}^3$$



Linear transformations
are fully determined by how they act on a basis.

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$$\text{If } T\left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right) = \begin{pmatrix} 2 \\ 0 \end{pmatrix} \quad \text{and} \quad T\left(\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right) = \begin{pmatrix} 0 \\ -1 \end{pmatrix}$$

then

$$T\left(\begin{pmatrix} 3 \\ 4 \end{pmatrix}\right) =$$

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then

$$T\left(\begin{pmatrix} 3 \\ 4 \end{pmatrix}\right) = T\left(3 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + 4 \begin{pmatrix} 0 \\ 1 \end{pmatrix}\right)$$

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$$\text{If } T\left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right) = \begin{pmatrix} 2 \\ 0 \end{pmatrix} \quad \text{and} \quad T\left(\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right) = \begin{pmatrix} 0 \\ -1 \end{pmatrix}$$

then

$$T\left(\begin{pmatrix} 3 \\ 4 \end{pmatrix}\right) = T\left(3 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + 4 \begin{pmatrix} 0 \\ 1 \end{pmatrix}\right) = 3T\left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right) + 4T\left(\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right)$$

Linear transformations
are fully determined by how they act on a basis.

$$\text{If } T\left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right) = \begin{pmatrix} 2 \\ 0 \end{pmatrix} \quad \text{and} \quad T\left(\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right) = \begin{pmatrix} 0 \\ -1 \end{pmatrix}$$

then

$$\begin{aligned} T\left(\begin{pmatrix} 3 \\ 4 \end{pmatrix}\right) &= T\left(3 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + 4 \begin{pmatrix} 0 \\ 1 \end{pmatrix}\right) = 3T\left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right) + 4T\left(\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right) \\ &= 3 \begin{pmatrix} 2 \\ 0 \end{pmatrix} + 4 \begin{pmatrix} 0 \\ -1 \end{pmatrix} \end{aligned}$$

Linear transformations
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Linear transformations
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$$T\left(\begin{pmatrix} 3 \\ 4 \end{pmatrix}\right) = T\left(3 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + 4 \begin{pmatrix} 0 \\ 1 \end{pmatrix}\right) = 3T\left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right) + 4T\left(\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right)$$

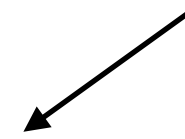
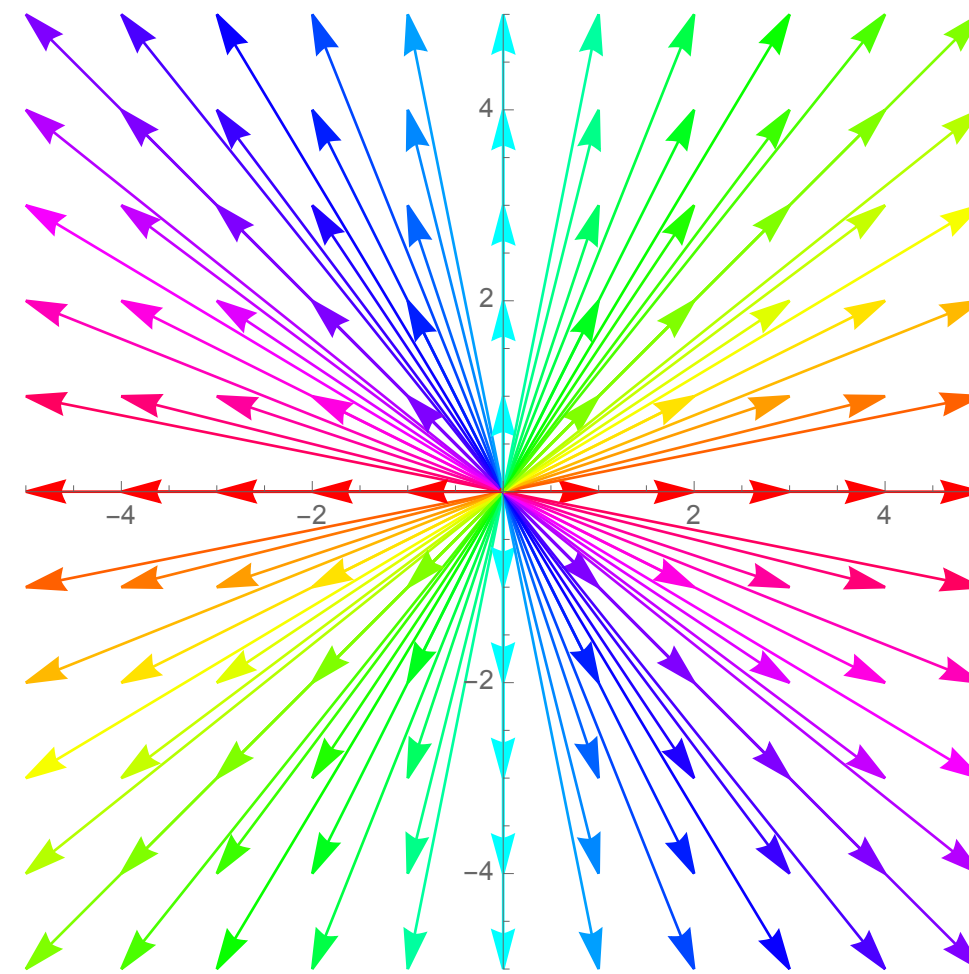
$$= 3 \begin{pmatrix} 2 \\ 0 \end{pmatrix} + 4 \begin{pmatrix} 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 6 \\ -4 \end{pmatrix}$$

$$T = \begin{pmatrix} 2 & 0 \\ 0 & -1 \end{pmatrix}$$

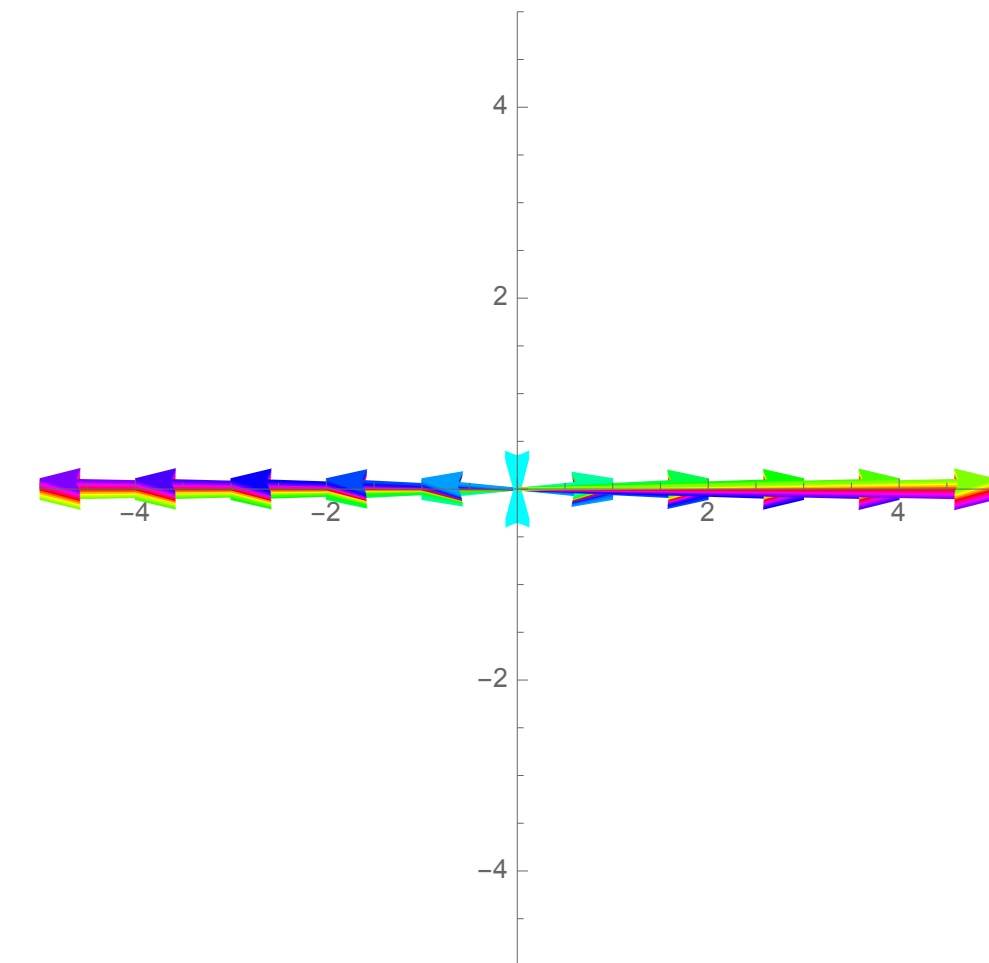
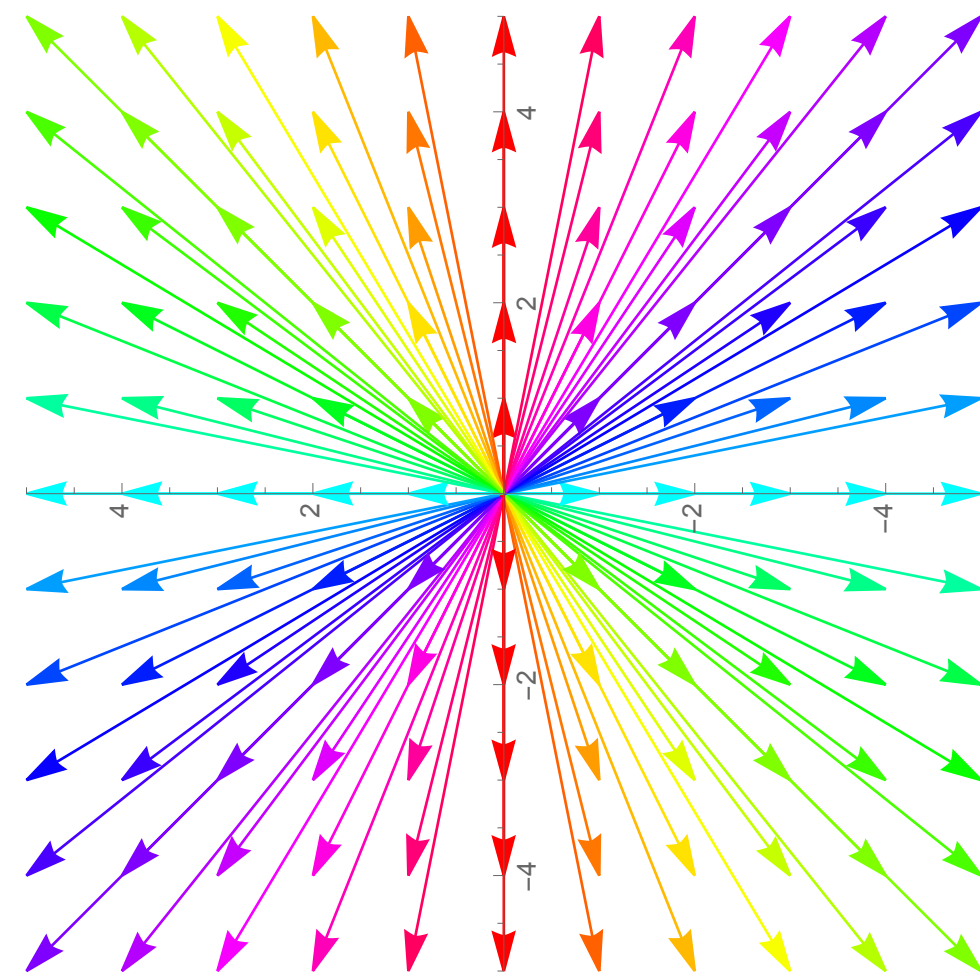
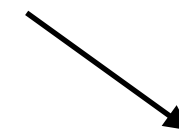
Linear transformations
are fully determined by how they act on a basis.

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \mapsto \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} 0 \\ 1 \end{pmatrix} \mapsto \begin{pmatrix} -1 \\ 0 \end{pmatrix}$$



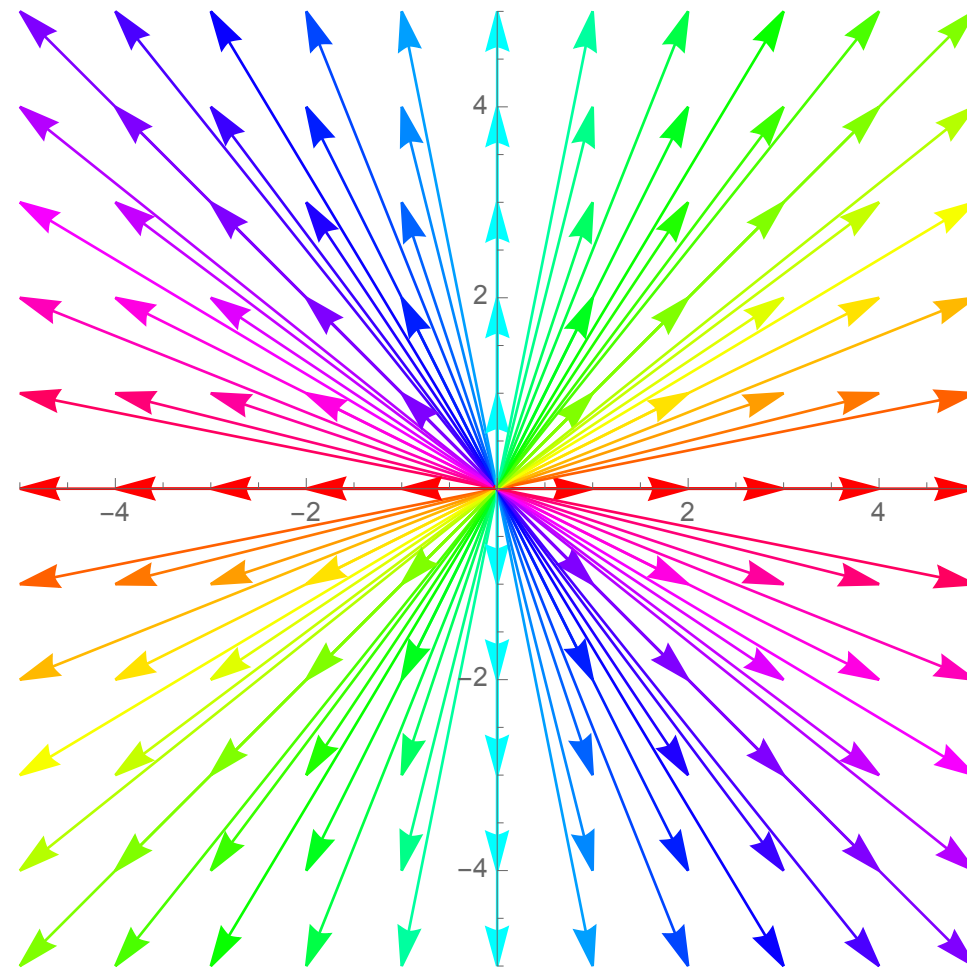
$$\mathbb{R}^2 \rightarrow \mathbb{R}^2$$



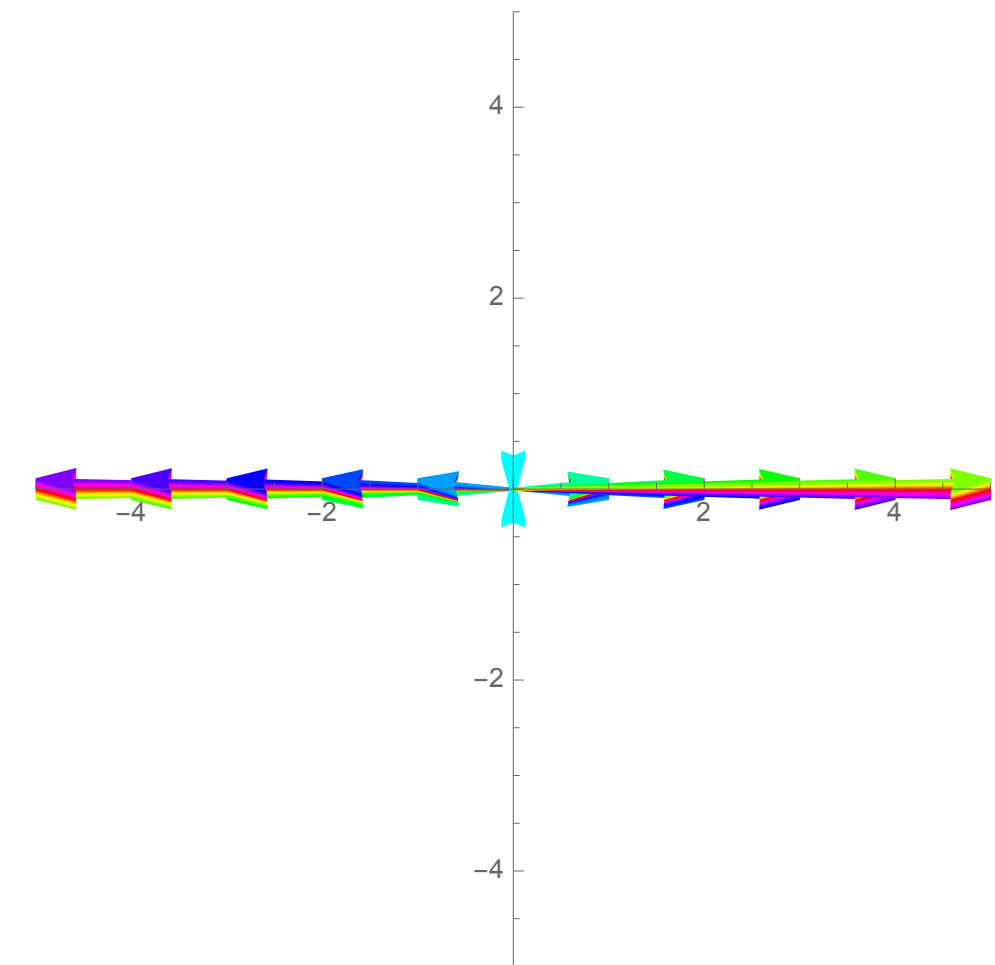
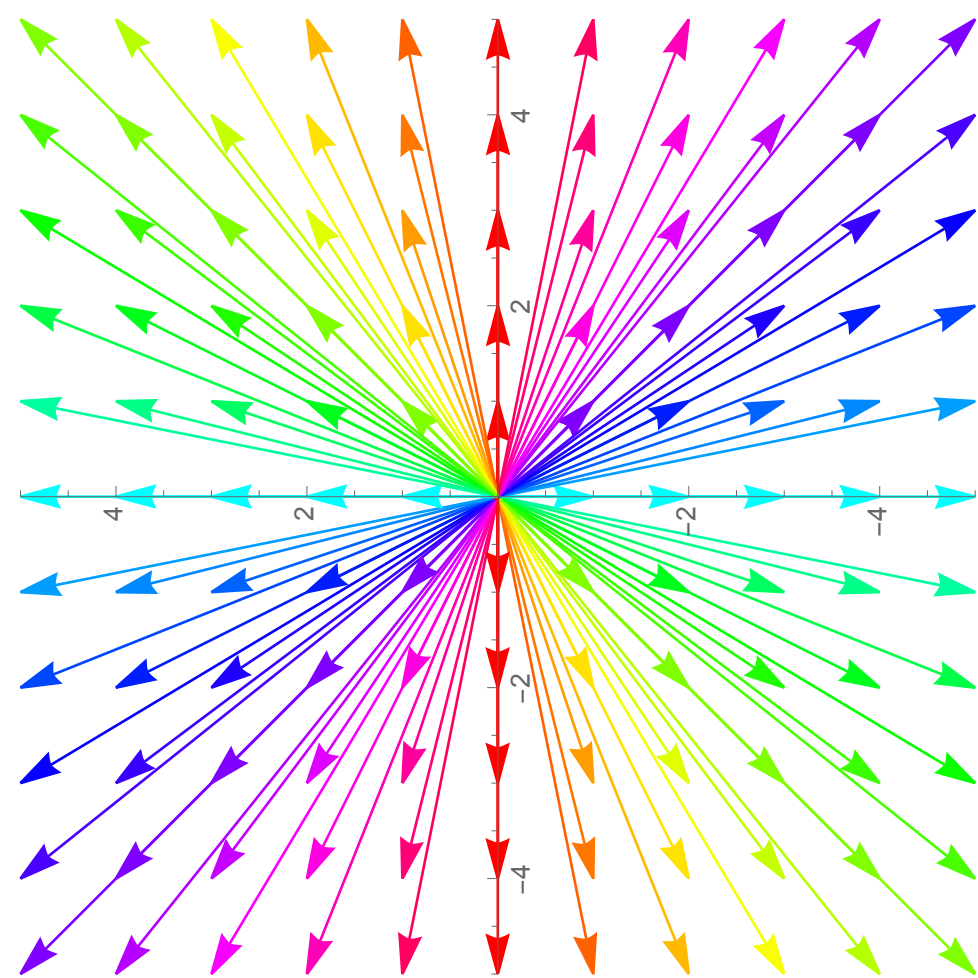
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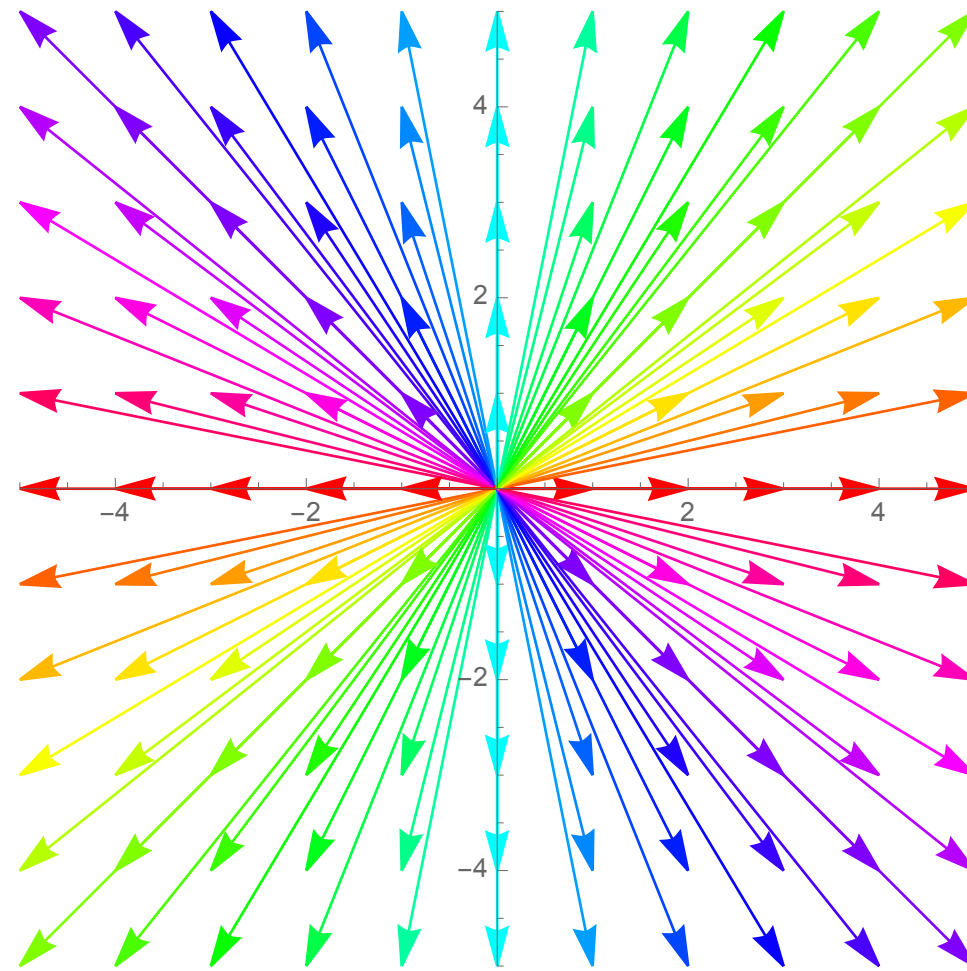
$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \mathbb{R}^2 \rightarrow \mathbb{R}^2$$



Linear transformations
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$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \mapsto \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

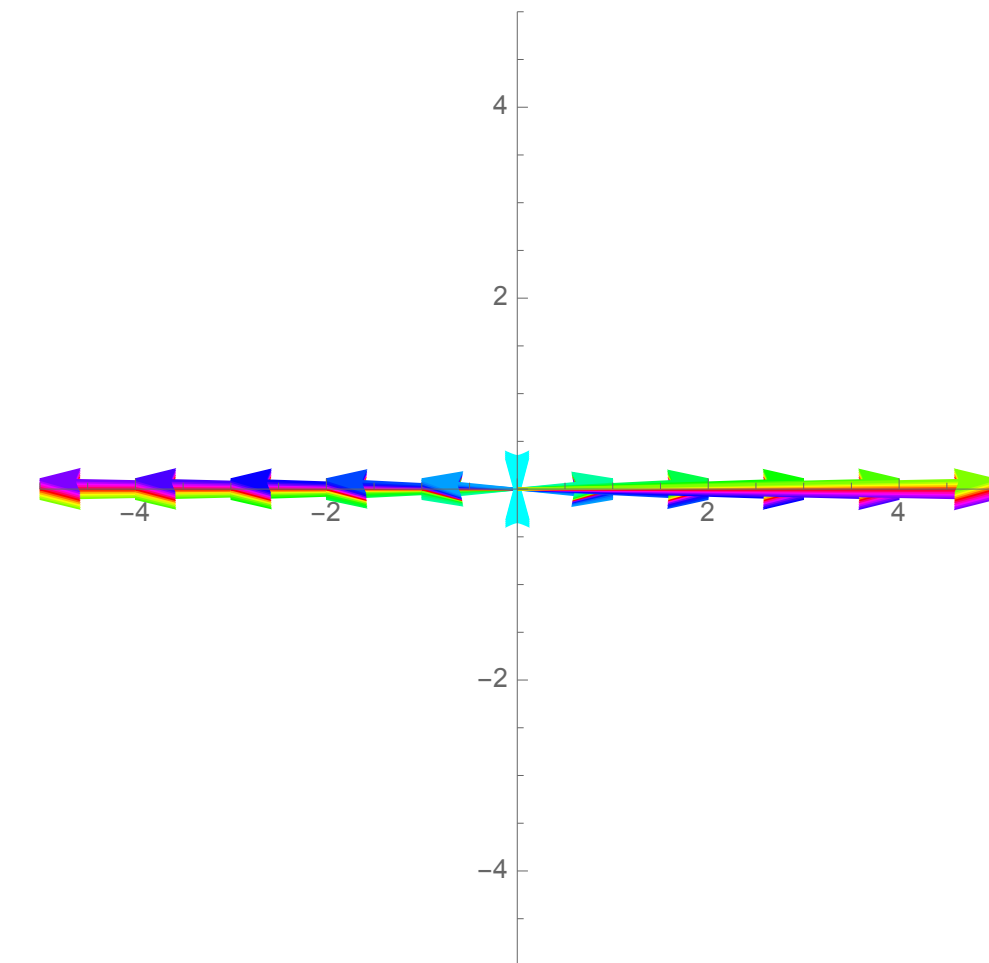
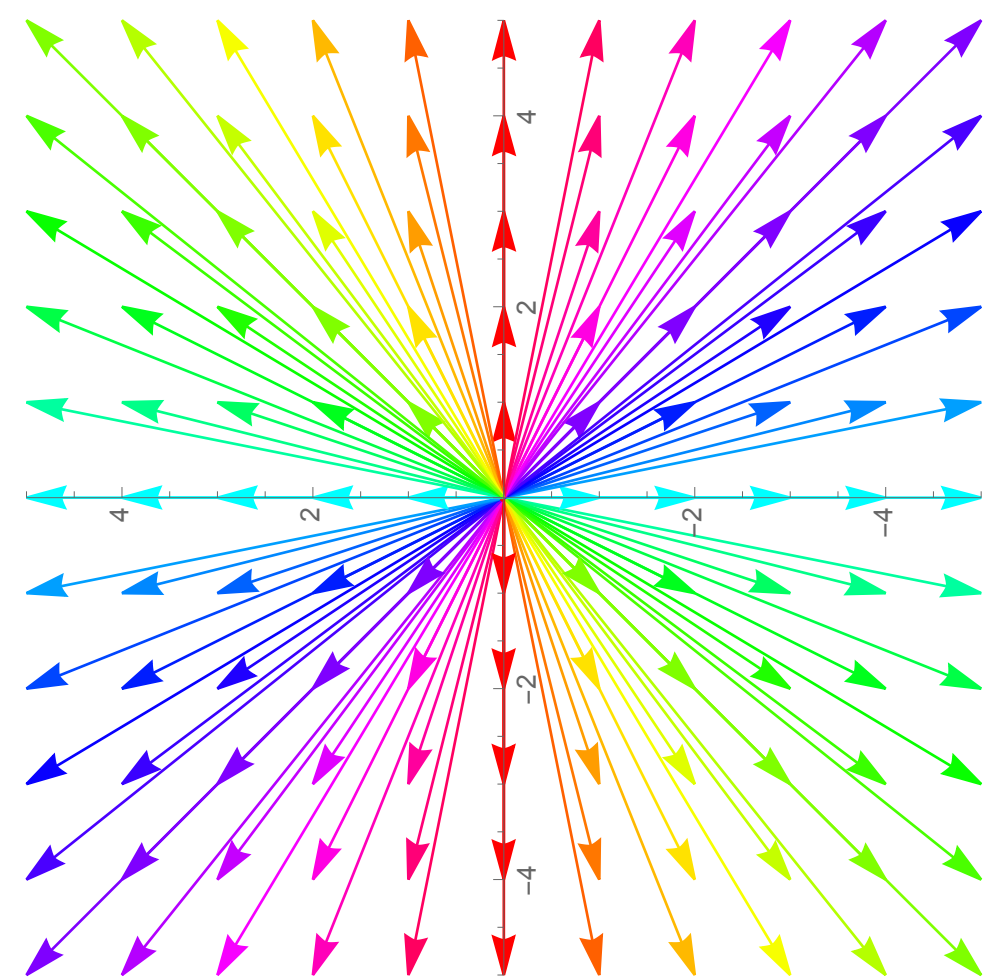
$$\begin{pmatrix} 0 \\ 1 \end{pmatrix} \mapsto \begin{pmatrix} -1 \\ 0 \end{pmatrix}$$



$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \mapsto \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

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$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \mathbb{R}^2 \rightarrow \mathbb{R}^2$$



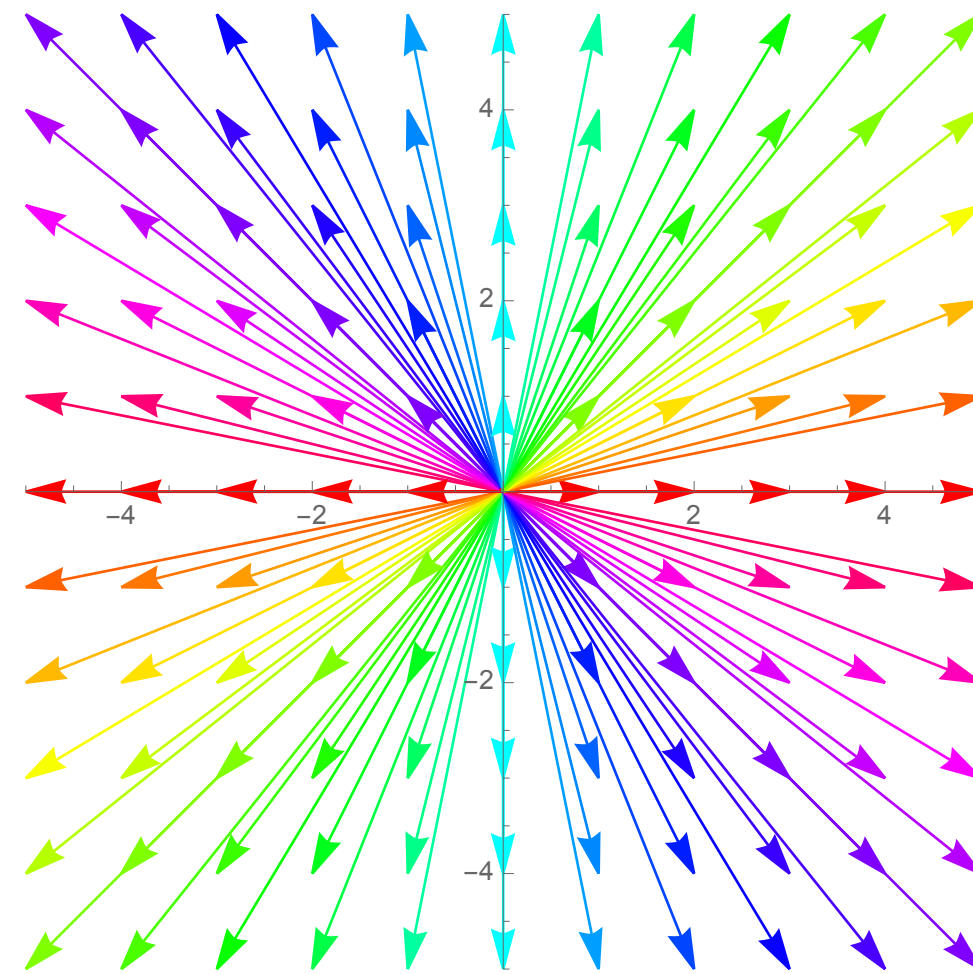
Linear transformations
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$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \mapsto \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

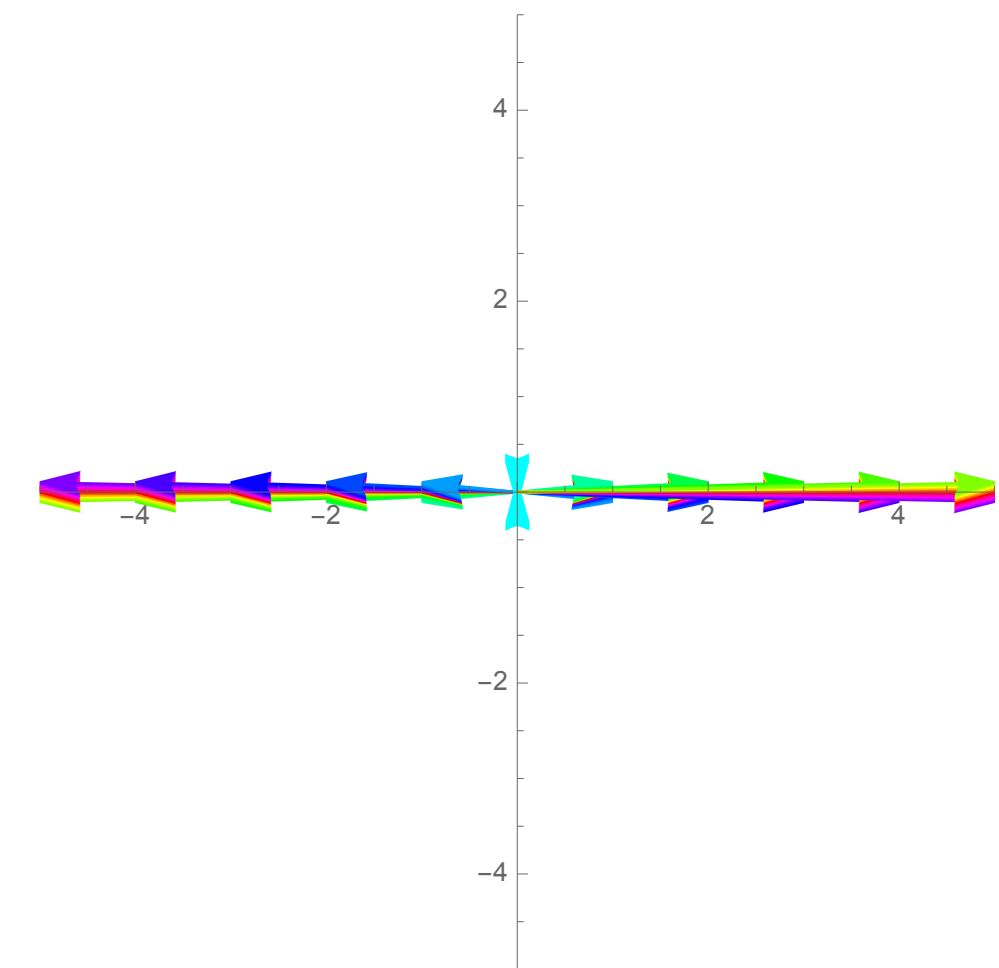
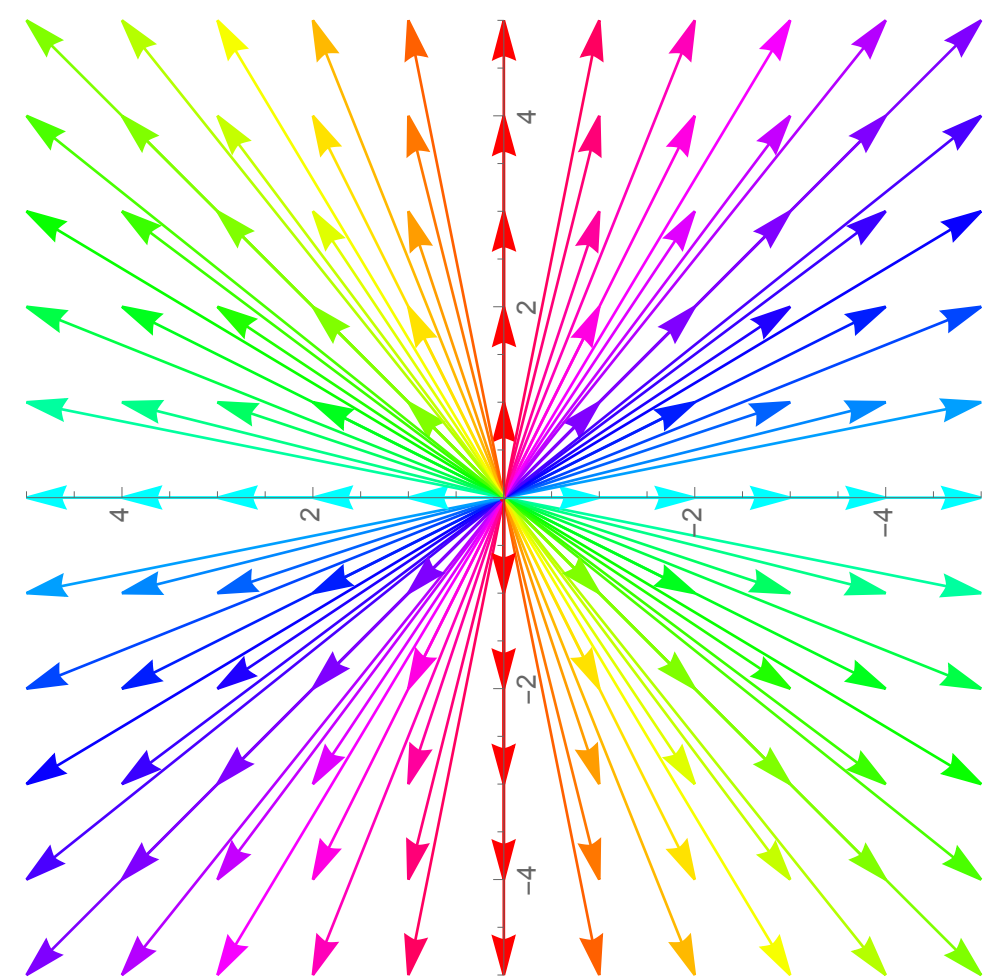
$$\begin{pmatrix} 0 \\ 1 \end{pmatrix} \mapsto \begin{pmatrix} -1 \\ 0 \end{pmatrix}$$

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$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \mathbb{R}^2 \rightarrow \mathbb{R}^2 \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$



Matrix multiplication is defined
exactly to capture the composition of linear transformations

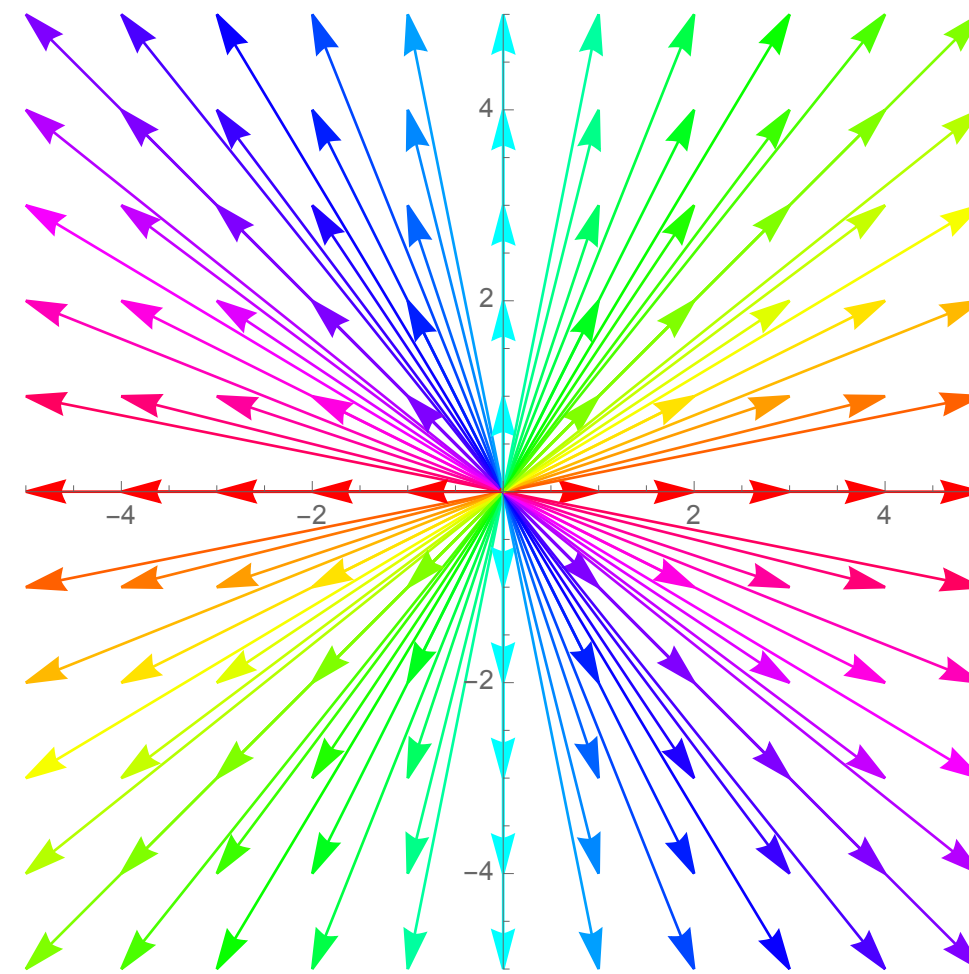
Linear transformations
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$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \mapsto \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

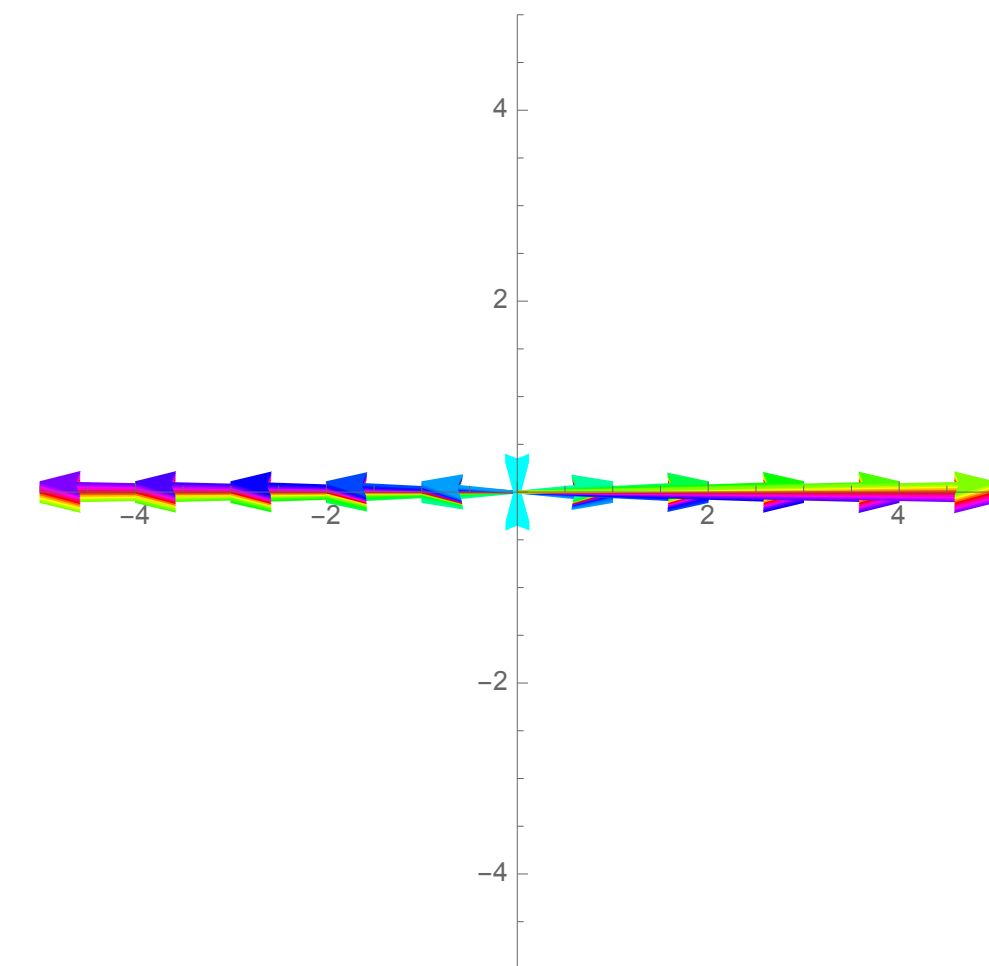
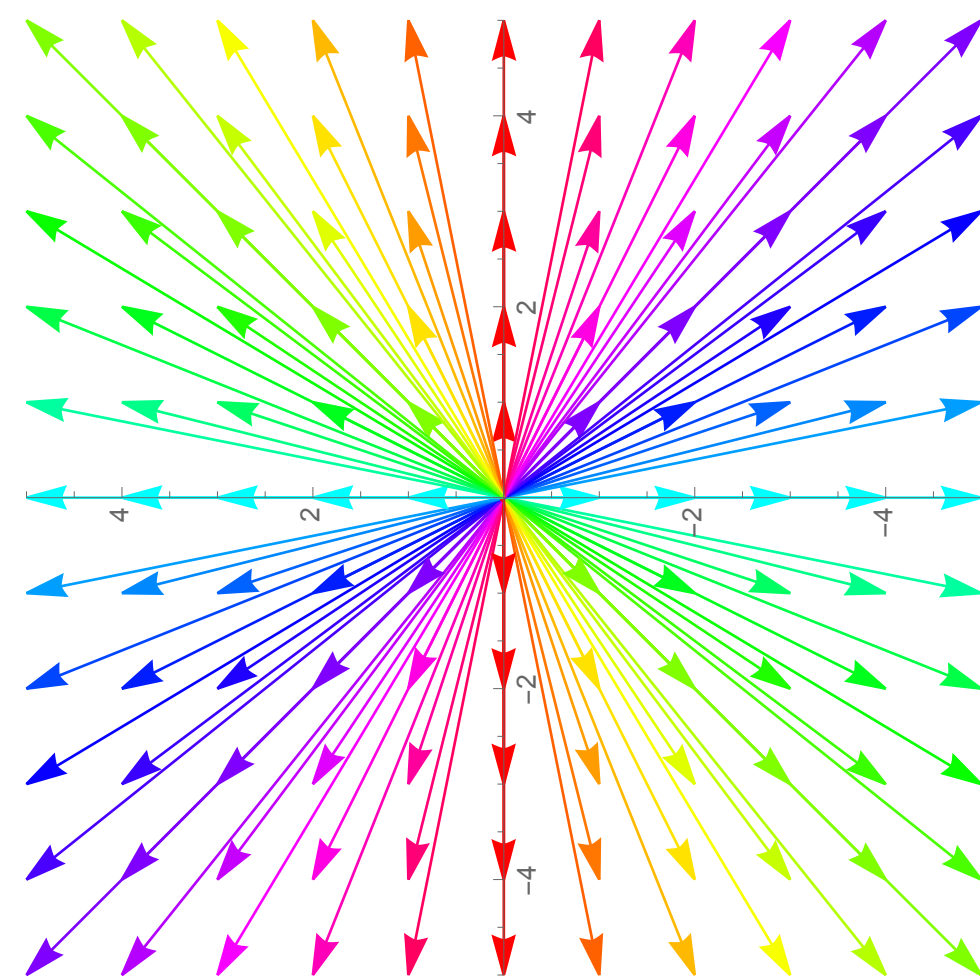
$$\begin{pmatrix} 0 \\ 1 \end{pmatrix} \mapsto \begin{pmatrix} -1 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \mapsto \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

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$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \mathbb{R}^2 \rightarrow \mathbb{R}^2 \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$



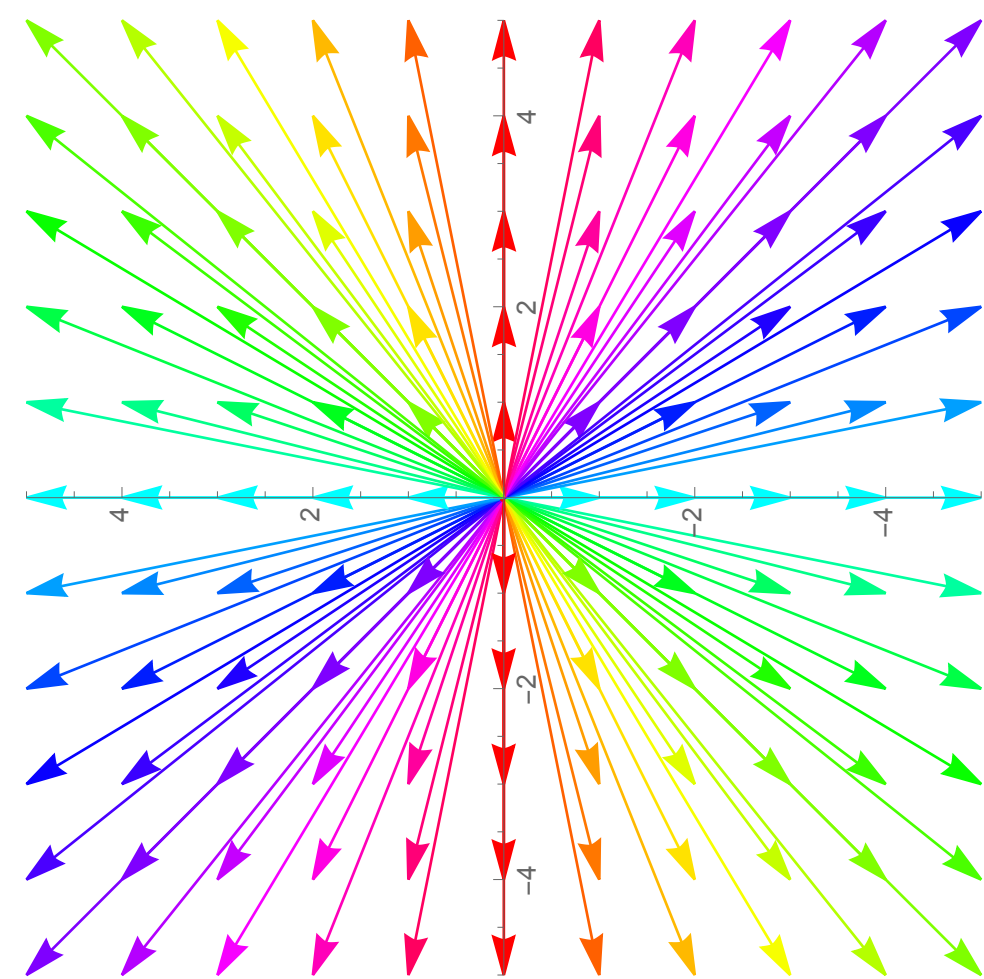
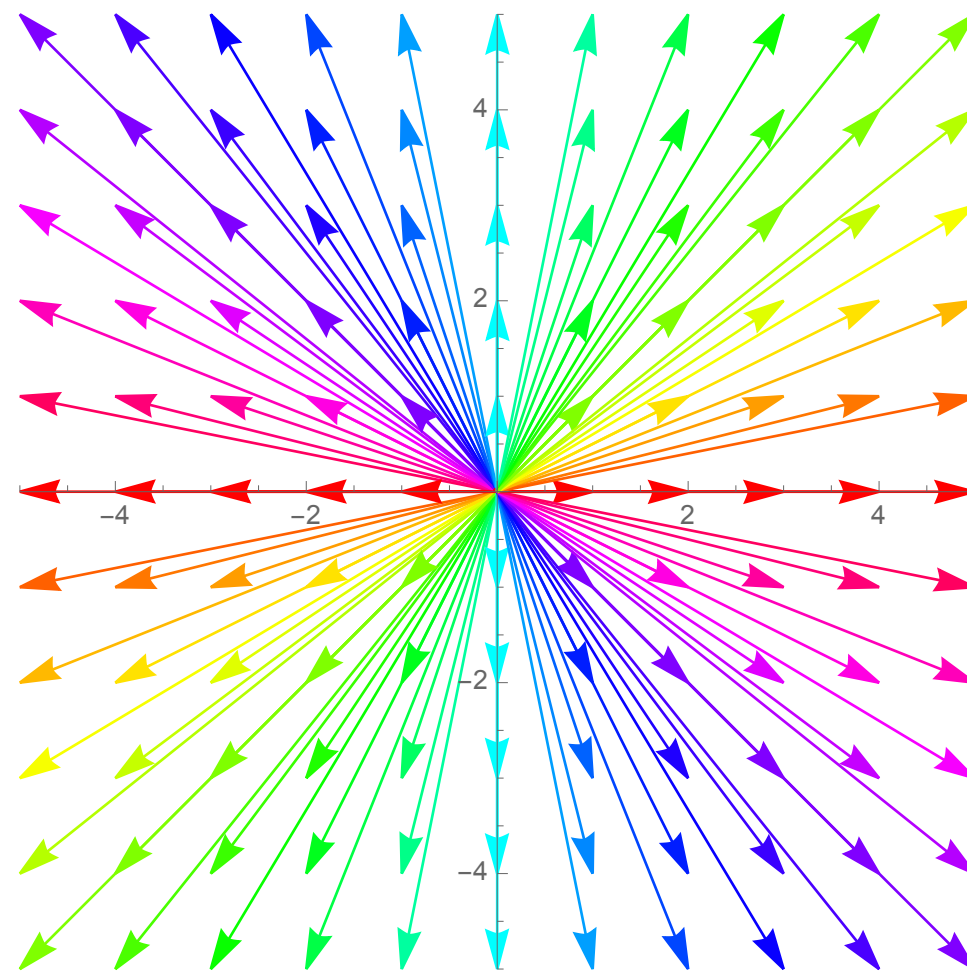
Linear transformations
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$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \mapsto \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

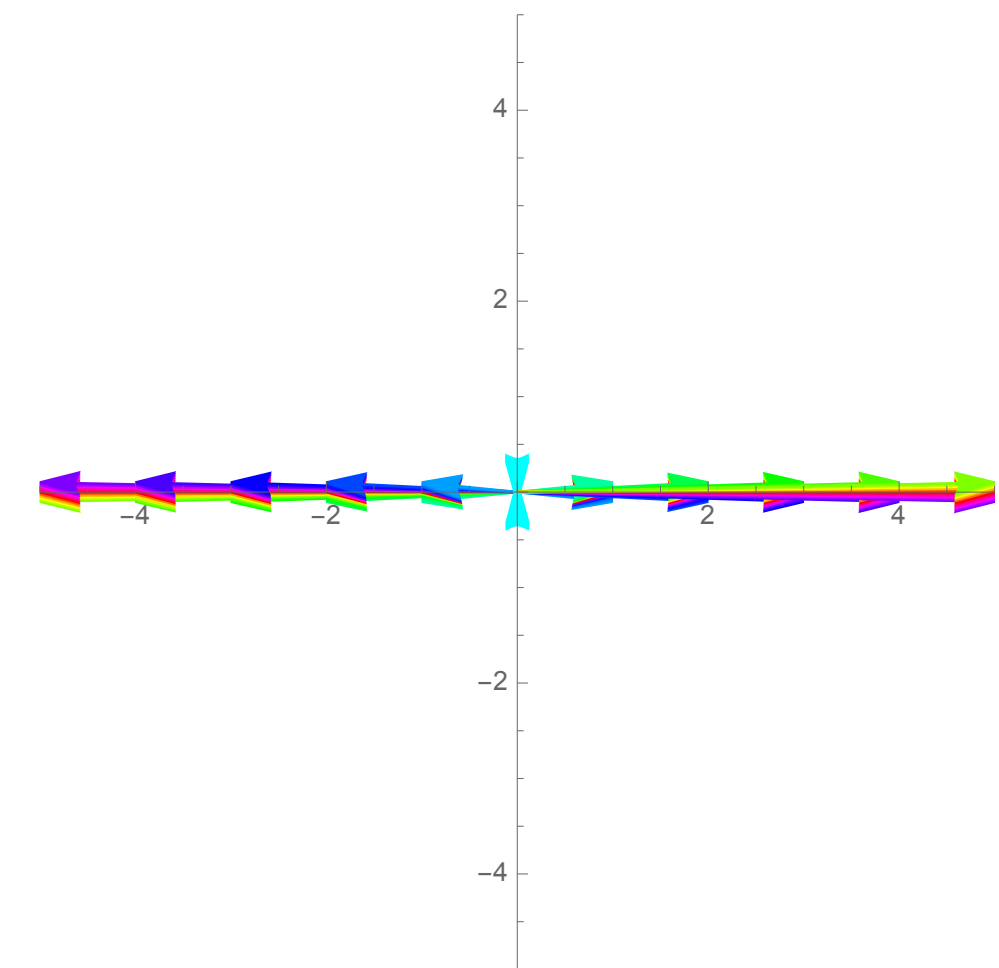
$$\begin{pmatrix} 0 \\ 1 \end{pmatrix} \mapsto \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$



$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \mathbb{R}^2 \rightarrow \mathbb{R}^2 \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

invertible

not invertible



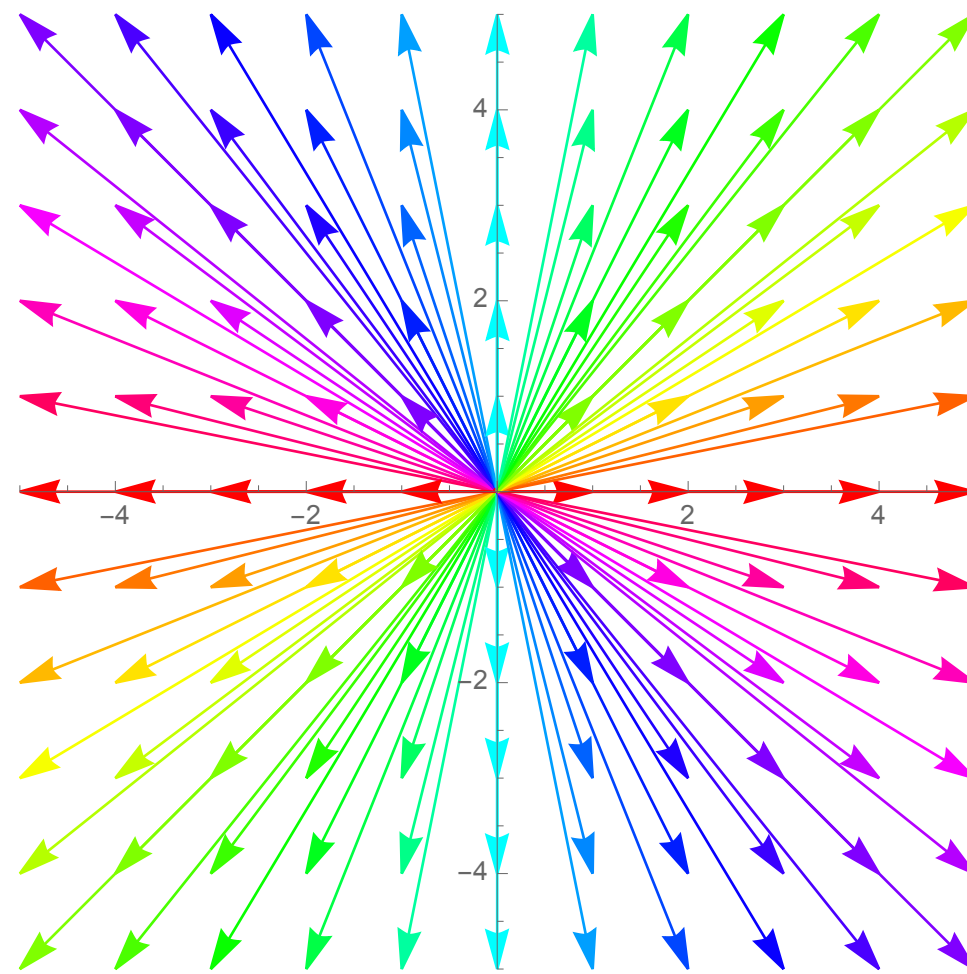
Linear transformations
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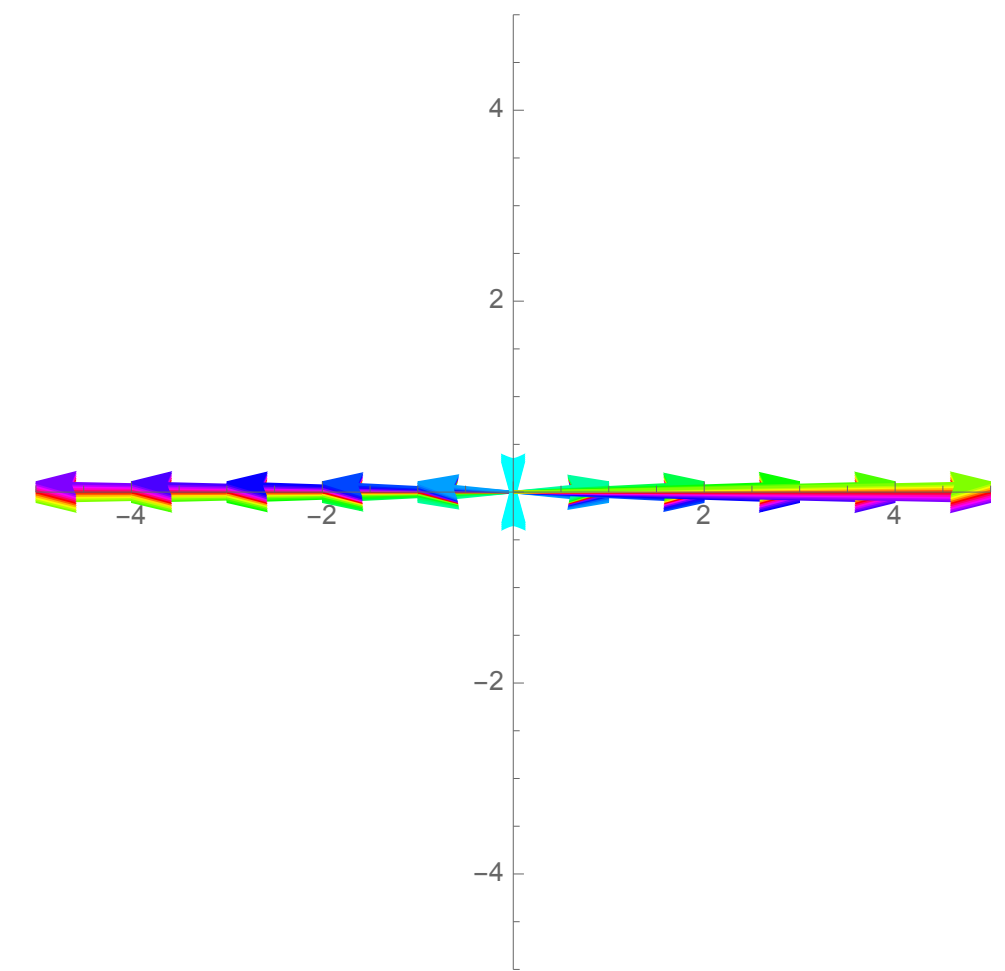
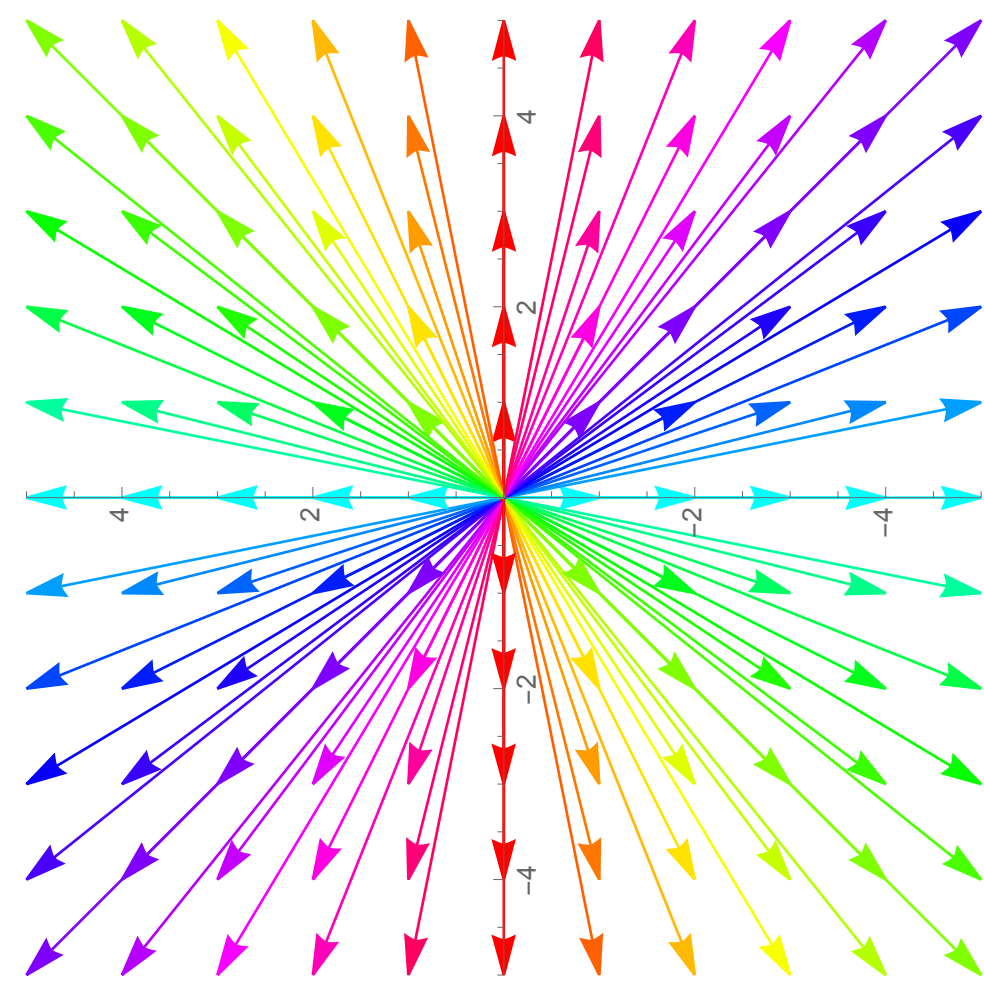


$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \mathbb{R}^2 \rightarrow \mathbb{R}^2 \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

When is a linear
transformation invertible?

invertible

not invertible



The Invertible Matrix Theorem. Let A be a square $n \times n$ matrix with entries in $\mathbb{F}$, and let $T : V \rightarrow V$ be the linear transformation $T(x) = Ax$. The following statements are equivalent.

- (1) A is invertible.
- (2) T is invertible.

- (3) A is row equivalent to I_n .
- (4) A has n pivots. (Prop. 2.3.6)

- (5a) $Ax = b$ has a unique solution for all b in $\mathbb{F}^n$. (Thm. 1.6.8)
- (5b) T is a bijection.
- (5c) The columns of A form a basis for $\mathbb{F}^n$. (Cor. 1.6.9)

- (6a) $Ax = 0$ has only the trivial solution. ($\text{Nul } A = \{0\}$)
- (6b) T is one-to-one.
- (6c) The columns of A are linearly independent.

- (7a) $Ax = b$ is consistent for all b in $\mathbb{F}^n$.
- (7b) T is onto.
- (7c) The columns of A span $\mathbb{F}^n$. ($\text{Col } A = \mathbb{F}^n$)

- (8) $\det(A) \neq 0$.
- (9) $0 \in \mathbb{F}$ is not an eigenvalue of A . (*not until Chapter 4*)

- Ch 1: vector spaces, bases, linear transformations, matrices
- Ch 2: solving linear systems, row reduction, inverse matrices
- Ch 3: determinants
- Ch 4: eigenvalues and eigenvectors
- Ch 5: inner products, orthogonality
- Ch 6 & 9: matrix decompositions